SAC 202 PROBLEMATIC SOILS AND THEIR MANAGEMENT (2+0)

THEORY NOTES





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SAC 202 – Problematic Soils and their management (2+0)

Soil is a natural finite resource base which sustains life on earth. It is a three phase dynamic system that performs many functions and ecosystem services and highly heterogeneous. Soil biota is the biological universe which helps the soil in carrying out its functions. Often soil health is considered independently without referring to interlinked soil functions and also based on soil test for few parameters. Physical condition of soil and biological fertility are overlooked in soil health management which needs revisiting of soil users. Recognizing the importance of soil health in all dimensions, 2015 has been declared as the International Year of Soils by the 68th UN General Assembly. Food and Agriculture organization of the United Nations has formed Global soil partnership with various countries to promote healthy soils for a healthy life and world without hunger. India, the second most populous country in the world faces severe problems in agriculture. It is estimated that out of the 328.8 m ha of the total geographical area in India, 173.65 m ha are degraded, producing less than 20% of its potential yield (Govt. of India, 1990). Soil heterogeneity is the reasons for the diverse nature of cropping and production pattern. Soil heterogeneity is the case where soil in a relatively small area varies greatly in texture, fertility, topography, moisture content, drainage etc. If it exists in large scale due to the parent material or manmade activities, then the problem of soil suitability to agriculture arises. Soil consists of a solid phase (minerals and organic matter) as well as a porous phase that holds gases and water.

The soils which possess characteristics that make them uneconomical for the cultivation of crops without adopting proper reclamation measures are known as problem soils.

Lecture 1. Soil health – Definition – Soil quality indices – Physical indicators

Sustaining soil health is the imminent need of the hour

Soil health is defined as the continued capacity of soil to function as a vital living system, which include sustaining biological productivity of soil, maintain the quality of surrounding air and water environments, as well as promote plant, animal, and human health. Soil function include sustaining biological diversity, activity and productivity, regulating water and solute flow, filtering, buffering, degrading organic and inorganic materials, storing and cycling nutrients and carbon, providing physical stability and support.

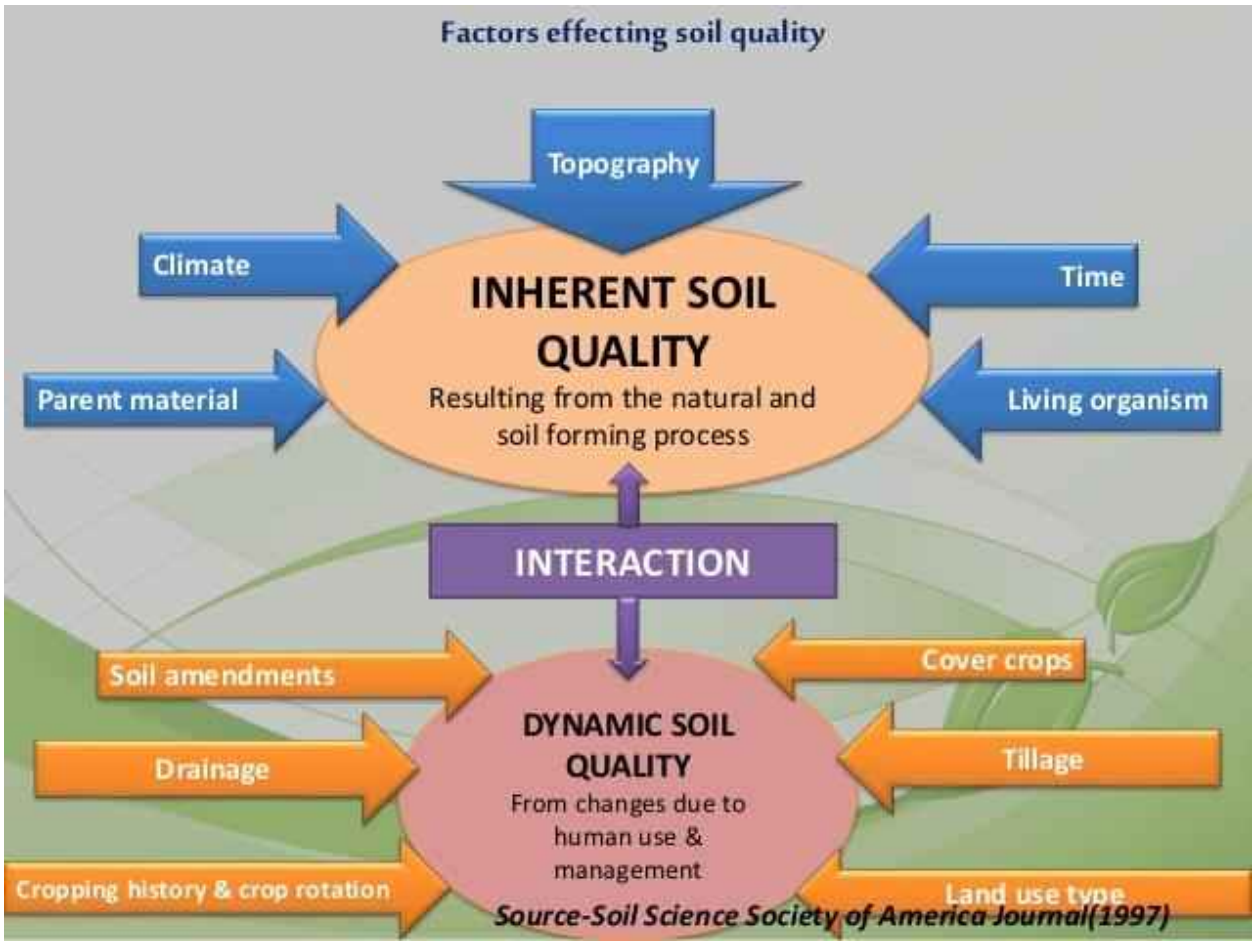
The concept of soil quality emerged in literature in the early 1990s and the first official application of the term was approved by the soil science society of America Ad hoc committee on soil quality. Soil quality was defined as the capacity of a reference soil to function within natural or managed ecosystem boundaries to sustain plant and animal productivity, maintain or enhance water and air quality and support human health and habitation.

What is soil health?

Doran and Parkin defined a soil quality as the capacity of a soil function, within ecosystem and land use boundaries, to sustain productivity, maintain environmental quality and promote plant and animal health.

In general soil health and soil quality are considered synonymous and can be used as interchangeably. The National Resources Conservation Service defines soil quality or soil health similarly, but add inherent and dynamic soil properties

Soil health deals with both the inherent and dynamic soil quality. The former relates to the natural (genetic) characteristics of the soil (e.g., texture), which are the result of soil-forming factors. They are generally cannot easily be amended. On the other hand, the dynamic soil quality component is readily affected by management practices and relates to the levels of compaction, biological functioning, root proliferation, etc which is of most interest to growers because good management allows the soil to come to its full potential.



Important soil functions related to crop production include

- Regulating water
- Sustaining plant and animal life
- Filtering and buffering potential pollutants
- Cycling nutrients
- Physical stability and support
- Pest and weed suppression

Characteristics of a healthy soil

- Good soil tilth
- Sufficient depth

- Sufficient but nor excess supply of nutrients
- Small population of plant pathogens and insect pests
- Good soil drainage
- Large population of beneficial organisms
- Low weed pressure
- Free of chemicals and toxins
- Resistant to degradation

Soil quality index

The USDA Natural Resources Conservation Service defines soil quality as the capacity of specific kind of soil to function, within natural or managed ecosystem boundaries to sustain plant and animal productivity.

Soil quality index is a function of the following parameters:

Soil quality Index (SQI) = f (SP, P, E, H, ER, BD, FQ, MI)

SP- Soil properties, P = Potential Productivity E- Environmental factors H – Health of animal and Human ER- Erodability, BD – Biological diversity FQ = Food quality safety, MI – Management inputs

Maintaining the functions of soil is thus central to the achievement of sustainable development.

Assessment of soil quality

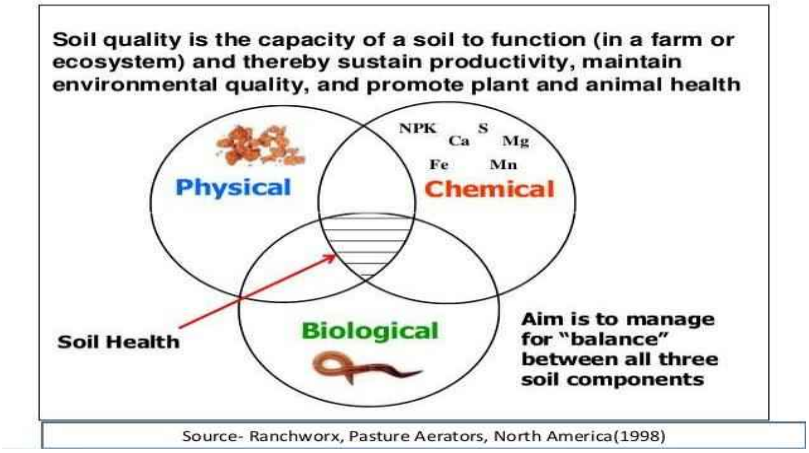
Soil quality or health cannot be determined by measuring only crop yield, water quality, or any other single outcome but with indicators which are measurable properties of soil or plants that provide clues about how well the soil can function. Soil quality cannot measured directly so we evaluate indicators.

Indicators are measurable properties of soil or plants that provides clues about how well the soil can function. Indicators can be physical, chemical and biological characteristics.

Useful indicators means

- correlate well with ecosystem processes
- integrate soil physical, chemical, and biological properties & processes
- be accessible to many users

- be sensitive to management & climate
- be components of existing databases
- be interpretable



Source- Ranchworx, Pasture Aerators, North America(1998)

Soil quality assessments are conducted by evaluating *indicators*. Indicators can be physical, chemical, and biological properties, processes, or characteristics of soils. They can also be morphological or visual features of plants. Indicators are measured to monitor management induced changes in the soil.

Soil quality indicators are selected because of their relationship to specific soil properties and soil quality. For example, soil organic matter is a widely used indicator, because it can provide information about a wide range of properties such as soil fertility, soil structure, soil stability, and nutrient retention. Similarly, plant indicators, such as rooting depth, can provide information about the bulk density or compaction of the soil

Indicators can be assessed by *qualitative* and/or *quantitative* techniques. A qualitative assessment is the determination of the nature of an indicator. A quantitative assessment is the accurate measurement of an indicator. For example, if erosion is the indicator being evaluated, a qualitative assessment would be the observation of rills and gullies in the field, indicating that erosion is occurring.

A quantitative assessment would measure the amount of erosion occurring in the field. In another example, a qualitative assessment of infiltration would be the observation of excessive runoff water from a field. A quantitative assessment would measure the infiltration rate.

Qualitative assessments have an element of subjectivity and, thus, are best done by the same person over time to minimize variability in the results. Indicators measured with a

quantitative method have a precise, numeric value. Therefore, different people conducting the same measurement should be able to produce very similar results.

The minimum data set, which shows the relationship of each indicator to soil health concerns. Compiling a minimum data set helps to identify locally relevant indicators and to evaluate the link between indicators selected and significant soil and plant properties for the region.

Table 1. Example of a Minimum Data Set of Indicators for Soil Quality

Indicator	Relationship to Soil Health
Soil organic matter (SOM)	Soil fertility, structure, stability, nutrient retention, soil erosion, and available water capacity
PHYSICAL	
Soil structure	Retention and transport of water and nutrients, habitat for microbes, and soil erosion
Depth of soil and rooting	Estimate of crop productivity potential, compaction, and plow pan
Infiltration and bulk density	Water movement, porosity, and workability
Water holding capacity	Water storage and availability
CHEMICAL	
pH	Biological and nutrient availability
Electrical conductivity	Plant growth, microbial activity, and salt tolerance
Extractable nitrogen (N), phosphorus (P), and potassium (K)	Plant available nutrients and potential for N and P loss
BIOLOGICAL	
Microbial biomass carbon (C) and N	Microbial catalytic potential and repository for C and N
Potentially mineralizable N	Soil productivity and N supplying potential
Soil respiration	Microbial activity measure

(Adapted from: Doran et al, 1996; Larson and Pierce, 1994; and Seybold et al, 1997)

Physical indicators: It provide information about soil hydrologic characteristics, such as water entry and retention, that influences availability to plants. Some indicators are related to nutrient availability by their influence on rooting volume and aeration status. Other measures tell us about erosional status. Indicators include measures of

- **Aggregate Stability**
- **Available Water Capacity**
- **Bulk Density**
- **Infiltration**
- **Slaking**
- **Soil Crusts**
- **Soil Structure and Macropores**

1. Aggregate stability

Soil aggregates are groups of soil particles that bind to each other more strongly than to adjacent particles. Aggregate stability refers to the ability of soil aggregates to resist disintegration when disruptive forces associated with tillage and water or wind erosion are applied. Wet aggregate stability suggests how well a soil can resist raindrop impact and water erosion, while size distribution of dry aggregates can be used to predict resistance to abrasion and wind erosion.

Why it is important

- Changes in aggregate stability may serve as early indicators of recovery or degradation of soils. Aggregate stability is an indicator of organic matter content, biological activity, and nutrient cycling in soil.
- Generally, the particles in small aggregates (< 0.25 mm) are bound by older and more stable forms of organic matter. . Greater amounts of stable aggregates suggest better soil quality. When the proportion of large to small aggregates increases, soil quality generally increases.

2. Available Water Capacity

The amount of water in soil is based on rainfall amount, what proportion of rain infiltrates into the soil, and the soil's storage capacity. Available water capacity is the maximum amount of plant available water a soil can provide. **It is an indicator of a soil's ability to retain water and make it sufficiently available for plant use.**

Available water capacity is the water held in soil between its field capacity and permanent wilting point. Field capacity is the water remaining in a soil after it has been thoroughly saturated and allowed to drain freely, usually for one to two days. Permanent wilting point is the moisture content of a soil at which plants wilt and fail to recover when supplied with sufficient moisture. Water capacity is usually expressed as a volume fraction or percentage, or as a depth (in or cm).

Why it is important: Soil is a major storage reservoir for water. Water availability is an important indicator because plant growth and soil biological activity depend on water for hydration and delivery of nutrients in solution. Runoff and leaching volumes are also determined by storage capacity and pore size distribution.

Available water capacity is used to develop water budgets, predict droughtiness, design and operate irrigation systems, design drainage systems, protect water resources, and predict yields.

3. Bulk Density

Bulk density is an indicator of soil compaction. It is calculated as the dry weight of soil divided by its volume. Bulk density is typically expressed in g/cm3.

Why it is important: Bulk density reflects the soil's ability to function for structural support, water and solute movement, and soil aeration.

Table 1. General relationship of soil bulk density to root growth based on soil texture.		
Soil Texture	**Ideal bulk densities for plant growth (g/cm3)**	**Bulk densities that restrict root growth (g/cm3)**
Sandy	< 1.60	> 1.80
Silty	< 1.40	> 1.65
Clayey	< 1.10	> 1.47

4. Infiltration

Infiltration is the downward entry of water into the soil. The velocity at which water enters the soil is infiltration rate. Infiltration rate is typically expressed in inches per hour. Water from rainfall or irrigation must first enter the soil for it to be of value.

Why it is important: Infiltration is an indicator of the soil's ability to allow water movement into and through the soil profile. Soil temporarily stores water, making it available for root uptake, plant growth and habitat for soil organisms.

5. Soil Crusts

Structural soil crusts are relatively thin, dense, somewhat continuous layers of non-aggregated soil particles on the surface of tilled and exposed soils.

Structural crusts develop when a sealed-over soil surface dries out after rainfall or irrigation. Water droplets striking soil aggregates and water flowing across soil breaks aggregates into individual soil particles. Fine soil particles wash, settle into and block surface pores causing the soil surface to seal over and preventing water from soaking into the soil. As the muddy soil surface dries out, it crusts over.

Structural crusts range from a few tenths to as thick as two inches. A surface crust is much more compact, hard and brittle when dry than the soil immediately beneath it, which may be loose and friable.

Why it is important: A surface crust indicates poor infiltration, a problematical seedbed, and reduced air exchange between the soil and atmosphere. It can also indicate that a soil has a high sodium content that increases soil dispersion when it is wetted by rainfall or irrigation.

6. Soil Structure and Macro pores

Sand, silt and clay particles are the primary mineral building blocks of soil. Soil structure is the combination or arrangement of primary soil particles into aggregates. Soil structure affects water and air movement through soil, greatly influencing soil's ability to sustain life and perform other vital soil functions.

Soil pores exist between and within aggregates and are occupied by water and air. Macro pores are large soil pores, usually between aggregates, that are generally greater than 0.08 mm in diameter. Macro pores drain freely by gravity and allow easy movement of water and air. They provide habitat for soil organisms and plant roots can grow into them. With diameters less than 0.08 mm, micropores are small soil pores usually found within structural aggregates. Suction is required to remove water from micropores.

Why it is important: Important soil functions related to soil structure are: sustaining biological productivity, regulating and partitioning water and solute flow, and cycling and storing nutrients. Soil structure and macro pores are vital to each of these functions based on their influence on water and air exchange, plant root exploration and habitat for soil organisms.

Granular structure is typically associated with surface soils, particularly those with high organic matter. Granular structure is characterized by loosely packed, crumbly soil aggregates and an interconnected network of macro pores that allow rapid infiltration and promote biological productivity. Structure and pore space of subsurface layers affects drainage, aeration, and root penetration. Platy structure is often indicative of compaction.

2. Soil Quality Indices - Chemical and biological indicators

Chemical indicators can give you information about the equilibrium between soil solution (soil water and nutrients) and exchange sites (clay particles, organic matter); plant health; the nutritional requirements of plant and soil animal communities; and levels of soil contaminants and their availability for uptake by animals and plants. Indicators include measures of:

- **Electrical Conductivity**
- **Soil Nitrate**
- **Soil Reaction (pH)**
- **Reactive carbon**

Soil pH

What it is: Soil pH generally refers to the degree of soil acidity or alkalinity. Chemically, it is defined as the log10 hydrogen ions (H+) in the soil solution. The pH scale ranges from 0 to 14; a pH of 7 is considered neutral. If pH values are greater than 7, the solution is considered basic or alkaline; if they are below 7, the solution is acidic.

Why it is important

- Soil pH affects the soil's physical, chemical, and biological properties and processes, as well as plant growth.
- The nutrition, growth, and yields of most crops decrease where pH is low and increase as pH rises to an optimum level Many crops grow best if pH is close to neutral (pH 6 to 7.5) although a few crops prefer acid or alkaline soils.
- In acid soils, calcium and magnesium, nitrate-nitrogen, phosphorus, boron, and molybdenum are deficient, whereas aluminum and manganese are abundant, sometimes at levels toxic to some plants.
- Phosphorus, iron, copper, zinc, and boron are frequently deficient in very alkaline soils. Bacterial populations and activity decline at low pH levels, whereas fungi adapt to a large range of pH (acidic and alkaline). Most microorganisms have an optimum pH range for

survival and function (At very acid or alkaline pH levels, organic matter mineralization is slowed down or stopped because of poor microbial activity linked to bacteria.

- Nitrification and nitrogen fixation are also inhibited by low pH.
- The mobility and degradation of herbicides and insecticides, and the solubility of heavy metals are pH dependent. The effects of soil pH on cation availability influence aggregate stability since multivalent cations, such as calcium ions, act as bridges between organic colloids and clays. Some diseases thrive when the soil is alkaline or acidic.

2. Reactive carbon

Reactive carbon (RC), also known as permanganate oxidizable carbon (POxC) or active carbon, is a fraction of the soil organic matter (SOM) pool of C that is oxidizable in the presence of potassium permanganate in solution. Carbon oxidized by this compound includes the C most readily degradable by microorganisms as well as that bound to soil minerals, making RC interpretation somewhat difficult. Because of this association to the mineral fraction, RC is considered a chemical indicator, not a biological indicator.

The residence time of RC is estimated to be 2 to 5 years, in contrast to recalcitrant C (e.g., humus) that has a turnover time of several hundreds to thousands of years. Reactive carbon originates from the various fractions of SOM. These fractions include fresh organic material, soil microbial biomass, particulate organic matter, and other easily metabolized organic compounds, such as carbohydrates (sugars) and proteins (amino acids), as well as C loosely bound to soil minerals.

Why it is important

Reactive carbon may be used as an indicator of the change produced by cropping and soil management practices that manipulate SOM content.

Soil Electrical Conductivity

Soil electrical conductivity (EC) measures the ability of soil water to carry electrical current. Electrical conductivity is an electrolytic process that takes place principally through water-filled pores. Cations (Ca^{2+}, Mg^{2+}, K^{+}, Na^{+}, and NH_4^{+}) and anions (SO_4^{2-}, Cl^{-}, NO_3^{-}, and

HCO_3) from salts dissolved in soil water carry electrical charges and conduct the electrical current. Consequently, the concentration of ions determines the EC of soils.

Why it is important

In agriculture, EC has been used principally as a measure of soil salinity. However, in non-saline soils, EC can be an estimate of other soil properties, such as soil moisture, soil depth, and nutrient content (e.g., nitrate concentration). EC is expressed in deci Siemens per meter (dS/m). Soils with EC values higher than 4mmhos/cm (4 dS/m) measured on their saturated extract are saline, and growth of crops sensitive to salinity is restricted on these soils. During recent years, censors were used to map electrical conductivity and deduct nitrate concentration; Smith and Doran (1996) reported nitrate increases with increasing EC.

Soil nitrate

Soil Nitrate Nitrate (NO_3^-) is a form of inorganic nitrogen (N) naturally occurring in soils. Sources of soil NO_3^- include decomposing plant residues and animal manure/compost, chemical fertilizers, exudates from living plants, rainfall, and lightning. Eventually, nitrate ions immobilized by microorganisms (nitrate taken up by microorganisms) are converted into organic forms and released back to the soil in plant-available forms when dead soil organisms are fed upon or decompose. In well drained soils, ammonium (NH_4^+) and ammonia (NH_3) are converted into NO_3^- by very specific populations of aerobic bacteria. This process is known as nitrification.

Another biological N transformation is denitrification, which is the conversion of NO_3^- into nitrous oxide (N_2O), nitrogen dioxide (NO_2), and nitrogen gas (N_2) that often occurs in anaerobic soils, such as waterlogged soils and wetlands. Even when nitrifying bacteria are very active in the outer parts of aggregates in well aerated soils, denitrification may still occur in anaerobic microsites inside the aggregates.

Nitrate is very soluble in water and can be easily transported by runoff and other surface and subsurface flows to rivers and lakes or moved downward to ground water.

Soil biological Indicators

Indicators can tell us about the organisms that form the soil food web that are responsible for decomposition of organic matter and nutrient cycling. Information about the numbers of organisms, both individuals and species, that perform similar jobs or niches, can indicate a soil's ability to function or bounce back after disturbance (resistance and resilience). Indicators include measures of:

- **Earthworms**
- **Particulate Organic Matter**
- **Potentially Mineralizable Nitrogen**
- **Respiration**
- **Soil Enzymes**
- **Total Organic Carbon**

Earthworms

Earthworms are native to non-glaciated areas of North America, but non-native species from Europe and Asia also exist here. Earthworms are classified into three groups based on their habitat. Litter-dwellers live in the litter, ingest plant residues, and may be absent in plowed, litter-free soil. Mineral soil-dwellers live in topsoil that is rich in organic matter. They burrow narrow channels and feed on a mixture of soil and plant residues. Deep soil-burrowers (night crawlers) dig long, large burrows into deep soil layers. They carry with them plant residues for consumption. Earthworm cast is digested material that is excreted back into the soil. Cast is enriched with nutrients (N, P, K, and Ca) and microorganisms during its passage through the worm’s digestive system. Fresh cast is a site of intense microbial activity and nutrient cycling. Earthworms contribute nutrients to the soil and improve porosity, tilth, and root development. They are measured in number/m2.

Why it is important:

- Earthworms play a key role in modifying the physical structure of soils by producing new aggregates and pores, which improves soil tilth, aeration, infiltration, and drainage.

- Earthworms produce binding agents responsible for the formation of water-stable macro-aggregates. They improve soil porosity by burrowing and mixing soil.
- As they feed, earthworms participate in plant residue decomposition, nutrient cycling, and redistribution of nutrients in the soil profile. Their casts, as well as dead or decaying earthworms, are a source of nutrients.

Particulate Organic Matter

Particulate organic matter (POM) fraction referred as all soil organic matter (SOM) particles less than 2 mm and greater than 0.053 mm in size (Cambardella and Elliot, 1992). POM is biologically and chemically active and is part of the labile (easily decomposable) pool of soil organic matter (SOM). Studies have shown that POM accounts for few to large amounts of soil C (20 % and more) in some soils of Eastern Canada and the USA depending upon agroecosystems and management practices

Why it is important:

- It is a source of food/energy for microorganisms and soil animals as well as nutrients for plant growth.
- Particulate organic matter enhances aggregate stability, water infiltration and soil aeration; it increases cation exchange capacity and buffering pH. It also binds environmental pollutants such as heavy metals and pesticides.
- Particulate organic matter may play an important role in the suppression of soil borne diseases (e.g. damping off of cucumber) by compost. This may be explained by the fact that POM is an important source of food/energy in the compost for microorganisms responsible of disease suppression.

Soil respiration

Carbon dioxide (CO_2) release from the soil surface is referred to as soil respiration. This CO_2 results from several sources, including aerobic microbial decomposition of soil organic matter (SOM) to obtain energy for their growth and functioning (microbial respiration), plant root and faunal respiration, and eventually from the dissolution of carbonates in soil solution.

Soil respiration is one measure of biological activity and decomposition. The rate of CO_2 release is expressed as CO_2-C lbs/acre/day (or kg/ha/d). It can be measured by simple field methods or more sophisticated field and laboratory methods. During the decomposition of SOM, organic nutrients contained in organic matter (e.g., organic phosphorus, nitrogen, and sulfur) are converted to inorganic forms that are available for plant uptake. This conversion is known as mineralization. Soil respiration is also known as carbon mineralization.

Why it is important

- Soil respiration reflects the capacity of soil to support soil life including crops, soil animals, and microorganisms. It describes the level of microbial activity, SOM content and its decomposition.
- In the laboratory, soil respiration can be used to estimate soil microbial biomass and make some inference about nutrient cycling in the soil.
- Soil respiration also provides an indication of the soil's ability to sustain plant growth. Excessive respiration and SOM decomposition usually occurs after tillage due to destruction of soil aggregates that previously protected SOM and increased soil aeration.

Soil enzymes

Soil enzymes increase the reaction rate at which plant residues decompose and release plant available nutrients. The substance acted upon by a soil enzyme is called the substrate. For example, glucosidase (soil enzyme) cleaves glucose from glucoside (substrate), a compound common in plants. Enzymes are specific to a substrate and have active sites that bind with the substrate to form a temporary complex. The enzymatic reaction releases a product, which can be a nutrient contained in the substrate.

Sources of soil enzymes include living and dead microbes, plant roots and residues, and soil animals. Enzymes stabilized in the soil matrix accumulate or form complexes with organic matter (humus), clay, and humus-clay complexes, but are no longer associated with viable cells.

Why it is important:

- Enzymes respond to soil management changes long before other soil quality indicator changes are detectable.
- Soil enzymes play an important role in organic matter decomposition and nutrient cycling .Some enzymes only facilitate the breakdown of organic matter (e.g., hydrolase, glucosidase), while others are involved in nutrient mineralization (e.g., amidase, urease, phosphatase, sulfates).
- With the exception of phosphatase activity, there is no strong evidence that directly relates enzyme activity to nutrient availability or crop production.
- The relationship may be indirect considering nutrient mineralization to plant available forms is accomplished with the contribution of enzyme activity.

Total Organic Carbon

Total organic carbon (TOC) is the carbon (C) stored in soil organic matter (SOM). Organic carbon (OC) enters the soil through the decomposition of plant and animal residues, root exudates, living and dead microorganisms, and soil biota. SOM is the organic fraction of soil exclusive of non decomposed plant and animal residues. SOM is a heterogeneous, dynamic substance that varies in particle size, C content, decomposition rate, and turnover time.

Soil Organic Carbon (SOC) is the main source of energy for soil microorganisms. SOM contains approximately 58% C; therefore, a factor of 1.72 can be used to convert OC to SOM. There is more inorganic C than TOC in calcareous soils. TOC is expressed as percent C per 100 g of soil.

Why it is important

- SOC is one of the most important constituents of the soil due to its capacity to affect plant growth as both a source of energy and a trigger for nutrient availability through mineralization.

- SOC fractions in the active pool are the main source of energy and nutrients for soil microorganisms. Humus participates in aggregate stability, and nutrient and water holding capacity.
- OC compounds, such as polysaccharides (sugars) bind mineral particles together into microaggregates.
- Glomalin, a SOM substance that may account for 20 % of soil carbon, glues aggregate together and stabilizes soil structure making soil resistant to erosion, but porous enough to allow air, water and plant roots to move through the soil.
- Organic acids (e.g., oxalic acid), commonly released from decomposing organic residues and manures, prevents phosphorus fixation by clay minerals and improve its plant availability, especially in subtropical and tropical soils.
- An increase in SOM, and therefore total C, leads to greater biological diversity in the soil, thus increasing biological control of plant diseases and pests.

Potentially mineralizable nitrogen

PMN can be defined as the fraction of organic nitrogen converted to plant available (or mineral) forms under specific conditions of temperature, moisture, aeration, and time. Determining levels of PMN can provide an estimate of available N in the soil.

PMN originates mainly from microbial biomass and plant and animal tissues the main source of the organic nitrogen pool. It represents the fraction of nitrogen easily decomposable by soil microorganisms and is considered an indirect measure of nitrogen availability during the crop growing season

Lec-3&4. Wastelands – Characteristics and Distribution in India

When a large percentage of the total land resources is lying unproductive or is being utilized at much less than its optimum potential the soil are called as wastelands. The word meaning of wastelands is ruined lying unused, devasted, vain, useless, worthless. In earlier time, the village common land which was not used for agriculture was called as wasteland.

American society of soil science defined wasteland as a piece of land not capable of producing materials or services of suitable value. A few authors' defined wastelands are those lands which produces less than 20 % of economic potential. According to other authors wastelands include overgrazed wastelands, degraded forests, eroded valleys, hilly slopes, over irrigated saline and alkali soils, drought pastures and waterlogged marshy lands where crop production is not economical.

Wasteland Survey and Reclamation Committee, Ministry of food and Agriculture, govt, of India (1961) has defined wastelands as those lands which are either not available for cultivation , uncultivated lands excluding fallow, barren lands, cultural wastelands , land under miscellaneous trees and permanent pastures.

Bhumla and Khare (1987) gives a working definition of wastelands. According to them "Those lands are wastelands (a) which are ecologically unstable, (b) whose top soil has nearly been completely lost, and (c) which have developed toxicity in the root zones or growth of most plants, both annual crops and trees".

The National Wastelands Development Board's (NWDB) Technical Task Group's Report (1986) after considering various definitions of wasteland defines wastelands as that land which is presently lying unutilized due to different constraints.

Kinds of wastelands

National wastelands development board classifies wastelands into two categories:

1. Cultivable wastelands
2. Uncultivable wastelands

I. Cultivable / Cultural wastelands:

Culturable wastelands include the lands which have the potential for the development of vegetative cover and are not being used due to different constraints of varying degrees.

The development of different types of vegetation cover in different categories of cultivable wastelands will depend upon the land capability defined by the factors such as relief, depth of soil, soil qualities, climate, present land-use, ownership pattern and tenurial system etc.
Categories of cultural wastelands

Gullied land/ravinous land

Gullied land- the gullies are the result of the localized surface runoff affecting the friable unconsolidated material resulting in the formation of perceptible channels resulting in undulating terrain having a maximum depth of 3m.

Ravinous land - this category of land contains system of gullies 77 running more or less parallel lo each other in deep alluvium and entering a nearby river flowing much lower than the surrounding table lands as are found along the courses of many rivers. The ravines are formed by gullies developed along river courses

The Ravinous land are categorized into three categories

1. Shallow – 3-6 meters deep
2. Medium – 6-9 meters deep
3. Deep - more than 9 meters deep

Surface waterlogged and marshy lands

This is the land where water table is at /or near the surface and water stands for most of the year. Water logging occurs due to the rise of sub-soil water-table and this happens almost all the over-irrigated areas.

Marsh is the land which permanently or periodically is inundated by water and is characterized by vegetation that includes grasses and reeds. Marshes are classified into salt, brackish and fresh water categories depending on the salinity of water.

Salt affected lands

Presence of excess soluble and exchangeable sodium in salt affected soil has adverse impact on plant growth. Alkali soil has more than 15% exchangeable sodium in which carbonate and bicarbonate salts of sodium predominate. In saline soils chlorides and sulphates of sodium, calcium, magnesium and sometimes though rarely nitrates and K are found.

Shifting cultivation Area

Such land is the result of cyclic land use consisting of felling of trees and burning of forest areas for growing crops. This results in extensive soil losses leading to land degradation.

Strip lands:

These lands are long narrow strips of lands or either side of roads, railway lines and canal banks. Such types of lands are used for tree cultivation

Degraded/under-utilized notified forest land

Lands as notified under the forest act and those lands with various types of forest cover in which denudation of vegetation cover is less than 20 % of canopy cover, are classified as degraded land.

Degraded pastures/grazing land

All those grazing lands in non-forest areas, whether or not they are permanent pastures or meadows, which have become degraded due to lack of proper soil conservation and drainage measures fall under this category.

II. <u>Unculturable wastelands</u>

Uncultural wastelands which cannot be used for vegetation are classified as

a) Brown rocky / stony / shut of rocks – The upper surface of these soils are covered with rocks, stones, boulders and other geological formations. Such land surface is not capable of supporting any living organism.

b) Steep sloppy areas – Very steep slopy lands tend to erosion and landslides

c) Snow covered and / or glacier lands – The lands above snowline are permanently covered by snow and glacier. These lands are unavailable for growing of any vegetation.

Wastelands can be classified into following categories

- **Gullied or ravenous land** - It formed due to localized overland flow of water. Water flow can erode the loose soil and create minor channels called Gullies. Network of gullies produce a ravenous land affected by drainage. In India, about 0.02 M. sq.km of land comes under this category. Madhya Pradesh and Rajasthan are the states having more ravine lands, followed by Uttar Pradesh.

- **Land with or without scrub-** The land with or without scrub is generally prone to deterioration due to erosion. Erosion is more active on elevated portions of land. This category covers a maximum of 0.194000 m. sq.km. About 19 % of this type exists in Madhya Pradesh followed by Maharastra. Gujarat has about 11 %.

- **Waterlogged land or Marsh land** -Wetlands is a collective term for marshes, swamps and bogs.

- **Land of soil salinity/alkalinity** - Land of soil salinity or alkalinity severely affects the growth of plants due to excessive salts or exchangeable sodium. About 0.020 m. sq.km has come under this category. Gujarat, Uttar Pradesh and Tamil Nadu are the states mainly having this problem.

- **Land of shifting cultivation** - Barren lands exposed with rocks and stones need more attention for development. In India about 0.065 M. sq.km of land are Barren lands. States like Jammu & Kashmir, Andhra Pradesh and Rajasthan have more Barren lands.

- **Underutilized Degraded Notified Forestland -**Barren lands exposed with rocks and stones need more attention for development. In India about 0.065 M. sq.km of land are Barren lands. States like Jammu & Kashmir, Andhra Pradesh and Rajasthan have more Barren lands.

- **Degraded Pasture/ Grazing Land** -The value of pasture land is largely determined by its productivity. It is important to delineate the pasture land for grazing.

- **Degraded Land of Plantation Crops**-The maximum is seen in Himachal Pradesh followed by M.P and Jammu & Kashmir
- **Desertic sands or Coastal sands** - Sandy areas are expected along the coasts and riverine areas. Stabilized accumulation of sand or alluvium may affect the plains or coasts. About 0.05 M. sq.km of land is covered by this category. Rajasthan is the main state followed by Assam.

- **Mining or industrial waste lands -** Dumping of mine wastes and tailings and release of effluents over adjacent areas create a land unsuitable for any purpose. India, about 0.0012

M.sq.km of land belongs to this category. Bihar, Madhya Pradesh, Rajasthan and Tamil Nadu are the states having such wasteland.

- **Barren land with rock/ stony wastes/ Steep sloping lands** - Frequent occurrence of landslides may not allow the land for appropriate use. Grading of slopes may help in planting tropical crops. Nearly 0.008 M.sq.km of land comes under steep slopes. It is a problem in Jammu & Kashmir, Himachal Pradesh, Maharashtra and Uttar Pradesh.

- **Snow covered lands** -The land which cannot be used directly is the snow-covered land. Glaciers are extreme environments. About 0.05 M.sq.km of land in India are snow covered. Jammu and Kashmir, Uttar Pradesh, Himachal Pradesh and Arunachal Pradesh have these types of lands.

Causes of wastelands formation

a) Deforestation

b) Over-cultivation

c) Over grazing

d) Unskilled irrigation

e) Improper developmental activities such as dumping of wastes, mine wastes

Wasteland distribution in India

Shri. B.B. Vohra, Chairman, National committee on Environmental Planning states that about 18 million hectare of 305 million hectares land is under other non agricultural uses. Another 21 million hectares are barren and uncultivable because of being snow bound or rocky in nature. Thus only 266 million hectares of land is available for agricultural works, crop production and other services.

Table 23.10: Aerial Estimation of Wasteland, (Statewise) in 2000

			(Area in ha)
1.	**Rajasthan**	Salt affected Land	497.640
		Gullied or Ravinous Land	8766.120
		Undulating Upland with or without scrub	7656
		Sandy Area	130193.938
		Barren Hill Ridge or Rock outcrop	7770.840
		Total	154884.538
		Percentage of State area	45.258
2.	**Jammu & Kashmir**	Salt affected Land	38.280
		Undulating Upland with or without scrub	76.560
		Jhum or Forest Blank	2373.360
		Barren Hill Ridge or Rock outcrop	153.120
		Snow covered or Glacial Area	70054.141
		Total	72695.461
		Percentage of State area	32.710
3.	**Tripura**	Undulating Upland with or without scrub	382.800
		Jhum or Forest Blank	2641.320
		Snow covered or Glacial Area	229.680
		Total	3253.800
		Percentage of State area	31.056
4.	**Himachal Pradesh**	Undulating Upland with or without scrub	459.360
		Jhum or Forest Blank	995.280
		Snow covered or Glacial Area	13589.800
		Total	15044.440
		Percentage of State area	27.022
5.	**Manipur**	Undulating Upland with or without scrub	306.240
		Jhum or Forest Blank	5014.680
		Barren Hill Ridge or Rock outcrop	344.520
		Total	5665.440
		Percentage of State area	25.340
6.	**Sikkim**	Undulating Upland with or without scrub	765.600
		Snow covered or Glacial Area	1077.840
		Total	1843.440
		Percentage	25.256
7.	**Mizoram**	Jhum or Forest Blank	3981.120
		Snow covered or Glacial Area	497.640
		Total	4478.760
		Percentage of State area	21.238

			(Area in ha)
8.	**Meghalaya**	Jhum or Forest Blank	4555.320
		Barren Hill Ridge or Rock outcrop	191.400
		Total	4746.720
		Percentage of State area	21.106
9.	**Nagaland**	Jhum or Forest Blank	3253.800
		Total	3253.800
		Percentage of State area	19.686
10.	**Gujarat**	Salt affected Land	1110.120
		Gullied or Ravinous Land	191.400
		Jhum or Forest Blank	76.560
		Sandy Area	1492.920
		Barren Hill ridge or Rock outcrop	842.160
		Total	3713.160
		Percentage of State area	8.396
11.	**Arunachal Pradesh**	Undulating Upland with or without scrub	191.400
		Jhum or Forest Blank	5512.320
		Snow covered or Glacial Area	7732.560
		Total	13436.280
		Percentage of State area	16.076
12.	**Uttar Pradesh**	Salt affected Land	19407.961
		Gullied or Ravinous Land	8842.680
		Undulating Upland with or without scrub	880.440
		Jhum or Forest Blank	2794.440
		Sandy Area	918.720
		Barren Hill Ridge or Rock outcrop	880.440
		Snow covered or Glacial Area	7043.520
		Total	40768.201
		Percentage of State area	13.846
13.	**Kerala**	Undulating Upland with or without scrub	803.880
		Jhum or Forest Blank	3483.480
		Barren Hill Ridge or Rock outcrop	918.720
		Snow covered or Glacial Area	114.840
		Total	5320.920
		Percentage of State area	13.690
14.	**Tamil Nadu**	Salt affected Land	574.200
		Gullied or Ravinous Land	4210.800

		(Area in ha)
	Undulating Upland with or without scrub	6507.600
	Jhum or Forest Blank	3330.360
	Sandy Area	1914
	Barren Hill Ridge or Rock outcrop	1071.840
	Total	17610.800
	Percentage of State area	13.538
15. **Odissa**	Salt affected Land	153.120
	Undulating Upland with or without scrub	12326.160
	Jhum or Forest Blank	6622.440
	Sandy Area	382.800
	Total	19484.506
	Percentage of State area	12.506
16. **Karnataka**	Salt affected Land	267.960
	Gullied or Ravinous Land	2220.240
	Undulating Upland with or without scrub	12517.560
	Jhum or forest Blank	3100.680
	Barren Hill Ridge or Rock outcrop	3828
	Total	21934.440
	Percentage of State area	11.436
17. **Andhra Pradesh**	Salt affected Land	880.440
	Gullied or Ravinous Land	4517.040
	Undulating Upland with or without scrub	18642.359
	Jhum or Forest Blank	880.440
	Sandy Area	1033.560
	Barren Hill Ridge or Rock outcrop	4708.440
	Total	30662.270
	Percentage of State area	11.076
18. **Madhya Pradesh**	Gullied or Ravinous Land	21283.680
	Undulating Upland with or without scrub	13436.280
	Jhum or Forest Blank	7005.240
	Barren Hill Ridge or Rock outcrop	2411.840
	Total	54137.040
	Percentage of State area	9.966
19. **Maharashtra**	Salt affected Land	1186.680
	Gullied or Ravinous Land	7388.040
	Undulating Upland with or without scrub	19637.641
	Jhum or Forest Blank	1454.640

			(Area in ha)
		Barren Hill Ridge or Rock outcrop	995.280
		Total	30662.281
		Percentage of State area	9.962
20.	**Goa, Daman and Diu**	Undulating Upland with or without scrub	306.240
		Jhum or Forest Blank	38.280
		Total	344.520
		Percentage of State area	9.034
21.	**Haryana**	Salt affected Land	1110.120
		Gullied or Ravinous Land	191.400
		Jhum or Forest Blank	76.560
		Sandy Area	1492.920
		Barren Hill Ridge or Rock outcrop	842.160
		Total	3713.160
		Percentage of State area	8.396
22.	**Bihar**	Gullied or Ravinous Land	1875.720
		Undulating Upland with or without scrub	8727.840
		Jhum or Forest Blank	2373.360
		Barren Hill Ridge or Rock outcrop	2335.080
		Total	15312.000
		Percentage of State area	8.806
23.	**Punjab (Including Chandigarh)**	Salt affected Land	1301.520
		Sandy Area	2603.040
		Total	3904.560
		Percentage of State area	7.730
24.	**West Bengal**	Salt affected Land	1263.240
		Undulating Upland with or without scrub	4249.080
		Jhum or Forest Blank	535.920
		Barren Hill Ridge or Rock outcrop	306.240
		Total	6354.480
		Percentage of State area	7.232
25.	**Assam**	Undulating Upland with or without scrub	574.200
		Jhum or Forest Blank	2373.360
		Total	2947.560
		Percentage of State area	3.752

Table 2: Category-wise wastelands of India

Category	Area (sq km)	% of total geographical area covered
Gullied and/ or ravinous land	20553.35	0.65
Land with or without scrub	194014.29	6.13
Waterlogged and marshy land	16568.45	0.52
Land affected by salinity/alkalinity-coastal/inland	20477.38	0.65
Shifting cultivation area	35142.20	1.11
Under-utilized/degraded notified forest land	140652.31	4.44
Degraded pastures/grazing land	25978.91	0.82
Degraded land under plantation crop	5828.09	0.18
Sands-Inland/coastal	50021.65	1.58
Mining/industrial wasteland	1252.13	0.04
Barren rocky/stony waste/sheet rock area	64584.77	2.04
Steep sloping area	7656.29	0.24
Snow covered and/or glacial area	55788.49	1.76
Total wasteland area	638518.31	20.17

Making wastelands productive

The lands which are unsuitable for cultivation due to one or more limitations can be used for raising fire wood plantations and for developing pasture lands. Wastelands for Agricultural use would not be economically used on the account of the following reasons

1) Investment in wastelands reclamation for agriculture use appears to be non – remunerative and non beneficial without irrigation.
2) For greater security investment , short term forestry schemes with fast growing species and grassland husbandry have already established their creditability

3) Major reclamation programmes of wastelands cannot bank or help on irrigation except in areas adjoining river and streams in the region.

At present, our country is under great stress of increased population and low income. At present it is very essential to reorient our entire farming economy by adopting advanced and new developed agricultural systems instead of expanding and developing them through quite long years old age farming practices and system.

Lec 5. Soil related constraints- physical - Highly permeable, Slowly permeable and fluffy soils and management technologies

Highly permeable soils

Occurrence

Sandy soils containing more than 70 per cent sand fractions occur in coastal areas, river delta and in the desert belts. Such soils do occur in Coimbatore, Trichy, Kanyakumari, Tuticorin, Thanjavur and Tirunelveli districts and in part of coastal areas in Tamil Nadu. A total area of 24, 12, and 086 ha in Tamil Nadu were affected by excessively permeable soils (23:97 % of total geographic area)

Impact of excessively permeable soils on Soil health

The nature of excessive permeability of the sandy soils results in very poor water retention capacity, very high hydraulic conductivity and infiltration rates. Excessively permeable soils are those having high amount of sand exceeding 70- percent. Due to this, the soils are inert and unable to retain nutrient and water. These soils being devoid of finer particles and organic matter, the aggregates are weakly formed, the non-capillary pores dominating with very poor soil structure. So whatever the nutrients and water added to these soils are not utilized by the crops and subjected to loss of nutrients and water. In addition, it is not providing anchorage to the crops grown. The physical properties of the soil are as follows:

Sand	- > 70 per cent
Clay	- >15 per cent
Water holding capacity	- 20 cm/hr

Management technology

- The soils should be ploughed uniformly.
- Twenty four hours after a good rainfall or irrigation, the soil should be rolled 10 times with 400 kg stone roller of 1 m long or an empty tar drum filled with 400 kg sand at optimum moisture (13 %)
- Then shallow ploughing should be given and crops can be raised.

1. Application of clay soil up to a level 100 t ha^{-1} based on the severity of the problem and availability of clay materials.

2. Application of organic materials like farm yard manure, compost, press mud, sugar factory slurry, composted coir pith, sewage sludge etc.
3. Providing asphalt sheet, polythene sheets etc. below the soil surface to reduce the infiltration rate.
4. Crop rotation with green manure crops like Sunhemp, sesbania, daincha, kolinchi etc.

Improving water and nutrient holding capacities

1. Application of organics like compost, Farm Yard Manure, Green manures and Green leaf manures,
2. Soil breeding (adding tank silt or soil from other field to increase the clay and silt contents of sandy soil)
3. Frequent irrigation with low quantity of water
4. Frequent split application of fertilizers
5. Application of slow release fertilizers

Slow permeable soils/Impermeable soils and their management

Occurrence

In Tamil Nadu, the area under slow permeable soils is around 7,54,631 ha (7.5% of TGA)

Causes

Slow permeable soil is mainly due to very high clay content, infiltration rate < 6cm/day, entry of water into the soil profile reduced so more runoff which eventually leads to soil erosion and nutrient removal. Since the capillary porosity is high it leads to impeded drainage. Poor aeration and reduced conditions. Nutrient fixation and toxicity of elements ultimately leads to poor crop growth and in certain cases leads to complete death of crops.

Remedial measures

(i) Incorporation of organics

Addition of organics namely FYM/composted coir pith/press mud at 12.5 t ha^{-1} found to be optimum for the improvement of the physical properties besides, it facilitates water movement to the root zone.

(ii) Formation of ridges and furrows

For rainfed crops, ridges are formed along the slopes for providing adequate aeration to the root zone. Interception drainage channels should be provided to carry the excess water to rice fields located at lower end of the slope.

(iii) Raised and sunken beds formation in between adjacent raised beds

The bulk density was found to be reduced due to increase in non capillary pores in upper 10 cm layer of raised bed besides increase in yield of crops by forming raised and sunken beds.

(iv) Formation of broad beds

To reduce the amount of water retained in black clay soils during first 8 days of rainfall, broad beds of 3-9 m vide should be formed either along the slope or across the slope with drainage furrows in between broad beds.

v) Providing drainage open/ subsurface

vi) Huge quantity of sand /red soil application to change the texture

vii) Contour /compartmental bunding to increase the infiltration

viii) Application of soil conditioners like vermiculite/ jalasakthi to reduce runoff and erosion

<u>Fluffy paddy soils</u>

Occurrence

In Tamil Nadu, fluffy paddy soils are prevalent in Cauvery delta zone and in many parts of the state. It is formed due to the continuous rice-rice cropping sequence. In Tamil Nadu about 25,919 ha of land if affected by fluffiness (0.26 % of TGA)

Causes

The traditional method of preparing the soil for transplanting rice consists of puddling which results in substantial break down of soil aggregates into a uniform structure less mass. The solid and liquid phases of the soil are thus changed. Under continuous flooding and submergence of the soil for rice cultivation in a cropping sequence of rice-rice-rice, as in many parts of Tamil Nadu, the soil particles are always in a state of flux and the mechanical strength is lost leading to the fluffiness of the soils. This is further aggravated by in situ incorporation of rice stubbles and weeds during puddling.

Impact of fluffiness

Sinking of draught animals and labourers is one of the problems during puddling in rice fields in many parts of Tamil Nadu. This has been thus, an invisible drain of finance for the farmers due to high pulling power needed for the bullocks and slow movement of labourers during the puddling operations. Further the fluffiness of the soil lead to very low bulk density and thereby leading to very rapid hydraulic conductivity and in turn the soil does not provide a good anchorage to the roots and the potential yield of crops is adversely affected.

Methodology

- The irrigation should be stopped 10 days before the harvest of rice crop
- After the harvest of rice, when the soil is under semi-dry condition proctor moisture level), compact the field by passing 400 kg stone roller or an empty tar drum filled with 400 kg sand 8 times.
- The usual preparatory cultivation is carried out after compaction.

Lec 6. Soil crusting, soil compaction, sub soil hard pan, sand dunes and shallow soils – characteristics and management

Soil surface crusting- Management technologies

Occurrence

In Tamil Nadu, surface crusting is prevalent in Trichy, Thanjavur, Pudukottai, Cuddalore and Sivagangai districts and mostly in red soil areas. In Tamil Nadu, the area under soil crusting is around 4,51,584 ha (4.49% TGA)

Causes

Surface crusting is due to the presence of colloidal oxides of iron and aluminium in Alfisols which binds the soil particles under wet regimes. On drying it forms a hard mass on the surface.

Impact on soil properties

- Prevent germination of seeds
- Retards/inhibits roots growth
- Results in poor infiltration
- Accelerates surface run off
- Creates poor aeration in the rhizosphere
- Affects nodule formation in leguminous crops

Remedial measures

- When the soil is at optimum moisture regime, ploughing is to be done.
- Lime or gypsum @ 2 t ha^{-1} may be uniformly spread and another ploughing given for blending of amendment with the surface soil.
- Farm yard manure @10 tha^{-1}or composted coir pith @ 12.5 t ha^{-1} or other organics may be applied to improve the physical properties of the soils, after preparation of land to optimum tilth.
- Scraping the surface soil by tooth harrow will be useful.
- Bold grained seeds may be used for sowing on the crusted soils.
- More number of seeds/hill may be adopted for small seeded crops.
- Sprinkling water at periodical intervals may be done wherever possible.
- Resistant crops like cowpea can be grown.

Sub soil hard pan- Management technologies

Sub soil hard pan is commonly found in red soils. Though soil is fertile, crops cannot absorb nutrients from the soil which leads to reduction in crop yields. In Tamil Nadu, it is prevalent in Coimbatore, Erode, Dharmapuri, Trichy, Cuddalore, Villupuram, Pudukottai, Sivagangai, Madurai and Salem districts particularly under rainfed conditions. In Tamil Nadu, the area under subsoil hardpan is around 10, 54,661 ha (10.48% TGA)

Causes

The reasons for the formation of sub surface hard pan in red soils is due to the illuviation of clay to the sub soil horizons coupled with cementing action of oxides of iron, aluminium and calcium carbonate.

Impact on soil physical properties

The sub soil hard pan is characterized by high bulk density(>1.8 Mg m^{-3}) which in turn lowers infiltration, water holding capacity, available water and movement of air and nutrients with concomitant effect on the yield of crops.

Chiseling technology to overcome the sub soil hard pan

Chisel plough is used to break the sub soil hard pan in red soil. It is a tractor drawn heavy iron plough which goes up to 45 cm depth.

Methodology

The field is to be ploughed with chisel plough at 50 cm interval in both the directions. Chiseling helps to break the hard pan in the sub soil besides it ploughs up to 45 cm depth. Farm yard manure or press mud or composted coir pith at 12.5 t ha^{-1} is to be spread evenly on the surface. The field should be ploughed with country plough twice for incorporating the added manures. The broken hard pan and incorporation of manures make the soil to conserve more moisture.

Shallow soils

In Tamil Nadu, shallow soils occur over an area of around 1,16,509 ha (1.16 % TGA).

Causes

The reasons for the formation of Shallow soils is due to the presence of parent rocks immediately below the soil surface (15-20 cm depth)

Impact

The shallow soil restricts root elongation and spreading. Due to shallowness less volume of soil is available exhaustive soil nutrients

Management

- ❖ Growing shallow rooted crops.
- ❖ Frequent renewal of soil fertility
- ❖ Growing crops that can withstand shallowness(Mango, West Indian cherry, country goose berry, fig ,tamarind , ber and cashew etc)

Sand Dunes and their stabilization-

- Direct result of wind erosion
- **Defined** as the sand particles after being transported by wind, are accumulated at some point on the land surface.

Properties of sand dunes

- Soil varies from place to place depends on parent material
- Soil non coherent, structure less, highly erodible
- Particle size varies from 0.1-0.2 mm
- Infiltration rate is 10-60 cm/hr
- Low CEC
- Low Organic matter content
- Low basic cation
- Low available P&K
- Low water holding capacity 6-9% by weight
- *Low available moisture content*
- *Rapid permeability*
- *Lack of clay & hence poor physical & chemical properties*
- *Poor moisture retention properties*
- *Very high soil temperature*

Ravines

- Word means ‘deep gorge’
- It is a land associated with a system of more or less parallel gullies in deep alluvium & draining into a major river or tributary
- last stage of water erosion

Formation of ravines

- Splash erosion
- Sheet erosion
- Rill erosion
- Susceptible Soft alluvium
- Rivers cut a deep channel from table land
- Integration of adjacent gullies

Properties of ravine soil

- kankar pan
- pH: 7-8 Nuetral-midly alkaline
- Parent material: Basaltic rocks; broken into small fragments
- Immature soils
- No significant profile development

Topography: Zones of ravines

1. Upper table land zone-with <3% slope

2.Transition zone-very steep slope

3.Deeper zone

Nutrient status

- Organic matter- Very low
- Nitrogen - Very low
- Phosphorus - Low-medium
- Potash - Medium-high
- Zinc - Low (below .05ppm)
-

Major Factors for ravine formation

- DeforestationIllicit felling, uncontrolled grazing, fire
- Unscientific agricultural practices
- Low annual rainfall Soft alluvium soil susceptible to scouringExtreme runoff

Mining

Types of disturbance – Mining -Loss of land

- Loss of surface top soil
- Surface water pollution
- Ground water pollution
- Sedimentation
- Fire hazards
- Air pollution
- Change in land use pattern

Reclamation through afforestation

- Site preparation
- Trial plots
- Suitability study
- Pits – 45 cm^3 – 60 cm^3
- Spacing closure – 1x1, 2x2, 1x2

Choice of species

- *Eucalyptus camaldulensis*
- *Eucalyptus tereticornis*
- *Acacia nilotica*
- *Acacia leucopholea*
- *Grevilearobusta*
- *Agave species*
- *Cenchursciliaris*

Coal Mined Areas

- Complete devastation of vegetation
- Silting in reservoirs
- Reversed heat radiation
- High velocity runoff
- Loss of forest areas
- Erosion and flooding
- Land and water pollution

Choice of Species

- *Eucalyptus camaldulensis*
- *Eucalyptus tereticornis*
- *Azadirachtaindica*
- *Dalbergiasisoo*
- *Acacia nilotica*
- *Acacia leucopholea*
- *Agave species*

Soil compaction

In the compaction process, soil particles are packed together in a closer state of contact indicated by change in bulk density and porosity. Under a dynamic load sufficient to increase the density of soil, the following changes may take place

- Compression of soil solids
- Compression of liquid and gases
- Changes in liquid and gas contents in the pore space
- Re- arrangements of soil solids

Effects on soil quality

• It causes soil pore spaces to become smaller

• It reduces water infiltration rate into soil

• It decreases the rate that water will penetrate into the soil root zone and subsoil

• increases the potential for surface water ponding, water runoff, surface soil water logging and soil erosion

• It reduces the ability of a soil to hold water and air, which are necessary for plant root growth and function

• It reduces crop emergence as a result of soil crusting

• Impedes root growth and limits the volume of soil explored by roots

• It reduces crop yield potential Compacted soil will restrict root growth and penetration into subsoil. This situation can lead to stunted, drought stressed plants as a result of restricted water and nutrient uptake, which results in reduced crop yields.

Management

- Use direct seeding practices to increase soil organic matter content, which will optimize soil structure.
- Take advantage of the natural soil processes of “wetting-drying” and “freeze-thaw” cycles to minimize the effects of soil compaction.
- For irrigated areas, fall irrigation may ensure sufficient water for the freeze thaw effects.
- Use a combination of fibrous and tap rooted crops in a rotation to penetrate soils, develop deep root channels and add organic matter to soil.

Lec 7. Soil erosion and their Management

Soil erosion is the detachment, transportation and deposition of soil particle from one place to another due to action of wind, water-in-motion or by the beating action of the rain drops.

Extension of erosion in India: It is estimated that out of total geographical area of 328 m.ha, about 175 m.ha are affected seriously by water and wind erosion. Out of 175 m.ha 150 m.ha are affected due to shifting cultivation and 25 m.ha due to erosion in wastelands. About 6,000 m.tonnes of soil are eroded every year from about 80 m.ha of cultivated lands, losing about 84 m.tonnes of nutrients.

Classification of erosion: Erosion may be broadly classified in to two groups.

a) Geological erosion
b) Accelerated erosion

a) Geological Erosion (natural or normal erosion)

It represents erosion of soil in its natural condition without the influence of human being. The Geological erosion is longtime eroding process. The soil loss in this erosion is compensated by soil forming process by natural weathering. Ex: stream channels and valleys

b) Accelerated erosion

Accelerated erosion is in excess of geologic erosion. It is activated by natural and man's activities. The accelerated erosion is short time eroding process. It is also termed as man-made erosion. The soil loss in this erosion is faster than soil forming process by natural weathering. Ex: soil creep and rock slide.

The water and wind are the primary agents to cause the accelerated erosion. Accelerated erosion can further be classified as water erosion and wind erosion

Water erosion: it is defined as the movement of soil by the action of running water over the exposed land surface.

Soil erosion is the result of Erosivity, Erodibility, Detachability and Transportability.

Erosivity: It is the ability of rainfall to cause erosion. Physical characters of reference are intensity, amount and distribution of rain and size of rain drop.

Erodibility: Erodibility is the vulnerability or susceptibility of the soil to get eroded. It depends upon the physical characteristics and management of soil.

Mechanics of Water Erosion

There are three steps to accelerated erosion by water:

i. **Detachability:** detachment or loosening of soil particles caused by flowing water, freezing and thawing of the topsoil, and/or the impact of falling raindrops.
ii. **Transportability:** transportation of soil particles by floating, rolling, dragging and splashing.
iii. **Deposition** of transported particles at some place lower in elevation.

FACTORS RESPONSIBLE FOR EROSION

1. **Excessive deforestation**
   Valuable forest trees are destroyed and due to this erosion are stimulated.
2. **Improper agricultural practices**
   Erosion is more if slope is more. If cultivation is done along the slope erosion will occur, hence cultivation should take along contours or across the slope due to which water flow is intercepted and erosion is stopped.
3. **Over grazing**
   Grass forms a cover and helps in prevention of erosion. In barren soils splash of rainfall occurs and soil is displaced. Due to overgrazing, field is subjected to severe erosion.

Evil effects of water erosion

1) Loss of productive or fertile soil
2) Deposition of sand on productive land
3) Silting of lakes and reservoirs
4) Lowering of ground water table
5) Fragmentation of land

FACTORS INFLUENCING WATER EROSION

Major factors are climate, topography, vegetation and soil. This is represented as

E=f (C, T, V, S)

Climatic Factor (C):

Rainfall, Temperature, Wind, Relative Humidity and Solar radiation. Rainfall is the main detaching agent. The impact of raindrops has a pronounced effect on erosion. The raindrops detach the particles and make them available for transport.

Terminal velocity of rain drop

It is the rate of the falling rain drop which equals the resistance offered by air to the fall of the drop. The impact of rain drops/ unit area is determined by the number, size and velocity of the raindrops. The velocity of the raindrop depends on the size, height of fall, wind velocity and air resistance. For 1 mm diameter of rain the terminal velocity is 4.5 m sec^{-1}; 5 mm diameter rain the V_t is 9 m sec^{-1}. It has been observed that a single rain drop may splash wet soil as much as 60 cm height and 150 cm around the spot where the raindrop hits.

Topography (T):

Degree and length of topography are two main factors. A flat land has not the problem of soil erosion, while the sloping lands are predominantly affected by the erosion. . On larger slopes there is higher concentration of over land flow, higher velocity of flow which triggers the higher rate of soil erosion. The velocity of runoff is mainly influenced by topography.

Vegetation (V)*:* The effects of vegetation for erosion are

i) Interception of rainfall: A part of the rainfall intercepted by the canopy vegetation never reaches the soil but it is evaporated directly. This part of rainfall does not contribute with the run-off and thereby minimizes the dispersion of soil.

ii) Decrease in runoff velocities: Any vegetation cover is a hindrance to runoff water. A well distributed, close growing vegetation not only slows the water which travels down the slope but

also tends to prevent the concentration of water. Hence the erosive capacity of runoff is minimized.

iii) Root effects: The binding effect of the root system binds the surface layer of the soil aggregates and increases its resistance to erosion. When the root die, decay and increases pore space leads to absorption of water.

iv) Biological influences: The soil fauna is most active in soils having ample vegetative cover. The soil under a thick forest cover is permeated with earthworms and beetles. These channels increase the permeability of the soil, in addition increase the soil aeration which provides a better environment for the activity of beneficial bacteria.

v) Transpiration effects: Vegetation increases the storage capacity of soil by way of transpiration of large quantity of moisture from the soil

Soil (S): The physical properties of soil that influence soil erosion includes structure, texture, organic matter, moisture content, compactness as well as chemical and biological characteristics of soil. When infiltration rate is greater than the rainfall intensity, then whatever rain occurs over the land surface, is absorbed by the soil and thus surface runoff becomes zero. the infiltration rate mainly depends the permeability of soil profile. The permeability is more the runoff will be less and vice versa. The compacted soil has low infiltration rate and high runoff.

DIFFERENT FORMS OF WATER EROSION

Splash / rain drop erosion: It results from soil splash caused by the impact of falling raindrops. If raindrops strikes on the land covered with thick blanket of vegetation the drop breaks into a spray of clear water which slowly finds its way into soil pores. If the rain drop strikes bare soil considerable splashing occurs. These splashes gradually remove fine materials from soil and leave the land infertile by leaving behind sand and gravel particles. The raindrop sometimes fall at high speed of 50 kmph and the soil particle may be splashed to a height of 60 cm and move laterally to a distance of 150 cm.

Sheet erosion: It is the removal of fairly uniform layer of soil from land surface by the action of rainfall and runoff water.

Rill erosion: In this stage small finger like rills are formed under landscape. These rills are smoothened by working of small implements. Year after year the rills will increase not only in number but also in shape and size. These gets wider and deeper .Rill erosion is more serious in soils having shallow top soil. It may be regarded as a transition stage between sheet and gully erosion.

Gully erosion: It is an advance stage of rill erosion where unaltered rills gets deepened and widened every year and begins to attain the form of gullies. Gullies unless stopped or controlled in the rill stage will get deepened and widened. Cattle paths, tracks, furrows or other small depression along the slope favor concentration of runoff. Every time the rainwater surges down these gullies, they get deepened and widened.

FOUR STAGES OF GULLY DEVELOPMENT

1. **Formation stage;** It is forming in channels by downward scouring of the soils.
2. **Development stage;** It is a stage where the upstream movement in the gully head and enlargement of gully in depth and width takes place.
3. **Healing stage;** It is the stage where the vegetation begins to grow in the gullies.
4. **Stabilization stage;** Here the gully bed, reaches a stable condition. The gully walls reach slope and vegetative cover spreads over the gully surface.

Stream bank erosion

Stream bank erosion is the scouring of soil material from the stream bed and cutting of the stream banks by the force of running water. Rivers and streams often meander and change their course by cutting one bank and depositing sand and silt loads on the other. The damage is many folds during flash floods

Other forms of erosion

Apart from water and wind as erosive agents, there are several other agents like glacial, snow, organic contents anthropogenic etc, also available.

Glacial erosion: predominant in cold regions, where mean temperature is found below 0°c. Movement of soil or rock from one point to another by the action of the moving ice of a *glacier*.

Snow erosion: predominant in areas where permanent snows cover is found.

Organic erosion: soil erosion is caused by living organism. it is divided into two forms.

i. **Phytogenic erosion (root erosion):** is the destruction of soil, caused by plant root system.

ii. Zoogenic erosion: the soil particles are removed by the animals.

Anthropogenic erosion: the erosion is due to activities of human being

Tunnel erosion: (piping erosion, piping) it is being a specific form of underground erosion, in which washing of soil particles through sub surface corridors or passage is carried out by underground water.

EROSION CONTROL MEASURES - VEGETATIVE MEASURES

Soil conservation is the preservation of soil from deterioration and loss by using it within its capabilities, and applying the conservation practices needed for its protection and improvement. Certain lands are suited to cultivation, while others are not. The natural limitations on the use of land constitute the basis for all soil conservation measures. Soil conservation planning should generally be done on a watershed basis.

Watershed

The watershed is the total lands are above a given point on a water way that contributes run off water to the flow at that point. Soil conservation work should start at the head of such topographical units and proceed in the same way the water flows to the downstream.

Land use capability classification:

The land use capability classification is a systematic arrangement of different kinds of land, to produce on a virtually permanent basis. The classification is arranged according to a number of categories. Standard land capability classes based on land slope and their proposed land use are given as follows.

Table: Land capability classification based on land slope and soil conservation

Class of Land	Slope in %	Adapted land use and soil conservation measures
A-Land suitable for cultivation		
Class I	0-1	Any crop with proper crop rotation and green manuring to maintain soil fertility
Class II	1-3	Contour farming, contour strip cropping and cover cropping, contour bunding or terracing
Class III	3-5	Intensive agronomical measures such as contour cropping, contour strip cropping and cover cropping. Terracing or contour bunding
Class IV	5-8	Contour bunds or terraces and intensive agronomic measures. Mostly soil building and maintaining crops are to be grown.
B. Land not suitable for cultivation		
Class V	8-12	Permanent pasture with controlled grazing
Class VI	12-18	Pasture, grasses and forestry. Grazing should be restricted
Class VII	18-25	Forest with restricted felling. Contour trenching as conservation measure
Class VIII	> 25	Forests with complete closure to grazing and felling of trees

A. Biological measures

Methods adopted for erosion control by using crop or vegetation, tillage practices and land and water management are known as biological measures. These methods can be adopted for lands having a slope up to 2 %.

1) **Contour cultivation**

The agricultural operations like tillage, planting and other intercultural operation are done along the contour. The contour cultivation reduces the velocity of overland slopes and erosion. Crops like maize, sorghum, pearl millet is normally grown in rows are suited for contour cultivation.

2) **Strip cropping**

Different crops are grown in alternative strips, across the slope such that they serve as barriers for soil erosion. There are four types of strip cropping such as

(i) Contour strip cropping

Growing in alternative strips of erosion resisting and erosion permitting crops along the contour is termed as contour strip cropping.

(ii) Field strip cropping

Field strip cropping consists of strips of uniform width running generally perpendicular to the direction of the erosive force, but do not confirm to nay contour

(iii) Buffer strip cropping

Strips of grass or legume crop laid out between contour strips of crops in the regular rotations.

(iv) Wind strip cropping

The strips are laid at right angles to the prevailing wind direction, irrespective of the land slope

3) **Cropping systems**

The sequence of crops grown in an area over a period of time is termed as cropping system. This system should be such that the soil is covered with vegetation during heavy rainfall period.

4) **Crop rotation**

Crops like groundnut, soybean, green gram etc., are commonly used in crop rotation. These crops are having soil binding capacity and act as erosion resisting crop.

5) **Mixed cropping**

Mixed cropping is the system of showing more than one crop together on the same land. Mixed crops varying in root systems helps in utilizing the plant nutrients in the profile. Erosion permitting and erosion resisting crops are raised in the same land in mixed cropping.

6) **Tillage practices**

The effect of tillage on erosion is a function upon the factors such as aggregation, surface sieving and infiltration. This practice consists of different ploughing operations using disc, harrow and cultivators.

7) **Mulch tillage/stubble mulching**

It is a soil and crop management practice that utilizes the residual mulches of the preceding crop by leaving a large quantity of vegetative residue on surface of the land. The mulch breaks the rain drops and thus dissipates their energy.

8) **Cover cropping**

The process of allowing growth of crops having economical value so as to afford productive cover to the bare space in the cultivated lands, without and at the same time interfering or suffocating the growth of the main crops grown in the field is called cover cropping. Example: Leguminous crops, sweet potato, pumpkin and groundnut.

WATER EROSION CONTROL MEASURES-MECHANICAL MEASURES FOR PLAIN - CONTOUR BUND- BROAD BED AND FURROW

Mechanical measures play a vital role in controlling and preventing soil erosion on agricultural lands. They are adopted to supplement the agricultural practices (biological methods). The mechanical measures include bunding, terraces etc.,

CONTOUR BUNDING

Contour bunding is the construction of small bund across the slope of the land on a contour so that the long slope is cut into a series of small ones and each contour bund acts as a barrier to the flow of water, thus making the water to walk rather than run, at the same time impounding water against it for increasing soil moisture. Contour bunts divide the length of the slope; reduce the volume of runoff water, and thus preventing or minimizing the soil erosion. Contour bunds are constructed in relatively low rainfall areas, having an annual rainfall of less than 600 mm, particularly in areas having light textured soils. For rolling and flatter lands having slopes from 2 to 6% contour bunding is practiced.

GRADED BUNDS

Graded bunds are constructed in medium to high rainfall areas having an annual rainfall of 600 mm and above and in soils having poor permeability (or) those having crust forming tendency (black soils), and in the lands having slopes between 2 % and 6 %. These bunds are provided with a channel if necessary.

BROAD BASE TERRACE

A broad base terrace is a broad surface channel or embankment constructed across the slope of the rolling land for reducing runoff erosion and for moisture conservation. The broad base terrace has a ridge of 25 to 50 cm high and 5m to 9m wide with gently sloping on both sides and a channel along the upper side, constructed to control erosion by diverting runoff at a non-erosive velocity.

BROAD BED AND FURROW SYSTEM

The broad bed and furrow system combines of erosion control with surface drainage. The beds could be of different dimensions to suit the cropping patterns adopted. Normal bed width adopted is 150cm with 30cm width furrows at the end of the beds

CONTOUR STONE WALL

In this, cut stones of size around 20 cm are dry packed across the hill slope to form a regular shape of random rubble masonry without mortar. After construction it should be stable enough so that a man can walk on it. They are constructed in lands having slopes between 10 and 16% and above.

BENCH TERRACING

Bench terracing is one of the most popular mechanical soil conservation practices adopted by farmers of India and other countries for ages. On sloping and undulating lands, intensive farming can be only adopted with bench terracing. It consists of construction of step like fields along contours by half cutting and half filling.

TRENCHES

Contour (or) staggered trenches is adopted in high rainfall hilly areas of lands with slopes steeper than 33 % or any slope with badly eroded soil. Length of staggered or contour trench will be 3 m to 3.65 m while the inter space between trenches in the same row will be only 2.4 m to 3 m.

WIND EROSION

Water erosion: it is defined as the movement of soil by the action of blowing wind across cultivated bare fields especially if soil is loose, dry and finely divided. It is estimated that about 17.2 m ha of land is suffering from wind erosion. Wind erosion has been serious in Western Rajasthan and adjoining areas.

Results (evil effects) of wind erosion

1. loss of soil fertility
2. formation of sand dunes (A sand wave of approximately triangular cross section formed by wind; found in deserts or near lakes and oceans)
3. extension of deserts
4. textural change
5. air pollution
6. sedimentation

Wind erosion is predominant in **arid and semi-arid regions,** where annual rainfall is less.

Conditions favorable for wind erosion

1. dry period
2. absence of a protective cover on the ground
3. a broad, flat or undulating topography.

Factors influencing wind erosion

Climate: the climatic factors which affect the wind erosion are precipitation, temperature, wind velocity and density of prevailing wind. Wind erosion is more pronounced when mean annual rainfall is comparatively less. When there is less rainfall, the moisture content in the soil is less

and thus, the soil particles are found in loose conditions, as a result they are easily detached by the blowing winds and causing into soil erosion due to wind.

Wind-speed: Wind erosion phenomena will increase proportionately in the presence of strong, regular prevailing winds

Soil texture: Loamy sand, rich in particles between 10 and 100 microns in size, is the most vulnerable soil (Bagnold 1937). More clayey soil is much stickier, better-structured, and hence more resistant. Coarse sand and gravelly or rocky soils are also more resistant, since the particles are too heavy to be removed by wind erosion. The optimum size for wind erosion is about 80 microns.

Soil structure: The less structure-improving matter a soil has on the surface (organic matter, iron and free aluminium, lime), the more fragile it will be, while the presence of sodium or salt often leads to formation of a layer of dust on the surface, which fosters wind erosion.

State of the soil surface: If the soil surface is stony, forming a "pavement", the risks of wind erosion are lower - as, for example, in *regs*.

A rough surface, left by cloddy tillage or ridges perpendicular to the prevailing wind, slows down the wind at ground level, thus reducing saltation.

Vegetation: Stubble and crop residues in the soil cut wind-speed at ground level.

Soil moisture increases cohesion of sand and loam, temporarily preventing their erosion by wind.

Mechanics of wind erosion

The occurrence of wind erosion could be described under following three distinct phases

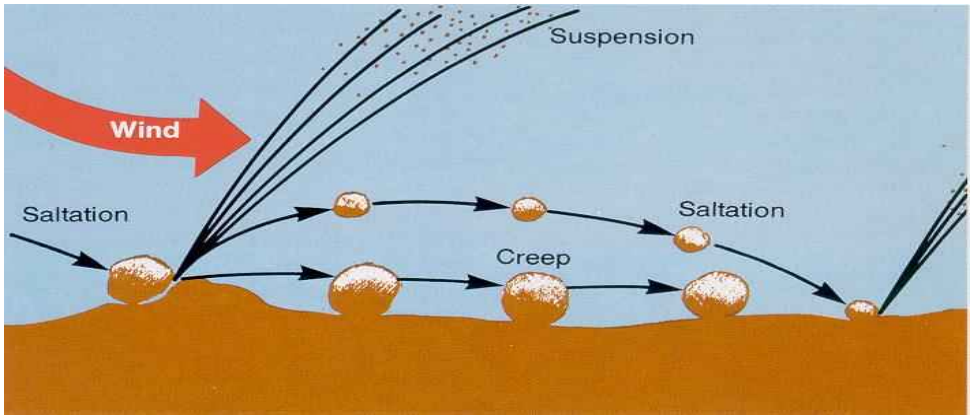
1) initiation of soil movement
2) transportation of soil particles
3) deposition of soil particles

Initiation of soil movement

Three types of particles movement are given below

a. **Suspension** – movement of very fine particles < 0.1 mm diameter –ex: dusty storms. According to stokes law, which states that the speed of the freely falling body through a fluid, is directly proportional to the square of the particles diameter, a small particle has low settling velocity and once lifted up, it remains suspended in the air for a long period of time.
b. **Saltation** – this is most effective among all three forms of soil movement. It is the movement of middle size particles – 0.05 mm to 0.5 mm in a series of low bounces over soil surface.
c. **Surface creep** – movement of soil particles by rolling along the land surface. Size range of particles 0.5 mm to 1 or 2 mm diameter.

Saltation is responsible for transporting the maximum portion of soil along the surface of ground.



Transportation

Transportation of soil particles is directly influenced by the particle size, gradation of particles, wind velocity and distance across eroding areas. It has been found that the quantity of soil moved through wind, varies as cube of the excess wind velocity over and above the constant threshold velocity. Transportation is directly proportional to the square root of the soil particle diameter. It also increases as the gradation of the soil particle decreases.

Deposition

Deposition of soil particles depends upon the particles weight and wind velocity, in which particles weight is directly related to the gravitational force. The deposition of soil particles occur, **w**hen gravitational force is greater than the resisting forces holding the particles in the air and when wind velocity decreased due to surface obstructions.

WIND EROSION CONTROL MEASURES

1. Vegetation and Vegetative Measure

Vegetation is generally the most effective means of wind erosion control. When used for this purpose, vegetation may be grouped into inter tilled crops, close growing crops and woody plants such as shrubs and trees. Crop residues are also of considerable value in preventing soil blowing. Plant covers and residues are capable of taking a portion of the direct force of the wind from the soil particles on the immediate surface.

In general, close growing crops are more effective for erosion control than inter tilled crops. The effectiveness of such crops as legumes, grasses and small grains is dependent quite largely on the stage to be covered with a dense cover crop during the summer.

(a) Strip Cropping

Strip cropping for wind erosion control consists of a systematic alternate arrangement of erosion susceptible and erosion resistant crops in relatively narrow strips whose length runs perpendicular to the direction of the prevailing erosion winds. The less erodible strips of stubble or other crops have some effect in reducing wind velocity across adjacent more erodible strips.

The major benefit is, however, trapping of saltating soil particles and thereby controlling soil avalanching. Therefore, the narrower the strip the more effective they are in reducing erosion. It has been estimated that in order to keep erosion to a tolerable limit the width of the erosion susceptible strip should not exceed 6 m in sandy soil, 8 m in loamy sand and 30 m in sandy loam.

(b) Crop Rotation

Good management techniques are important. A good crop rotation that will maintain soil structure and conserved moisture should be followed as for as possible. However, deficient moisture often limits the use of soil improving and soil conserving crop such as legumes and grasses. In some regions, emergency crops with low moisture requirements are often planted on summer fallow land before the season of high intensity winds.

(c) Stubble Mulching

Stubble mulching the practice of maintaining crop residues at the ground surface offers good protection from soil blowing. Crop residues like wheat, straw, stalks of pearl millet, maize, sorghum etc., are left on the ground. Residues of crops following harvest continue to protect the soil against wind erosion, just as does the growing crop. Such residues, if properly managed, can provide excellent control during periods between crops.

2. Tillage Practices to control Soil Blowing

Tillage machinery and tillage practices can either increase or reduce the soil blowing problem. Machines that tend to pulverize the soil or to diminish the vegetative cover increase soil blowing. Tillage equipment that will prove most effective is that which will do a good job of creating a cloddy surface and at the same time avoid burying the crop residue. The equipment should leave a rough, cloddy and a residue covered surface with residues well anchored and standing if possible. Deep ploughing to depths of 30 to 45 cm has become a common practice in the western countries to improve the wind resistance of sandy soil.

Residues, for example, are usually best handled with implements having sweep type furrow openers. Such implement move under the material, leaving it in an erect position, most effective against wind. Both bullock-drawn and tractor-drawn cultivators, equipped with sweep furrow openers are suitable.

3. Surface Roughness

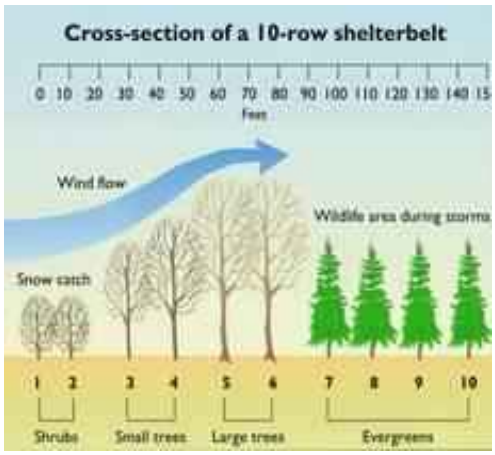
The rate at which soil can be moved by wind varies inversely with the roughness of the surface. Any form of ridging placed perpendicular to the direction of the prevailing erosive winds tends to reduce soil loss. Even ordinary plough furrows, if aligned at right angles to the prevailing wind, will have an appreciable effect in reducing soil movement. Roughness of bare land is made of surface irregularities and large clods. Therefore, ridging should be done at a soil moisture content that will produce clods. In the furrow the direction of air movement is opposite to that of the wind. The decrease in wind velocity and change in wind direction between the ridges cause soil deposition.

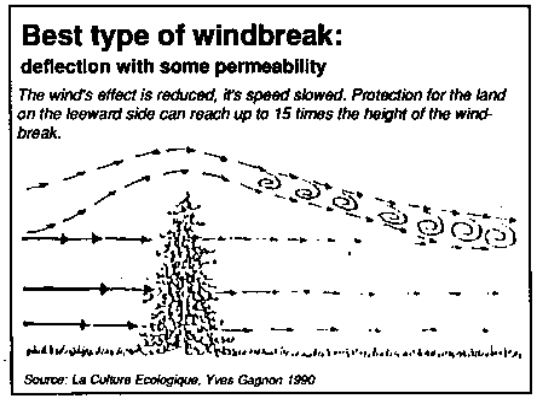
4. Conserving Moisture

The conservation of moisture, particularly in the arid and semi-arid regions, is of utmost importance for wind erosion control as well as for crop production. The means of conserving moisture fall into three categories: increasing infiltration, reducing evaporation and preventing unnecessary plant growth. In practice, these can be accomplished b such practices as contour farming, contour bunding, mulching and selecting suitable crops. The greater the amount of mulch on the surface the greater the quantity of moisture conserved.

5. Windbreaks and Shelterbelts

Windbreaks are such structures which break the wind-flow and reduce wind speed while shelterbelts are rows of trees or shrubs planted for protection of crop against wind. A row of trees and shrubs planted across the winds-direction is the most effective. It reduces the wind speed up to 60-80% on leeward side. The height of tall tree and length of wind break determine the extent of protection provided to soil. Windbreaks and shelterbelts provide the protective shelter against desiccating winds to extent of 5-10 times the height of the tall tree on windward side and up to 30 times on leeward side. For example a 10-11 meter tall windbreak when encountered by 45-50 km /hr wind, it reduces on windward side to 20-30 km /hr and to 10 km / hr on just leeward side. The windbreak reduces the wind speed on the leeward side at 200 m away up to 20 % only and in the area as at 300m away, there is no effect at all. Therefore at an interval of 300m, such windbreaks should be repeated.





Advantages of wind breaks or shelter belts

1. Reduction of wind velocity across fields
2. Reduction in soil blowing through wind erosion and dust storm.
3. Reduction of soil moisture evaporation and plant transpiration.
4. Less physical damage to agricultural crops grown which otherwise will be badly damaged by the beating action of the blowing and the sand.
5. The productivity of the land is improved and the land value goes up.
6. The wind-breaks provide food and cover to song birds and will be a good recreation spot.

Disadvantages of wind breaks or shelter belts

1. The wind-breaks or belts occupy a portion of agricultural lands. The loss in income by way of reduction in the cultivated area can however be compensated by the increased income through increased crop yield as well as the value of trees, planted.
2. Root effects on the agricultural crops adjoining the boundaries of the wind-break or belt. This could however be got over either by fertilizing the trees or roots pruning by digging a trench along boundaries or both.
3. Proper care and careful attention is needed during early years of establishment replacing casualities, watering, if necessary, etc.

Windward side: The windward side is the side that receives the wind.

Leeward side: The leeward side is the other side where the wind break / shelter belt blocks the winds.

Universal Soil Loss Equation

- **The Natural Resources Conservation Service (NRCS) uses the *Universal Soil Loss Equation* (RUSLE) to calculate soil loss by erosion as a function of 5 factors:**

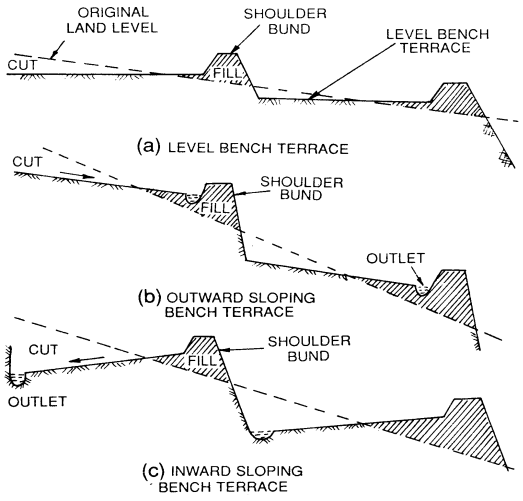
$$A = R \times K \times L \times S \times C \times P$$

  - **A - annual soil loss (tons/Ac/yr)**
  - **R - erosivity due to rainfall or runoff**
  - **K - erodibility of the soil**
  - **L - slope length**
  - **S - slope steepness**
  - **C - cropping and management factors**
  - **P - erosion control practices**

CONTOUR STONE WALL (10-16%)



BENCH TERRACING 16-33%



Trenches more than 33% slope.

Lec 9. Flooded Soil

Introduction

Under continual flooding, swamp soils develop characteristics which are fundamentally different from those of upland soils. Although it is not essential that the farmer have a complete technical understanding of the differences between the two types of soils, it is important to know about some basic properties of flooded soils so that management practices will be appropriate for local conditions.

Characteristics of Flooded Soils

Three major changes - physical, biological, chemical - occur when a soil is flooded.

1. **Physical Changes**

Upon flooding, the pore spaces (air spaces) in the soil become saturated with water. As a result, the soil swells, and hard clods soften and break into small aggregates. Puddling completely destroys the remaining structural aggregates (clods and clumps) and transforms the soil into a sludge, or soupy mixture. This slows the drying of the soil, since the exchange of air between the atmosphere and the soil is impeded, and since the water particles are held by soil particles and prevented from percolating downward and escaping.

2. **Biological Changes**

The absence of soil air (and particularly oxygen) in flooded, puddled soils causes a change in the varieties of microbes, or microscopic organisms which live in the soil. Microbes existing in the absence of oxygen are known as **anaerobic microbes**, and they tend to be much slower, less **efficient decomposers of organic matter than their aerobic cousins**. Consequently, the rate of **decay of organic matter tends to be slow in flooded soils**. Also, the end products produced by anaerobic decomposition differ' some are toxic to rice, particularly those released during the first two weeks after decomposition begins. This can be important in timing organic matter incorporations between plantings: if the farmer plants too early into a

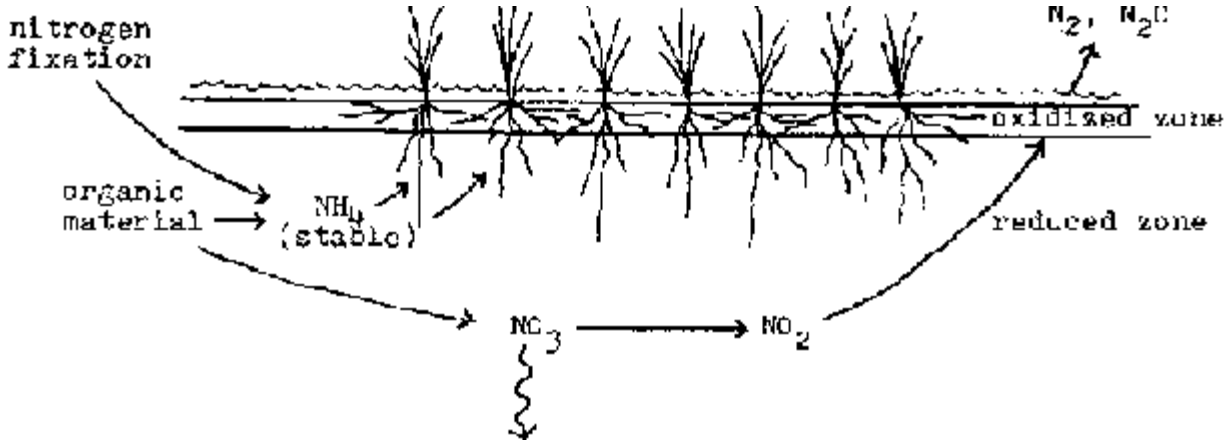
flooded plot containing plowed-under stubble and/or chaff, the toxicities produced during normal decomposition may stunt the growth of the rice.

3. **Chemical Changes**

Flooded soils develop two distinct chemical zones. (see figure below) The upper zone, a thin 1-10 mm, absorbs oxygen from the water, **turns brown in color**, and reacts to nitrogen like an unfolded soil. This zone is called **the oxidized zone**, in reference to its chemical condition of being oxidized. The lower zone, which extends down as far as the water, is extremely low in available oxygen, **turns dark blue or gray in color**, and takes on chemical properties quite different from those of the oxidized layer above. This lower zone is known as the **reduced zone**.

When a soil is flooded, the nitrogen in the incorporated plant (and animal) residues is changed to the ammonium form (NH_4), which is stable under flooded conditions and will later be used by growing rice plants. If the soil is allowed to dry thoroughly (e.g., when it is drained for plowing), a micro bacteriological change takes place during which the ammonium form of nitrogen is changed to the nitrate form (NO_3).

When the soil is later **re-flooded**, part of the nitrogen held in the nitrate form is changed into nitrogen gases (N_2, NO_3) and escapes into the air. Between **20-700 kg/ha of nitrogen can be lost through this process, known as denitrification**, so it is extremely important to keep the Plot thoroughly flooded at all times after initial irrigation has taken place.



Management of Problem Soils

All swamp farming systems are not created equal. Although swamps tend to look the same from the ground, often they vary considerably below the surface. Some swamps are shallow and sandy, others deep and peaty, still others filled with rock or clay deposits. Because soil characteristics significantly affect plant growth, the farmer should know what steps should be taken to minimize soil-related problems.

The key to management of problem soils lies in anticipating potential problems before they occur and taking the steps necessary to head them off. Problem soils can be attacked three ways' 1) through the selection of an appropriate variety, 2) through proper swamp development, and 3) through effective management practices. Never wait until after the crop is in the ground to begin thinking about soil problems, since by then it will be too late to change varieties or modify the water control system.

Always begin thinking about the soil early in the development process. The very first time you visit a farmer's swamp, take the time to dig a few scattered pits to reveal local soil characteristics. Question the farmer about the depth of the topsoil, about the type of underlying material, about the presence of undecomposed organic material, etc. With a little practice, you will learn to identify problem soils long before planting time, and consequently you and the farmer will be able to devise an effective strategy involving development, varietal selection, and crop management.

A. <u>Sandy Soils</u>

Many swamp systems in Sierra Leone are extremely shallow and sandy. The main problem with sandy soils is leaching: water percolates through sand very easily, carrying away nutrients and bringing in toxic materials. Effective water control is difficult, because sandy soils do not retain water well' as a result, drought stress is common among crops grown in sandy swamps. Nutrient deficiencies occur regularly, as evidenced by stunting, yellowing of the leaves, and low tillering.

To treat sandy soils:

1) Development practices

Develop the swamp so as to discourage leaching. There are several ways to slow movement of water through the soil'

-keep the main drain shallow (so it will not "suck" water from the plots).
-dig deep peripheral gutters (to intercept groundwaters percolating down from adjacent hillsides).

2) Varietal Selection

Select a variety which does well under relatively difficult conditions, i.e. a variety which can withstand an irregular water regime, nutrient deficiencies, and soil toxicities.

3) Management Practices

To minimize leaching and the damage it cases:

- Keep a constant, slow-moving or non-moving flood on the plots.
- spread out fertilization by applying top dressings in many small splits.
- To improve the structure and fertility of the soil, as well as to improve its ability to retain water, incorporate large amounts of organic material before and, if necessary, after the growing season.

B. Peaty Soils

Swamps which remain permanently flooded often contain large amounts of undecomposed organic matter (because the soil never gets a chance to dry, which means that aerobic decomposition never occurs). Peaty soils are characterized by high acidity and the presence of numerous toxic materials, both of which severely affect crop growth. Peaty swamps are easily recognized by matted clumps of undecomposed organic material, noxious gases, and mineral slicks. Rice grown in peaty soil is usually stunted, browning, and low tillering.

To treat peaty soils:

1) Development Practices

When developing the swamp, be sure to dig an effective drain. The main drain should be extremely deep to draw water not only from the surface of the plots, but from beneath the surface as well.

2) Varietal Selection

Select a variety which does well under conditions of high acidity. An iron toxicity-resistant variety is usually a good bet, or a native variety.

3) Management Practices

To enable aerobic decomposition of organic material in the soil:

- Continually drain the plots.
- Till the topsoil during the dry season to a depth of several feet, if possible (encourage the farmers to construct vegetable mounds).

To prevent incorporation of additional organic material into the soil, burn all grasses, weeds, and crop residues (stubble, chaff, etc.).

C. Clays

Clays have a high natural fertility, but often they resist giving up their nutrients to plant root systems. Furthermore, because of their dense structure (they are composed of extremely small, tightly-packed particles), they tend to be difficult to work. Therefore, improving the structure of clays should be the farmer's primary goal. When the clay is fairly dry (just before it starts to crack), begin tilling to increase aeration. Incorporate large amounts of organic material, and practice crop rotation to help maintain fertility.

III. Iron Toxicity

Iron toxicity is a soil-related condition caused by the presence of too much iron in lowland rice paddies. Iron toxicity results from a complex set of chemical imbalances in the soil and in the plant which in many instances severely inhibits yields. The condition occurs most commonly in sandy areas of swamps adjoining upland slopes, or in peaty areas where drainage is poor.

The symptoms of iron toxicity are numerous and varied, but the most typical indication is a marked bronzing of leaves and stems, generally occurring 4-7 weeks after transplanting. In addition, roots are stunted, coarse, and reddish brown or dark brown in color (a coating of iron oxides reduces root surface and decreases capacity to absorb soil nutrients). Plants are stunted

and tiller poorly. In extreme cases, black or brown spots appear on leaves and stems, and leaf edges turn dark brown and roll in toward the midrib.

The causes of iron toxicity are technically complex and hence difficult to describe succinctly. Very simply, the presence in the soil of excessive quantities of iron inhibits nutrient absorption by the plant and leads to severe nutritional imbalances which manifest themselves in the typical symptoms. The intake of phosphorus and potassium is especially inhibited, and it is the relative lack of these two primary nutrients (in conjunction with the relative abundance of nitrogen) which causes many of the symptoms.

Treatment of iron toxicity can take several forms. Swamp development measures which can help prevent development of a situation favorable to iron toxic conditions include deepening peripheral gutters in sandy swamps (to decrease percolation of iron compounds down from the uplands) and improving drainage in peaty swamps (the mineral slicks in peaty swamps are often associated with iron toxicity). Fertilization with large amounts of phosphorus and potassium can be effective, although application of nitrogenous fertilizer generally aggravates the symptoms (since nitrogen is relatively available to the plant already, and the presence of nitrogen in the absence of phosphorus induces the symptoms). The most effective treatment in any case is selection of iron-toxicity-resistant varieties. Certain varieties (often traditional varieties) are able to flourish in iron-toxic soils, and they offer the most practical and economical solution to the problem.

Lecture 10: Acid Soil and Acid sulphate soil – Genesis and Characteristics

Acid soil is a base unsaturated soil which has got enough of adsorbed exchangeable H^+ ions so as to give soil a pH lower than 7.0.

Genesis of acid soils

(i) Leaching due to heavy rainfall

Acid soils are common in all regions where rainfall or precipitation is high enough to leach appreciable amounts of exchangeable bases (Ca^{2+}, Mg^{2+}, Na^+ and K^+) from the surface soils and relatively insoluble compounds of Al and Fe remains in the soil. The nature of these soils is acidic and its oxides and hydroxides react with water (H_2O) and release hydrogen ions in solution and soil becomes acidic. As the soil gets gradually depleted of its exchangeable bases through constant leaching, it gets desaturated and the exchange complex is saturated with H^+ ions of the carbonic acid.

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

$$H_2CO_3 + CaCO_3 \longrightarrow Ca(HCO_3)_2 \text{ (Leachable)}$$

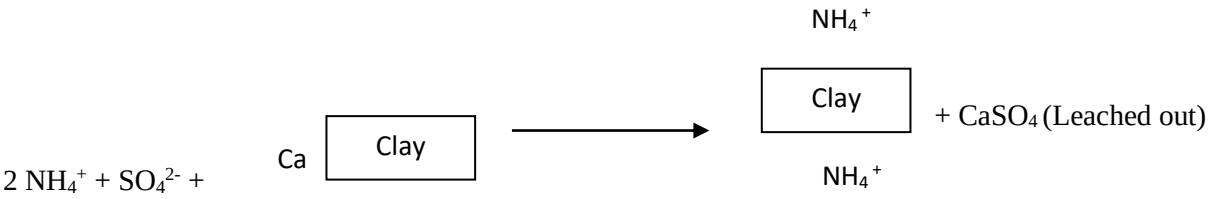
(ii) Acidic parent material

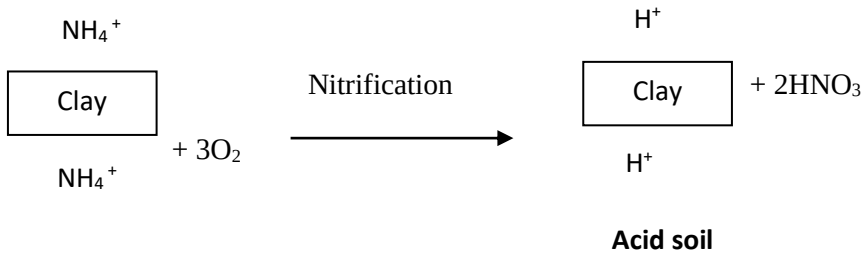
Some soils developed from acidic parent materials like granite and gneiss may develop soil acidity.

(iii) Acid forming fertilizers and soluble salts

The use of acid forming fertilizers like ammonium sulphate $(NH_4)_2SO_4$ and ammonium nitrate NH_4NO_3 increases soil acidity. Ammonium ions from $(NH_4)_2SO_4$ when applied to the soil replaces calcium (Ca^{2+}) ions from the exchange complex and the calcium sulphate formed is finally leached out.

$$(NH_4)_2SO_4 \rightleftharpoons 2\ NH_4^+ + SO_4^{2-}$$





Similarly sulphur also produces acid forming sulphate ion through oxidation. Divalent cations of soluble salts have a greater effect on lowering soil pH than the monovalent cations.

(iv) Humus and other organic acids

Decomposition of organic matter releases weak organic acids, which is capable of solubilizing bases resulting in increased acidity.

(v) Aluminosilicate minerals

At low pH values most of the aluminium (Al) is present as the hydrated aluminium ions (Al^{3+}) which undergoes hydrolysis and release hydrogen (H^+) ions in soil solution.

$$Al^{3+} + H_2O \rightleftharpoons Al(OH)^{2+} + H^+$$

$$Al(OH)^{2+} + H_2O \rightleftharpoons Al(OH)_2^{+} + H^+$$

$$Al(OH)_2^{3+} + H_2O \rightleftharpoons Al(OH)_3^{+} + H^+$$

$$Al(OH)_3^{4+} + H_2O \rightleftharpoons Al(OH)_4^{+} + H^+$$

(vi) Carbon dioxide (CO_2)

CO_2 evolved from root activity and microbial metabolism combines with H_2O to form H_2CO_3 resulting in soil acidity.

(vii) Hydrous oxides

The oxides of iron and aluminium under favourable conditions undergo stepwise hydrolysis with the release of hydrogen (H^+) ions in the soil solution and develop soil acidity.

(viii) Aluminium and iron polymers

The Al^{3+} ions displaced from clay minerals by cations are hydrolyzed to monomeric and polymeric hydroxyl aluminium complexes as shown below. Hydrolysis of these forms liberate hydronium ion thus lowering the soil pH.

$$Al(OH)_2(H_2O)_4 + H_3O^+ \rightleftharpoons Al(OH)_3(H_2O)_3 + H_3O^+ \text{ (pH: 6.5 – 8.0)}$$

$$Al(OH)_3(H_2O)_3 + H_3O^+ \rightleftharpoons Al(OH)_4^{+-}(H_2O)_2 + H_3O^+ \text{ (pH: 8.0 – 11.0)}$$

Similarly,

$$Fe(H_2O)_6 + H_2O \rightleftharpoons Fe(OH)_4(H_2O)_5 + H_3O^+$$

(Reaction occurs under more acidic conditions than Al)

Kinds of Soil Acidity

Soil acidity may be of the following two kinds

(i) Active acidity

It is defined as the acidity that develops due to hydrogen (H^+) and aluminium (Al^{3+}) ions in the soil solution. The magnitude of this acidity is limited.

(ii) Exchange acidity

It is defined as the acidity that develops due to adsorbed hydrogen (H^+) and Aluminium (Al^{3+}) ions on the soil colloids. The magnitude of this acidity is very high. It is also called potential / reserve acidity.

(iii) Total acidity

The summation of active and exchange acidity is called total acidity.

Total acidity = Active acidity + Exchange acidity.

Distribution of acid soils

Out of 157 M ha of cultivable land in India, 49 M ha are acidic, if which 36 M.ha have soil pH less than 5.6 and the rest 13 M ha have soil pH in the range of 5.6 to 6.5. In Tamil Nadu, acid soil occupies nearly 2.6 M.ha, predominantly found in the hilly areas of Nilgris and Kodaikanal, high rainfall zone of Kanyakumari districts and parts of Pudukottai, Sivagangai, Cuddalore and Vilupuram districts.

Broadly acid soils of India can be classified into seven distinct groups *viz.*, (i) Laterite (ii) Laterite and Lateritic red (iii) Mixed red, black and yellow (iv) Ferruginous red (v) Podsolic brown forest and forest soil (vi) Foot hill soils and (vii) Peat soils.

Problems of soil acidity

(i) Acid toxicity

The acid toxicity includes toxicities of acidic anions as well as H^+ ions.

(ii) Toxicity of nutrient elements

a. Iron and Manganese

The concentration of these two ions (Fe^{2+} and Al^{3+}) in soil solution depends upon the soil reaction. The nutrient elements viz., Mn^{4+} and Fe^{3+} reduce to Mn^{2+} and Fe^{2+} respectively and their concentration is increased to toxicity levels. Due to such toxic effects, a physiological disease of rice is found in submerged soils which is popularly known as ***browning disease*.**

b. Aluminium

The toxicity of aluminium tends to decrease with an increase in concentration of other cations such as calcium. Aluminium toxicity is a problem in both upland and lowland soils. Aluminium toxicity in soil affects plant growth in the following ways as

- It restricts the root growth
- It affects various plant physiological processes like cell division, formation of DNA and respiration
- It restricts the absorption and translocation of important nutrient elements from the soil to the plants like P, Ca, Fe, Mn etc.,
- It causes wilting of plants
- It also inhibits the microbial activity in the soil.

(iii). Nutrient availability

a. Non specific effects –It causes inhibition of root growth and thereby affects the nutrient availability.

b. Specific effects

- **Exchangeable bases:** Due to complementary ion effect, exchangeable bases are released preferentially in a fractional exchange. Deficiency of bases like Ca^{2+} and Mg^{2+} are found in acid soils
- **Nutrient imbalance**
  - Phosphorus reacts with iron, aluminium and manganese and produces insoluble phosphatic compounds rendering phosphorus unavailable to plants
  - Availability of phosphorus is reduced due to fixation of phosphorus by hydrous oxides of iron and aluminium or by adsorption
  - Molybdenum is very limited and is unavailable to plants
  - Availability of boron is also reduced due to adsorption on sesquioxides, iron and aluminium hydroxyl compounds.
  - Nitrogen, potassium and sulphur become less available in a soil with pH less than 5.5.

(iv). Microbial availability

Bacteria and Actinomycetes function better in soils having moderate to high pH values. They cannot function at a pH below 5.5. Nitrogen fixation in acid soils is greatly affected by lowering the activity of *Azotobacter* spp. Besides these, soil acidity also inhibits the symbiotic nitrogen fixation by affecting the activity of Rhizobium sp. Fungi can grow well under very acidic conditions and causes various diseases like root rot of tobacco, blight of potato etc., However potato scab (caused by Actinomycetes) is not prevalent under acidic environment.

Acid sulphate soil

Acid sulphate are drained coastal wetland soils that have become acid (pH<4) due to oxidation of the pyritic minerals in the soil. Undrained soils containing pyrites need not be acid and they are called potential acid sulphate soils.

Types of acid sulphate soils

Potential acid sulphate soils

ASS which have not been oxidized by exposure to air are known as potential acid sulfate soils (PASS). They are neutral in pH (6.5–7.5), contain unoxidized iron sulfides, are usually soft, sticky and saturated with water and are usually gel-like muds but can include wet sands and gravels have the potential to produce acid if exposed to oxygen

Actual acid sulphate soils

When PASS is exposed to oxygen, the iron sulfides are oxidized to produce sulfuric acid and the soil becomes strongly acidic (usually below pH 4). These soils are then called actual acid sulfate soils (AASS). They have a pH of less than 4, contain oxidized iron sulfides, vary in texture and often contain jarosite (a yellow mottle produced as a by-product of the oxidation process).

Occurrence in India

Soil with sufficient sulphides (FeS2 and others) to become strongly acidic when drained are termed acid sulphate soils or as the Dutch refer to those soils cat clays. When allowed to develop acidity, these soils are usually more acidic than pH 4.0. Before drainage, such soils may have normal soil pH and are only potential acid sulphate soils. Generally acid sulphate soils are

found in coastal areas where the land is inundated by salt water. In India, acid sulphate soil is, mostly found in Kerala, Orissa, Andhra Pradesh, Tamil Nadu and West Bengal.

Formation of Acid Sulphate Soils

Land inundated with waters that contain sulphates, particularly salt waters, accumulate sulphur compounds, which in poorly aerated soils are bacterially reduced to sulphates. Such soils are not usually very acidic when first drained in water. When the soil is drained and then aerated, the sulphide (S^{2-}) is oxidized to sulphate (SO_4^{2-}) by a combination of chemical and bacterial actions, forming sulphuric acid (H_2SO_4). The magnitude of acid development depends on the amount of sulphide present in the soil and the conditions and time of oxidation. If iron pyrite (FeS_2) is present, the oxidized iron accentuates the acidity but not as much as aluminium in normal acid soils because the iron oxides are less soluble than aluminium oxides and so hydrolyze less.

Characteristics

Acid sulphate soils contain a *sulphuric horizon* which has a pH of the 1: 1 soil: water ratio of less than 3.5, plus some other evidences of sulphide content (Yellow colour). Such strong acidity in acid sulphate soils results toxicities of aluminium and iron, soluble salts (unless leached), manganese and hydrogen sulphide (H_2S) gas. Hydrogen sulphide (H_2S) often formed in lowland rice soils causing akiochi disease that prevents rice plant roots from absorbing nutrients.

Management of Acid Sulphate Soils

Management techniques are extremely variable and depend on many specific factors *viz,* the extent of acid formation, the thickness of the sulphide layer, possibilities of leaching or draining the land etc. The general approaches for reclamation are suggested below:

Keeping the area flooded.

Maintaining the reduced (anaerobic). Soil inhibits acid development, the use of the area to rice growing. Unfortunately, droughts occur and can in short time periods cause acidification of these soils. The water used to flood the potential acid sulphate soils often develop acidity and injure crops.

Controlling water table. If a non-acidifying layer covers the sulphuric horizon, drainage to keep only the sulphuric layer under water (anaerobic) is possible.

Liming and leaching. Liming is the primary way to reclaim any type of acid soil. If these soils are leached during early years of acidification, lime requirements are lowered. Leaching, however, is difficult because of the high water table commonly found in this type of soil and low permeability of the clay. Sea water is sometimes available for preliminary leaching.

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Lec 11. Lime Requirement of Acid Soil, Liming Materials and Reclamation of Acid soil

Lime Requirement

Lime requirement of an acid soil is defined as the amount of liming material that must be added to raise the pH to some prescribed value. This value is usually in the range of pH 6.0 to 7.0, since this is an easily attainable value within the optimum range of most crop plants. Shoemaker *et al.* (1961) buffer method is used for the determination of lime requirement of an acid soil.

Principles of Liming Reactions

Lime reactions in soils depends upon the nature and the fineness of the liming materials. Lime is applied to soils in the form of ground limestone. Limestones can be classified as calcitic ($CaCO_3$), dolomite ($CaMg(CO_3)_2$) or a mixture of the two. Both are sparingly soluble in pure water but become soluble in water containing carbon di oxide.

$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

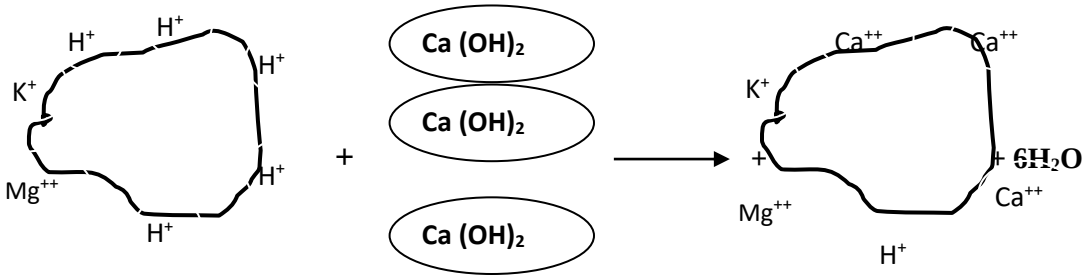
$$Ca(HCO_3)_2 \longrightarrow Ca^{2+} + 2HCO_3^- \downarrow$$

$$H^+ + HCO3^- \longrightarrow H_2CO_3 \rightleftharpoons H_2O + CO_2$$

(soil solution) (from lime)

In this way hydrogen ions (H^+) in the soil solution react to form weakly dissociated water and calcium (Ca^{2+}) ion from limestone is left to undergo cation exchange reactions. Acidity of the soil is thus neutralized and the percent base saturation of the colloidal material is increased.

The process of changing pH by the addition of lime ($Ca(OH)_2$) is illustrated below.



Acidic clay **Lime** **Neutral clay** **water**

Factors affecting liming reactions

(i) Moisture

Greater the moisture content, more rapid is the rate of reaction. As moisture increases, oxygen content decreases, CO_2 increases resulting in an increased rate of reaction.

(ii) Temperature

Lime and liming materials react more rapidly at high than at low temperatures. This effect is probably related to diffusion rates of end products away from the reaction sites.

(iii) Amount of exchange acidity

The amount of exchange acidity present in the soil affects reaction rate. If a soil has a high lime requirement and if sufficient quantity of limestone is added to neutralize the acidity present, the initial reaction will be quite rapid. But as the acidity becomes neutralized, the rate of reaction decreases and finally as neutrality is approached becomes almost negligible.

Liming materials

Liming materials may be defined as materials that are necessary for the neutralization of H^+ ions in soil solution. The materials commonly used are the oxides, hydroxides, carbonates and silicates of calcium or calcium and magnesium. In addition to these compounds, the accompanying anion must be one that will reduced the activity of hydrogen (H^+) ions and hence aluminium in the soil solution. These are called *Agricultural liming materials*.

Gypsum is not considered as a Liming Material

Gypsum is not considered as a liming material because on its application to an acid soil, it dissociates into calcium (Ca^{2+}) and sulphate (SO_4^{2-}) ions.

$$CaSO_4 \rightleftharpoons Ca^{2+} + SO_4^{2-}$$

The accompanying anion is sulphate and it reacts with soil moisture produces mineral acid (H_2SO_4) which also increases soil acidity instead of reducing soil acidity.

$$SO_4^{2-} + H_2O \longrightarrow H_2SO_4$$

Besides this, calcium (Ca^{2+}) in the gypsum after dissociation will result in replacement of adsorbed aluminium (Al^{3+}) in a localized soil zone with a significant lowering of soil pH. Hence gypsum does not qualify as a liming material.

Kinds of Liming Materials

Various kinds of liming materials are used for the correction of soil acidity. They are as follows.

(i) Oxides of lime

It is normally called burned lime or quick lime. Oxides of lime is more caustic than limestones. Burned lime is produced by heating limestone and dolomite as follows.

$$CaCO_3 + \text{Heat} \longrightarrow CaO + CO_2 \uparrow$$

Limestone

$$CaMg(CO_3)_2 + \text{Heat} \longrightarrow CaO + MgO + 2CO_2 \uparrow$$

(ii) Hydroxides of lime

It can be produced by adding water to burned lime and is called *slaked lime*.

$$CaO + H_2O \longrightarrow Ca(OH)_2 + CO_2$$

(Burned lime) (Slaked lime)

It is more caustic than burned lime (CaO). If it is kept open in the moist air, then combination of calcium hydroxide occurs as follows.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

$$Mg(OH)_2 + CO_2 \longrightarrow MgCO_3 + H_2O$$

(iii) Carbonates of lime

These are by products of certain industries and so the content of calcium and magnesium varies. The two important minerals found are calcite ($CaCO_3$) and dolomite ($CaMg(CO_3)_2$).

(iv) Slags

These are generally three types of slags that are found important.

a. **Blast furnace slag** – It is a by- product of the manufacture of Pig iron. As a liming material, this behaves as calcium silicate. The neutralizing value of blast furnace slags ranges from about 75-90%.

b. **Basic slag** – It is a by-product of the basic open hearth method of making steel from pig iron, which in turn is produced from high phosophorus iron ores. Impurities in the iron, including silica and phosphorus are fluxed with lime and the basic slags are produced. Its neutralizing value ranges from 60-70%.

c. **Electric furnace slag** – This is produced from the electric furnace reduction of phosphate rock during preparation of elemental phosphorus. This product is largely calcium silicate and is used as a liming material.

d. **Other liming materials** – Coral shell, chalk, wood ash, press mud, by-product material of paper mills, sugar factories, fly ash and sludge etc., are considered as liming materials and also used for the amelioration of soil acidity.

Efficiency of liming materials

The efficiency of liming materials can be judged on the basis of following important factors because of varying neutralizing capacity of different liming materials.

(i) Neutralizing value (N.V) or Calcium Carbonate Equivalent (CCE) –Calcium carbonate equivalent (CCE) is defined as the acid neutralizing capacity of an agricultural liming material expressed as a weight percentage of calcium carbonate.

$$\text{CCE of a liming material} = \frac{\text{Molecular weight of } CaCO_3}{\text{Molecular weight of a liming material whose E is to be determined}} \times 100$$

Liming materials	**Neutralizing value of CCE (%)**
Calcium oxide (CaO)	179
Calcium hydroxide ($Ca(OH)_2$)	136
Dolomite ($CaMg(CO_3)_2$)	108.7
Calcite ($CaCO_3$)	100
Basic slag ($CaSiO_3$)	86

(ii) Purity of liming materials – More pure the liming material, higher will its effectiveness for the amelioration of soil acidity.

(iii) Degree of fineness of liming materials – Finer materials increases the surface contact with the soil and the rate of reaction is high. The amount of finer fraction of liming materials will be required much less as compared to coarser fractions of the material to achieve a certain pH. The fineness is measured in terms of the ability of a material to pass through a sieve having 60 holes of equal size in one linear inch (60 mesh sieve).

Percent Effective Calcium Carbonate (ECC) or (Neutralizing Index)

The effective calcium carbonate (ECC) rating of a limestone or liming material the product of the calcium carbonate equivalent (CCE) and the fineness factor. The fineness factor is the sum of the product of the percentage of material in each of the three size fractions (60 mesh sieve, 20 mesh sieve and 8 mesh sieve) multiplied by the appropriate effectiveness factor.

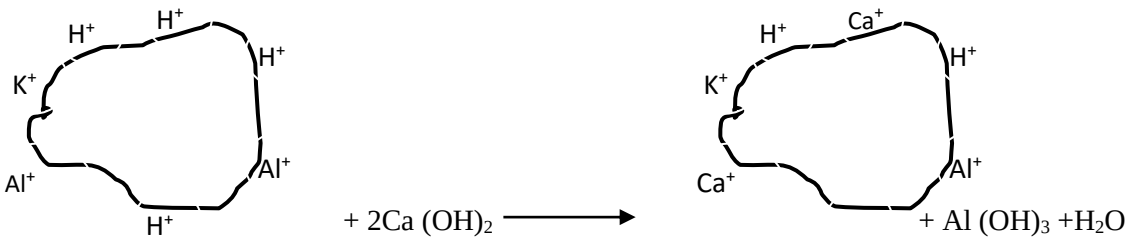
Percent ECC or N.I = CCE x Fineness factor.

Chemical reactions during the process of amelioration

(i) Oxides of lime (CaO) – When oxides of lime like CaO is applied to an acid soil, it reacts immediately as follows.



(ii) Hydroxides of lime (Ca $(OH)_2$) – When hydroxides of lime like Ca $(OH)_2$ is applied for the reclamation of an acid soil, the following chemical reaction takes place.

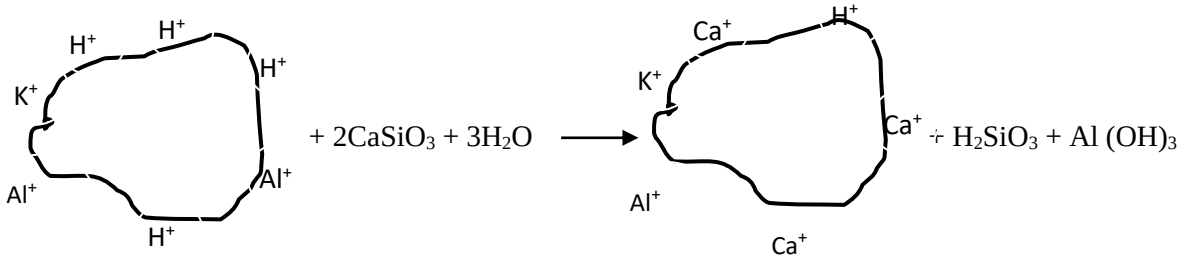


(iii) Carbonates of lime ($CaCO_3$) – When carbonates of lime like calcite is applied to an acid soil, a part of $CaCO_3$ undergoes solution and combines with H_2CO_3 to form soluble $Ca(HCO_3)_2$. The calcium bicarbonate in solution form reacts with the soil colloids with the evolution of CO_2 as follows.

$$CaCO_3 + H_2CO_3 \xrightarrow[H]{} Ca(HCO_3)_2$$

$$Ca(HCO_3)_2 + \text{soil} \xrightarrow[H]{} Ca(\text{Soil}) + H_2O + 2CO_2$$

(iv) Basic Slag – Basic slag can also be used as a liming material when it is applied to an acid soil the following chemical reaction takes place.



Besides these chemical amendments, some other management practices found effective for the amelioration of an acid soil includes (i) Use of basic fertilizers like sodium nitrate, basic slag etc., (ii) Reducing leaching of basic cations by following appropriate soil and water management practices and (iii) Acid tolerant crops like rice, potato, tea, wheat, sweet potato, maize, brinjal etc.,

Relative tolerance of crops to soil acidity

Crops	**Optimum pH range**
Cereals	
Maize, sorghum, wheat, barley	6.0-7.5
Millets	5.0-6.5
Rice	4.0-6.0
Oats	5.0-7.7
Legumes	

Field beans, soybean, pea, lentil etc.	5.5-7.0
Groundnut	5.3-6.6
Others	
Sugarcane	6.0-7.5
Cotton	5.0-6.5
Potato	5.0-5.5
Tea	4.0-6.0

Liming Factor

Liming Factor may be defined as the factor by which the actual amount of lime can be calculated from the estimated theoretical amount of lime. This factor varies from 1 to 3, depending on rate of limestone solution, plant uptake and leaching during the reaction period.

Methods of applying lime

The application of small amounts of lime of soil every year or twice a year is effective. Lime should be applied well ahead of the crop cultivation by broadcasting and should be mixed with the whole plough layer soil so that liming reaction can occur in a faster rate. When both the surface and subsurface soils are strongly acidic e.g. ultisols, incorporation up to a depth of about 30 cm (12 inches) is required.

Effects of over liming

When excessively large amount of lime is applied to an acidic soil, the growth of plants is affected due to the following causes.

(i) Deficiency of ion, copper and zinc
(ii) Phosphorus and potassium availability will be reduced
(iii) Due to high OH^- ion concentration, root development will be inhibited in association with tip swelling brought about by hydrations. Due to dehydrating properties of boron, it acts as a protective agent for excess OH^- ion concentration
(iv) Boron deficiency occurs
(v) Incidence of diseases like scab in root crops

All these effects can be reduced with the application of large amount of organic manures like well rotten farmyard manure, green manure crops, compost, phosphorus, boron or a mixture of micronutrient fertilizers to the soil.

Liming on Plant Nutrition

Direct Benefits

(i) Reduction in toxicity of aluminium and manganese. Reduced uptake of calcium (Ca^{2+}) and magnesium (Mg^{2+}) in the soil solution can also be alleviated with the application of lime.

(ii) Removal of hydrogen (H^{+}) ion toxicity which damages root membranes and causes detrimental effect for the growth of microbes like bacteria.

Indirect Benefits

(i) Phosphorus availability – At low pH values and at high pH values, phosphorus availability is greatly reduced due to precipitation as iron and aluminium phosphates, calcium or magnesium phosphates respectively. Liming programme should be planned so that the pH can be kept between 5.5 -7.0 in order to derive maximum benefit from the applied phosphatic fertilizer.

(ii) Micronutrient availability – The toxic effect of most of the micronutrients like Fe, Mn,Cu,Zn,B (except Mo) can be prevented by the application of lime.

(iii) Nitrification – Nitrification is enhanced by a liming to a pH of 5.5 to 6.5. Most of the organisms responsible for the conversion of ammonia to nitrates require large amount of active calcium.

(iv) Nitrogen fixation – The process of nitrogen fixation both symbiotic and non-symbiotic is favored by adequate liming.

(v) Soil Physical Condition – The structure of fine textured soil may be improved by liming. This is the result of an increase in the organic matter content and to the flocculation of calcium-saturated soils. Liming decreases the bulk density of soils, increases infiltration and percolation rates of water. Liming checks soil erosion.

(vi) Diseases - The amelioration of soil acidity by liming helps to control certain plant pathogens. Club root disease of cole crops can be reduced with the application of lime.

(vii) Efficiency of fertilizers – Liming increases the efficiency of different fertilizers especially nitrogenous and phosphatic fertilizers by modifying the soil reaction favourably.

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Lec. 12, 13, 14, 15 and 16. Genesis and Classification of Saline Sodic Soils

Salt affected soils occur most extensively in arid climates, but these soils are also found in coastal sea areas where soils are inundated by ocean or sea water. Soils are said to be **saline** if they contain an excess of soluble salts and **sodic or alkaline** if they contain an excess of sodium. Soils are said to be **saline-sodic** or **saline-alkali** if they contain appreciable amounts of both soluble salts and sodium.

Distribution of salt affected soils

Saline –sodic soils occur in all the continents of the world. Based on FAO / UNESCO soil map, about 952 million hectares of land in the world is under varying degrees of deterioration due to excessive accumulation of salts in the soil profile. In India about 7421000 ha are affected by salinity and sodicity.

Common sources of salts

1. Rocks and minerals

The salts primary originate as a result of hydrolysis, hydration, carbonation and oxidation – reduction of the easily weatherable minerals inherited from the parent materials viz., halite, gypsum, sulphides, calcite, dolomite, apatite, amphiboles, olivine, feldspars and primary layer silicates.

2. Arid and semi-arid climate

Salt affected soils are mostly formed in arid and semi-arid regions where low rainfall and high evaporation prevails. The low rainfall or precipitation in these regions is not sufficient to leach out the soluble weathered products and hence the salts accumulate in the soil. During rain, the salts dissolve in water and migrate downward. In the dry weather, the salts move up with the water and are brought upto the surface where they are deposited as the water evaporates. With alternate downward and upward movements of rain water, the salts get concentrated in the surface layer and form a ‘white efflorescence’.

3. Ground water

Groundwater contains large amounts of water-soluble slats which depends upon the nature and properties of the geological material with which water remains in contact. Where water table and evapotranspiration are high, salts move along with water upward through the capillary activity and the salts accumulate on the soil surface in the form of crystals.

4. Ocean or sea water

Ocean water containing 42 x 10^{15} tonnes of dissolved salts (of which 85.6 % is NaCl) is a significant direct source of salts in the soils of the low-lying coastal areas. Near the sea coast, the rain water may contain salts upto 200 mg / L, containing high concentrations of chlorides of Na and Mg, but inland rain water is dominated by sulphates and bicarbonates of Ca and Mg.

5. Irrigation water

The application of irrigation water without proper management (i.e., lack of drainage and leaching facilities) increases the water table and surface salt content in the soil. Salts can also accumulate due to flooding by waters from mining shafts or by industrial effluents.

6. Atmospheric accession of salts

Atmospheric accession of salts in the soil of inland areas can occur as a result of falling dried droplets of ocean water from the atmosphere (aerosol) along with rain water or as dry salt-dust.

7. Salts blown by wind

In arid regions near the sea, appreciable amount of salts are blown by wind year after year and get deposited on the surface soil. Due to low rainfall, the deposited salts are not washed back to the sea or leached to the lower soil horizon and thereby develops salinity in the soil. The salinity of Rajasthan is mostly developed through this source.

8. Excessive use of basic fertilizers

Use of basic fertilizers like sodium nitrate, basic slag etc., may develop soil alkalinity.

9. Marine rocks and deposits

Marine rocks and evaporites developed with the upliftment of many parts of the continents from the sea / ocean are also important sources of salts in the soil. E.g. Localized concentration of salts along the lower Himalayas and the Shivalik's, and a large part of the present Indo-Gangetic Plains, Western Rajasthan upto the Aravalli's and Kutch in India, which may be attributed to the upliftment of these ranges during the geological periods from under the sea.

10. Salt springs

Salt springs emerging from many regions can also add considerable quantities of salts to the soil, because volcanic gases and waters from the interior of the earth contain large quantities of chlorides.

Taxonomic Classification

According to the Taxonomic Soil Classification System (soil Survey Staff, 1975), the salt affected soils of the arid zone are classified under the order Aridisol. In some regions salt affected soils have also been classified under the orders Alfisol, Mollisol, Inceptisol and Vertisol.

Under this soil taxonomic system, the most common diagnostic horizon (epipedon) is Ochric and sub surface horizons are argillic, natric, cambic, calcic, gypsic and salic.

Important characteristics of salt affected soils

Characteristics	**Saline soil**	**Non-saline alkali soil**	**Saline- alkali soil**	**Degraded alkali soil**
Content in soil	Excess soluble salts of sodium	Absence of soluble salt and presence of excess exchangeable sodium on the soil complex	Soils are both saline and non-saline alkali soil. It is a stage of transition. Soil contains Na-clay as well as soluble salts	Hydrogen (H^+) ions in the upper layer and sodium (Na^+) in the lower layer
Exch. Cations	Exchangeable Ca	Exchangeable Na	--	--
Colour	White	Black	---	Black in lower layer
Dominant salts	Sulphates, chlorides and nitrates of sodium	Sodium carbonate	----	Sodium carbonate in lower layer
SAR	Less than 13	More than 13	More than 13	< 13 in the surface and > 13 in the lower horizon
ESP	Less than 15	More than 15	More than 15	More than 15

Soil pH	Less than 8.5	8.5-10.0	More than 8.5	pH about 6.0 in the surface soil and 8.5 in the lower layer constituting the main soil body
Physical condition of the soil	Flocculated condition, permeable to water and air. Soil structure optimum	Deflocculated condition, permeability to water and air is poor. Very poor soil structure	Flocculated or deflocculated depending upon the presence of sodium salts and Na-clay	Compact, low infiltration and permeability. Develops columnar structure
Morphological character	White crust on the soil surface	Silica gels and humus produces hard pan in the sub-soil	----	----
Organic matter content	Less than normal soils	Very low. These soils are almost barren (Usar)	Variable	Low
Total soluble salt content	More than 0.1 % High osmotic pressure to prevent absorption of moisture and plant nutrients from such soils.	Less than 0.1 %	More than 0.1 %	Less than 0.1 %
Soluble boron	Causes problem in this soil Safe : <0.7 ppm Marginal :0.7 – 1.5 ppm Unsafe : > 1.5 ppm			
Electrical conductivity of the saturation extract	> 4 dSm^{-1}	< 4 dSm^{-1}	> 4 dSm^{-1}	< 4 dSm^{-1}

Effects of salts on plant growth

The growth of plants is suppressed when the salt concentration is low, but may entirely prevented if it is too high. Seeds fail to germinate, plants die at early stages, show stunted growth and foliage becomes characteristically bluish-green. The entire field may look barren. Plant growth is affected due to changes in proportion of exchangeable cations, soil reaction, ionic balance in plants, physical properties of the soil and osmotic and specific ion toxicity effects.

(i) Osmotic stress

Under osmotic stress, plant cells continue to divide, but do not elongate, resulting in an increase in the number of cells per unit areas which accounts of the typically dark bluish green colour of the foliage.

(ii) Toxic effect of some ion

The ions which cause specific toxicities in plants are chlorides, sulphates, bicarbonates, sodium, magnesium and boron. Excessive intrusion of these ions in the roots and their movement in the plants through transpiration stream may lead to necrosis, burning of leaf tips and margins and eventually death. Excessive alkalinity causes caustic effect on the roots of plants.

(iii) Nutritional imbalances

Salinity induced cationic and nutritional imbalances occur within the plants. Examples include excessive absorption of Na^+ ions leads to decrease in the absorption of Ca^{2+}, Mg^{2+} and K^+ ions. High concentration of bicarbonates in soil solution can induce iron chlorosis and micronutrient deficiencies. Increase in soil salinity and alkalinity can also adversely affect soil biology, mineralization of organic matter and transformations of applied fertilizer nutrients.

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Saline Soil – Characteristics and Management

Saline soil

The process by which the saline soil is formed is called Salinization. Saline soils occur mostly in arid or semi-arid regions. Saline soils are practically non-existent in humid regions except when the soil has been subjected to sea water in river deltas and low lying lands near the sea.

Nature of soluble salts in saline soils

Soluble salts refers to those inorganic chemicals that are more soluble than gypsum ($CaSO_4 \cdot 2H_2O$), which has a solubility of 0.241 g per 100 ml of water at 0°C. Common table salt (NaCl) has a solubility nearly 150 times greater than gypsum. Most soluble salts in saline soils are composed of the cations sodium (Na^+), calcium (Ca^{++}), Magnesium (Mg^{++}) and the anions chloride (Cl^-), sulphate ($SO4^{2-}$) and carbonate (CO_3^{2-}) also occur, as do many other ions. Boron may also be present in saline soils.

Characterization of saline soil

(i) Soluble salt concentration in the soil solution

In saline soils, the soluble salt concentration in the soil solution is very high and as a result, the osmotic pressure of the soil solution is also very high. Plant growth is affected due to wilting and nutrient deficiency. Salt content of more than 0.1 % is injurious to plant growth.

(ii) Osmotic pressure

Osmotic pressure of the soil solution is closely related to the rate of water uptake and the growth of plants in saline soils. It should be assessed of field capacity soil moisture regimes.

Osmotic pressure = 0.36 x EC (dSm^{-1})

(iii) Electrical conductivity of the soil saturation extract

Measurement of EC of the soil saturation extract is also essential for the assessment of the saline soil for the plant growth and is expressed as dSm^{-1}.

< 2 – Salinity effects mostly negligible

2-4 – Yields of very sensitive crops may be restricted

4 –8 - Yields of many crops restricted

8-16 - Only tolerant crops yield satisfactorily

> 16 - Only a few tolerant crops yield satisfactorily

(iv) Concentration of water-soluble boron

The determination of water-soluble boron concentration is also another criterion for characterization of saline soils. The critical limits of boron concentration for the plant growth is

Boron concentration

< 0.7 - Crops can grow (safe)

0.7 -1.5 - Marginal

> 1.5 - Unsafe

(v) Soil texture

A sandy soil with 0.1 % salt would be saline enough to injure the growth of common crops, while a clayey soil with the same amount of salt may be just a normal soil in which the yields of even sensitive crops would not be affected.

Relationship between EC and TSS

For soil solutions with EC ranging from 0.1 to 5 dSm^{-1},

- Total dissolved solids (mg / L) is EC (dSm^{-1}) x 640
- Sum of soluble cations or anions = EC (dSm^{-1}) x 10

For soil solutions with EC ranging from 3 to 30 dSm^{-1}

OP (bars) = EC (dSm^{-1}) x (-0.36)

Where OP is the osmotic potential or the negative of osmotic pressure of solution which directly measures the effect of salinity on plant growth.

Problems of salt affected soils

- Soils are usually barren but potentially productive
- Wilting coefficient of saline soil is very high
- Amount of variable soil moisture is low

- Excessive salts in the soil solution increases the osmotic pressure of soil solution in comparison to cell sap. Thus it is more difficult for roots to extract moisture. During a dry period, salt in soil solution may be so concentrated as to kill the plants by pulling water from them (exosmosis). Due to high salt concentration, plants have to spend more energy to absorb water and to exclude salt from metabolically active sites
- Salt toxicity – When the concentration of soluble salts increases to a high level, then it produces toxic effect directly to plants such as root injury, inhibition of seed germination etc.,

Management of saline soils

1. Physical amelioration

The commonly followed physical methods for amelioration of salt affected soils include deep ploughing, sub-soiling, sanding, profile inversion and scrapping. Many farmers often resort to scrapping of surface by a few centimeters of salt and dump it some other place. Reclamation by this method cannot be achieved on a permanent basis.

2. Hydrotechnical amelioration

Reclamation of salt affected soils by this method involves the removal of salts from the saline soil through the process of leaching with water and drainage. Ponding continuously with sufficient depth of good quality water on the surface of the soil and ensuring downward movement of water is the most traditional method for leaching of salts. Land leveling is an important step before initiation of the leaching process. The passage of 1 metre leaching water per metre of soil depth under continuous ponded conditions normally removes approximately 80 % of soluble salts from the soil. Basin furrow method of leaching is considered to be more efficient.

Leaching requirement (LR) – Leaching requirement may be defined as the fraction of irrigation water that must be leached through the root zone to control soil salinity at any specified level. The leaching requirement is the ration of the equivalent depth of the drainage water and may be expressed as a fraction or as percent.

$$LR = \frac{D_{dw} \times 100}{D_{iw}} = \frac{EC_{iw} \times 100}{EC_{dw}}$$

LR - Leaching requirement expressed in percentage

D_{dw} - Depth of drainage water in inches

D_{iw} - Depth of irrigation water in inches

EC_{iw} - Electrical conductivity of the irrigation water in dSm^{-1}

EC_{dw} – Electrical conductivity of the drainage water in dSm^{-1}

Drainage system

The need for the development of either surface or sub-surface drains has to be worked out on the basis of the nature of the soil, ground water conditions, climate, cropping pattern, economic considerations etc.,

3. Cultivation of salt tolerant crops

Threshold salinity		
4 -8 dSm^{-1}	**2 – 4 dSm^{-1}**	**< 2 dSm^{-1}**
Date palm	Alfalfa	Turnip
Cowpea	Rice	Onion
Soybean	Tomato	Mung bean
Wheat	Cucumber	Berseem
Sorghum	Groundnut	Sugarcane
Sugarbeet		Maize
Cotton		Citrus
Barley		Potato
Cluster bean		Cabbage
		Celery

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Sodic Soil – Characteristics, Gypsum Requirement and Management

Exchangeable sodium percentage (ESP)

Sodium saturation in the soil is expressed as exchangeable sodium percentage (ESP) and as sodium adsorption ratio (SAR) in the soil solution, because of their significance in deteriorating physico chemical properties of the soil and adversely affecting the growth of plants.

ESP = 100 (Exchangeable Na^+ ions) / (CEC)

The value of SAR i.e. the relative proportion of Na^+ ions with respect to Ca^{2+}, Mg^{2+} ions in the soil solution as calculated by the Equation

$$SAR = [Na^+] / \sqrt{[Ca^{2+} + Mg^{2+}] / 2}$$

ESP = 100 (ESR) / (1+ESR), where ESR is the Exchangeable Sodium Ratio

Alkali soils

Soils have sufficient sodium saturation and alkalinity to adversely affect plant growth and crop productivity. Carbonates (CO_3^{2-} + HCO_3^-) of sodium are the dominant salts. The concentration of neutral salts (Cl^- and SO_4^{2-}) is much lower. Sparingly soluble gypsum, which may be present in many saline soils , is nearly absent in alkali soils. Alkali soils are characterized by pH more than 8.5, ESP more than 15 and EC lower than 4 dSm^{-1} at 25°C.

Black alkali soils

In alkali soil, because of high pH and Na_2CO_3 the finally decomposed organic matter is dissolved along with the water that imparts a dark black or brown colour to the soil.

Saline alkali soils

The US Salinity Laboratory Staff (1954) have defined saline-alkali soils as the soils with EC more than 4 dSm^{-1} at 25°C and ESP more than 15. These soils have been formed as a result of both salinization and alkalization processes. When high ECe is due to high concentration of chlorides and sulphates of sodium, the pH of these soils seldom exceeds 8.5. Clay particles in these soils remain

flocculated. But when EC is high because of soluble carbonates of sodium, the pH of these soils can be much higher than 8.5. With leaching of soluble salts of chlorides and sulphates of sodium, these soils behave like alkali soils.

Effects of alkali soils

(i) Dispersion of soil colloids

Under alkali conditions, the damage is not due to salt concentration. The sodium ion adsorbed by clay colloids causes de-flocculation and dispersion of clay which results in a loss of desirable soil structure and helps for the development of compact soil.

(ii) Other physical properties

Due to dispersion and compactness of soil, aeration, hydraulic conductivity, drainage and microbial activity are reduced.

(iii) Caustic influence

It is due to the high sodicity caused by sodium carbonate and sodium bi-carbonate

(iv) Concentration of hydroxyl ion

High hydroxyl ion has a direct detrimental effect on plants. Damage from hydroxyl ions occurs at pH 10.5 or higher.

(v) Specific ion effect

The presence of excess sodium may induce deficiencies of cations like calcium and magnesium due to exclusion of calcium and magnesium from the adsorption sites.

Management

Physical amelioration

Profile inversion can be adopted only under conditions where surface soil is good but the soil below some depth is sodic, saline or has undesirable characteristics. This is a very cumbersome method and is generally not followed by farmers.

Chemical amelioration

Alkali / sodic soils contain Na_2CO_3 and $NaHCO_3$, which upon hydrolysis produce alkalinity. The common amendments used for reclamation of alkali soils is

Amendment	**Soil Conditions**
Gypsum	Saline and alkali soils having pH up to 9.0
Sulphur, Iron Sulphate, Iron pyrite	Saline and alkali soils having pH from 8.0 -9.0

The chemical amendments can be broadly grouped into

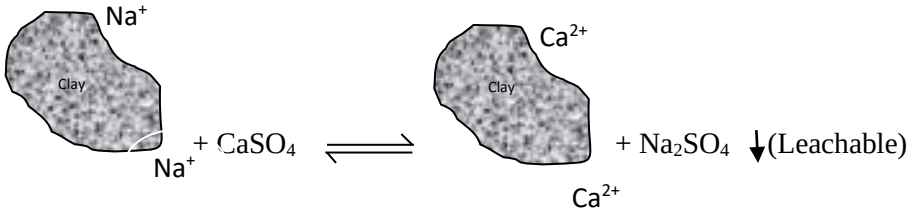
(i) Soluble sources of calcium – Gypsum, Calcium chloride, Phospho-gypsum
(ii) Sparingly soluble calcium salts – Calcite
(iii) Acids or acid – formers – Sulphur, Sulphuric acid, Sulphates of iron and aluminium, pyrites, lime-sulphur.

Chemical Reactions involving reclamation of alkali soils

(i) Gypsum ($CaSO_4.2H_2O$)

When gypsum is applied to sodic soils, loss of exchangeable sodium occurs and calcium takes the place of sodium on the exchange complex.

$$Na_2CO_3 + CaSO_4 \rightleftharpoons CaCO_3 + Na_2SO_4 \downarrow \text{(Leachable)}$$



(ii) Sulphur

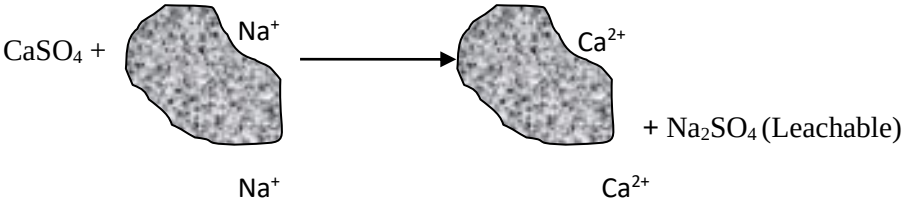
$$2S + 3O_2 \longrightarrow 2SO_3 \text{ (By the action of sulphur oxidizing bacteria)}$$

$$SO_3 + H_2O \longrightarrow H_2SO_4$$

$$H_2SO_4 + Na_2CO_3 \longrightarrow CO_2 + H_2O + Na_2SO_4 \text{ (Leachable)}$$

On calcareous alkali soil

$H_2SO_4 + CaCO_3 \longrightarrow CO_2 + H_2O + CaSO_4$

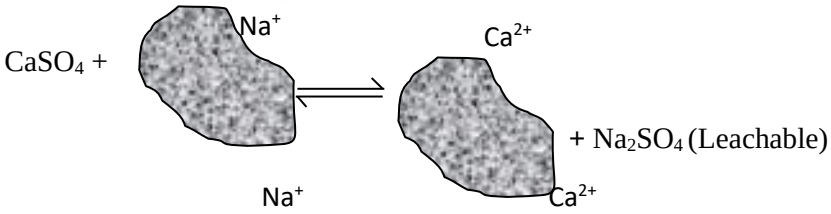


(iii) Iron pyrite (FeS_2)

$FeS_2 + O_2 + H_2O \rightleftharpoons FeSO_4 + H_2SO_4$

$FeSO_4 + 2H_2O \rightleftharpoons H_2SO_4 + Fe(OH)_2$

$H_2SO_4 + CaCO_3 \rightleftharpoons CaSO_4 + H_2O + CO_2$



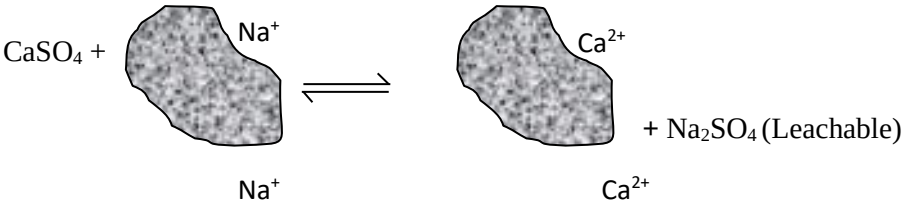
(iv) Iron Sulphate ($FeSO_4$)

When iron sulphate is applied to the soil, the following chemical reaction will occur.

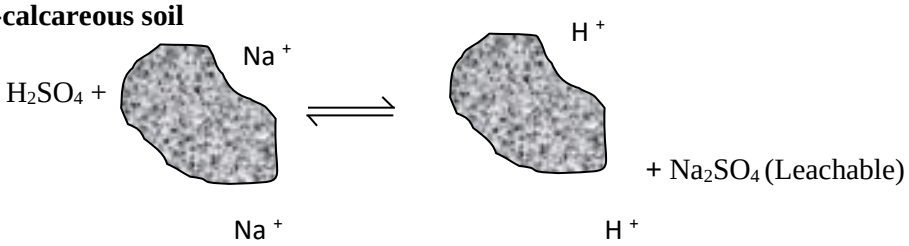
$FeSO_4 + 2H_2O \rightleftharpoons H_2SO_4 + FeO$

On calcareous alkali soil

$H_2SO_4 + CaCO_3 \rightleftharpoons CO_2 + H_2O + CaSO_4$



In non-calcareous soil



(v) Lime Sulphur (CaS_5)

When lime sulphur is applied to soils, the following reaction takes place.

$$CaS_5 + 8O_2 + 4H_2O \rightleftharpoons CaSO_4 + H_2SO_4$$

Then the reactions are similar as that of iron sulphate in calcareous and non-calcareous alkali soils.

Other methods -Salt precipitation

The elimination of salts and exchangeable sodium from soils by leaching is presently practiced but the leached salts have been washed into ground waters or streams resulting in ground water pollution. A new concept in managing the soils is by precipitation of salts. Instead of leaching the salts completely away, they can be leached to only 0.9 -1.8 m deep (3-6 ft), where much of the salt would form slightly soluble gypsum or carbonates during dry periods and not react any longer as soluble salts.

Gypsum Requirement

The main principle for the reclamation of sodic or alkali soils is to replace exchangeable Na by another action calcium (Ca^{2+}). Gypsum is considered the best and cheapest for the reclamation purpose. In India, the resources of gypsum are estimated to be more than 1000 million tones. Calcium (Ca^{2+}) solubilized from gypsum replaces sodium leaving soluble sodium sulphate (Na_2SO_4) in the water, which is then leached out. Gypsum requirement is expressed as me of Ca^{2+} per 100 g of soil.

Gypsum Requirement (GR) is the calculated amount of gypsum necessary to add to reclaim the soil. Gypsum requirements depends upon exchangeable sodium content to be replaced, exchange efficiency and the depth of soil to be reclaimed. This determination includes Ca^{2+} required to replace the exchangeable Na^+ ions plus that required to neutralize the alkalinity.

$$\text{Gypsum requirement i.e., me of } Ca^{2+} / 100 \text{ g soil} = \frac{\text{ESP (Initial)} - \text{ESP (Final)}}{100} \times \text{CEC}$$

ESP (initial) is obtained from the analysis of soil before reclamation or application of gypsum; ESP (final) is usually kept at 10 and CEC is the cation exchange capacity in C mol (p^+) kg^{-1} of the soil. Gypsum requirement obtained from the calculation should be multiplied by the purity factor of gypsum to arrive at the field requirement.

Also the gypsum requirement worked by these methods is based on 100 % replacement of Na^+ by Ca^{2+} ions. Under actual conditions, the efficiency is much lower. To compensate for the lack of a quantitative replacement, the GR worked out should be multiplied by 1.25. The quantities of different amendments to be applied in comparison with gypsum are given below.

Amendment	**Tonnes equivalent to 1 tonne of gypsum**
Gypsum ($CaSO_4$. $2H_2O$)	1.0
Sulphur	0.18
Lime - Sulphur	0.75
Sulphuric acid	0.57
Iron sulphate ($FeSO_4.7H_2O$)	1.62
Aluminium sulphate	1.27
Limestone ($CaCO_3$)	0.58

- Gypsum should be broadcasted and incorporated in the surface 0-10 cm soil by discing or by using a cultivar
- Mixing of gypsum in deeper soil decreases its effectiveness, as a fraction of gypsum reacts with soluble carbonates to precipitate Ca as $CaCO_3$
- Mined gypsum ground to pass through 2 mm sieve is cost effective and efficient.

Biological Amelioration

Organic materials and the action of plant roots improve biological activity in the soil. The decomposition of materials increases the concentration of CO_2 and organic acids in the soil which help in

mobilizing calcium by dissolving calcium compounds. This can be accomplished by green manuring, incorporation of crop residues, application of FYM, press mud and other organic materials.

Addition of crop residues and other organic materials have got a tremendous influence on maintaining soil physical, chemical & biological properties of soils. Organic amendments like straw, rice husk. Ground nut & safflower hulls, compost, poultry droppings, green manures, tree leaves & saw dust. They produce CO_2 (carbonic acid) & organic acids, they increase the solubility of calcite and lower the ESP.

Crop selection

Relative tolerance of crops to alkalinity or sodicity.

ESP (range)	**Crop**
2 -10	Deciduous fruits, nuts, citrus, avocado
10-15	Safflower, Blackgram, Peas, Lentil, Pigeonpea
16-20	Chickpea, Soybean
20-25	Clover, Groundnut, Cowpea, Onion, Pearl Millet
25-30	Linseed, Garlic, Cluster bean
30-50	Oats, Mustard, Cotton, Wheat, Tomatoes
50-60	Beets, Barley, Sesbania
60-70	Rice

Industrial by products as Amendments:

Phospho-gypsum	Used to provide soluble Ca directly/indirectly by dissolving soil lime
Press mud	
Molasses	
Acid wash	
Effluent from milk plants	

18. Polluted soils- industrial effluents- Characteristics, reclamation and management

Pollution is a serious problem ever since sewage and industrial effluents are disposed into water courses and on land. It has increased with the growth of industry as well as of population. Industries can conveniently be classified in to two groups *viz.* (i) Dry process industries and (ii) Wet process industries. Dry process industries (mostly engineering and assembling industries) do not use water in their process, while wet process industries use water either as raw material or in their process or for both. Only a little quantity of water is absorbed in the process and the rest being discharged as effluent. Practically all the volume of water used is calculated as the volume of effluent. In addition to the process waters, large volumes of cooling water are also discharged as waste. Boiler blow down is also an effluent contributing significant quantities to the bulk. These effluents contain a wider variety of materials of both organic and inorganic nature including toxic substances and are usually discharged with or without treatment into surface waters such as rivers, streams, lakes or into oceans or on land or into sewers. The solid wastes produced in these industries are not given so much importance as they are either recycled dumped or disposed of to remote places. Liquid industrial wastes are of great concern because of their harmful effects. The effects are described in detail below.

Effects of industrial effluents

Industries need a wide variety of raw materials and chemicals which are later discharged as effluents. Acids, alkalis, toxic metals, pesticides and other poisonous substances such as cyanide, dyes, oils, detergents, resins, rubbers are a few to mention. Heated effluents that impart thermal loading on receiving waters and effluents containing radio active materials are also of prime concern. Some of the effluents such as from tanning and meat packing may also contain pathogenic bacteria. The nature and extent of pollution depends on the materials present in the effluent and on the quantity discharged.

Effects of industrial effluents on water ecosystem

Color : The effluents contain dyes in higher concentrations which impart color to the receiving streams and they persist for longer distances. Photosynthesis of phytoplankton is affected by these colors.

pH value : The extreme alkalinity makes the receiving water unfit for any purpose. Further, it is deleterious to most of the aquatic life.

Suspended impurities : The colloidal and suspended impurities produce turbidity in the receiving waters. The turbidity and color along with the oil and scum create an unsightly appearance.

Depletion of oxygen : Natural substances such as starch and dextrin and inorganic substances such as sulfide and nitrite present in the effluent exert an immediate oxygen demand. The stream will then be devoid of oxygen and the aquatic life are affected adversely.

Toxic substances : Chromium, sulfide, chlorine and aniline dyes present in these wastes are directly toxic to fish and microbial organisms which carryout purification. Thus the self purification of the water body is affected.

Oils : Various oils (mineral) in the effluent interfere with the oxygenation of stream as they form a blanket on the surface and prevents the entry of oxygen at air/water interface.

Dissolved minerals : The mineral materials, mostly sodium salts increase the salinity of the water and consequently it becomes unfit for irrigation.

Effects on land

1. The excess content of sodium (60%) and boron (2 mg/l) are deleterious to crops.
2. The high sodium alkalinity combined with salinity impairs the growth of plants.
3. Texture of the soil is affected by sodium and penetration of roots is prevented.
4. Soil permeability is also affected by sodium and ultimately the soils will loss its productivity.
5. Suspended and colloidal impurities clog the pores and form a mat on the surface of soil preventing the passage of air, water etc.

LEATHER INDUSTRY

The leather processing industries are tanneries. Tannery is a major foreign exchange earning industry in India. But this industry appears to be the major source of pollution, especially in Tamil Nadu state which contributes more than 70% of this industry functioning in India. High concentration of salts (NaCl) and the heavy metal namely Chromium (Cr) are the major pollutants of tannery wastes. While the effluent is rich in NaCl, the sludge on average contained 20000 mg kg^{-1} of chromium. The improper disposal of tannery wastes lead to extensive contamination of land and water with salts and chromium, thereby the agricultural productivity have been drastically affected not only due to low yield but also due to poor quality of produce.

Chemistry of chromium

Chromium exists in both anionic and cationic state and the nature of the species may vary with charges in the soil characteristics. Among the various forms of chromium, the trivalent form Cr (III) and hexavalent form Cr V1 are the important transformation stages. Hexavalent species Cr (V1) is a strong oxidizer and result in oxygenated species, that are pH dependent, highly soluble and hence it is toxic than trivalent form of Cr (III). The dominant hexavalent species are $HC_4O_4^-$ (pH of 0 + 0 5.9) and CrO_4^{2-} (pH of >6). Trivalent Cr (Cr III) is the more stable form and less affinity for O_2, hence it is less toxic.

Remediation of chromium contaminated soils

Traditional techniques for remediating chromium contaminated water involve reduction of Cr (VI) to CR (III) by chemical(usually Fe^{2+}) or electrochemical means at pH > 5, followed by precipitation and finally filtration or sedimentation. A rapid developing area of biotechnology is “Bioremediation” the use of microbes of break down pollutants in the environment, particularly the soil.

Examples of chromate reducing microorganisms

Aerobic	Anaerobic

Agrobacterium radiobacter *Enterobacter cloacae*

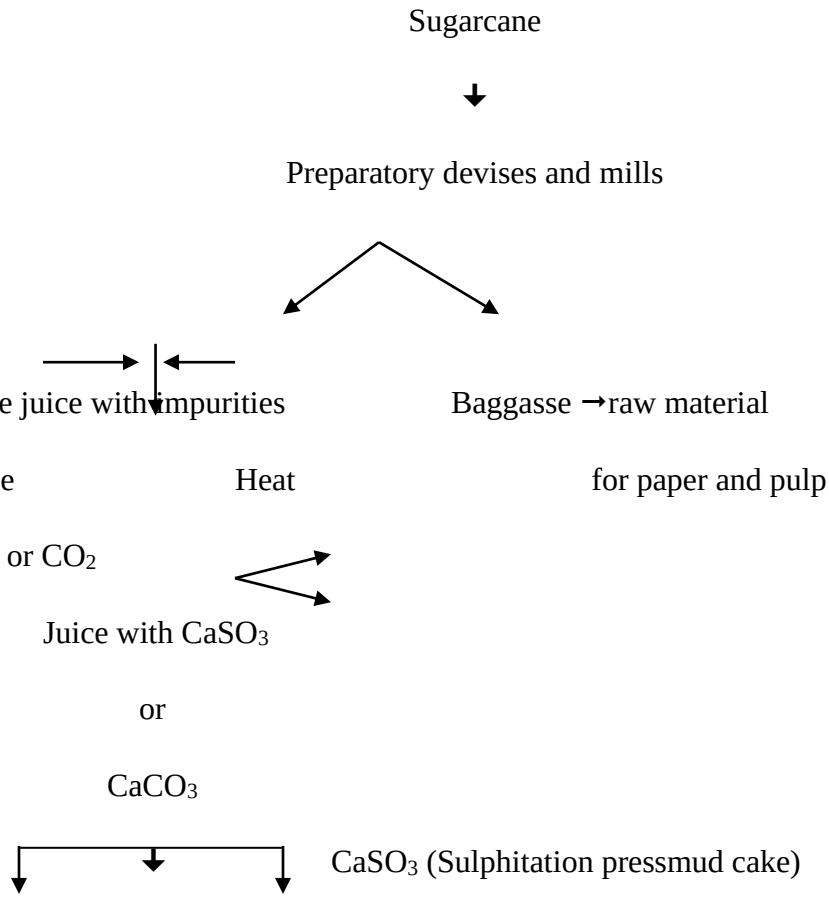
Bacillus subtilis *Escherichia coli*

Pseudomonas flourescens *Pseudomonas florescencens*

Escherichia coli *Agrobacterium radiobacter*

SUGAR AND DISTILLERY INDUSTRIES

In India, the sugar industry is the second largest agro processing industry after textiles. A typical sugar factory generates a large quantity of by products like baggasse, pressmud, molasses and spent wash from distillery. The following Fig. shows the processing of sugarcane in the sugar factory with its by products.



Precipitate

↓ $CaCO_3$ (Carbonation pressmud cake)

Filtration

↓

Clear juice

↓

Evaporation

↓

Sugar Molasses

Baggasse

It is the fibrous residue of the cane stalk left after crushing and extraction of juice. The release of baggasse as a by-product from the sugar industry is about 30% of the weight of cane crushed. From the sugar factories alone, there is a possibility of collecting about 45 million tonnes of baggasse.

Constituents

The main constituents of baggasse are cellulose (35-38%), hemicellulose (24-30%) and lignin (18-22%). Baggasse is a good absorbent of water, on weight basis it can absorb four times more water.

Recycling

- ✓ Baggasse consists mainly of cellulose, pentosans and lignin. The bagasse can be effectively utilized as a raw material for paper and pulp industries.
- ✓ It is now a days used as a primary raw material to commercially produce new sprint.

- ✓ It can also be used as fuel in high pressure boilers for cogeneration of power. Sugar factories in our country have the potential to produce over 400 MWs of power in this manner.
- ✓ Application of bagasse in soil reduces wind and water erosion and improves soil health.
- ✓ Bagasse compost can be prepared by mixing with livestock manure at the ratio of 10:1 (10 parts of bagasse, 1 part of livestock manure).
- ✓ Bagasse can also be used as a carrier for phosphate in soil application.
- ✓ Bagasse is one of the best substrates for the cultivation of oyster mushroom.

Molasses

Molasses is another important by product of the sugar industries. It is the mother liquid left after crystallization of sucrose from which further quantity of sucrose can not be removed economically. The yield of molasses per tonne is approximately 2.7 to 4.0 per cent. Sugar factories release about 7.5 million tonnes molasses every year. The molasses released from sugar industry is now being effectively utilized in distillery units. Therefore, there is no molasses waste accumulation now a days. It contains 50% total reducing sugars with 40% as fermentable sugars.

Pressmud

Sugarcane filter cake which is commonly known as pressmud is a solid residue by product obtained during clarification of sugar juice. Pressmud contains sugar, fibre coagulated colloids including cane wax, albuminoids, inorganic salts and soil particles. If the constituents of pressmud have been derived only from sugarcane, it does not contain any harmful or toxic substances deleterious to the soil or plants.

Utilization of pressmud cakes (PMC)

- ✓ A source of nutrients.
- ✓ Soil conditioner or as an ameliorant for acidic and saline sodic soils.
- ✓ A medium for raising sugarcane seedlings.

- ✓ A source of carrier material for legume inoculants.

Distillery industries

In India, alcohol (ethyl alcohol) is produced mainly by the fermentation of diluted cane molasses solution. After fermentation, the alcohol is separated by distillation and the voluminous residual liquid (1:15 = alcohol : waste liquid ratio) is discharged as distillery spent wash. Annually about 40.72 million kilo litres of spent wash is discharged from the Indian distilleries.

Characteristics of distillery effluent

Distillery effluent is acidic in nature (pH 3.5-4.0), dark coloured and contains high levels of organic carbon, N, P, K, Ca, Mg and S apart from small amounts of micronutrients viz. Zn, Cu, Fe, Mn etc. Apart from this, it has high electrical conductivity and high amounts of BOD and COD which cause pollution problems when disposed in land and water. The dark brown colour is due to the presence of meloidin of cane molasses which are not effectively decomposed by yeast, methane bacteria or activated sludge. The bad odour is due to the presence of skatole, indole and other sulphur compounds.

Treatment and disposal of distillery effluent

The distilleries are under legal obligation to treat spent wash so as to satisfy the tolerance limits prescribed by ISI. The pollution control board insists that BOD and COD are to be reduced to specified level before it is discharged either in land or irrigation purposes.

Modes of treatment

- ✓ Anaerobic digestion by open / closed
- ✓ Anaerobic digestion and methane gas generation.
- ✓ Dilution and use for irrigation
- ✓ Concentration and conversion into feed and fertilizer
- ✓ Concentration and incineration.

Of all the above methods concentration and incineration method above is able to totally eliminate the soil and water pollution. But this method is proved to be costlier. In all the above processes, the nutrient elements which are originally present will find a place when recycled back to the soil.

Scope on the use of spent wash in agriculture

Since the spent wash is highly acidic in nature with a pH range of 3.5-4.0, it can be utilized as an amendment for reclamation of sodic soil. Several investigations on this aspect has revealed that the spent wash applied at the rate of 5 lakh litres/ha reduces the pH of alkali as well as non-saline calcareous sodic soils.

Reclamation technology of sodic soil

- ✓ Plough the land
- ✓ In sloppy / uneven land form compartments of 25 cents each
- ✓ Apply distillery spent wash evenly @ 5 lakh litres /ha.
- ✓ Wait for 5-7 days.
- ✓ Impound good quality water upto 10-15 cm above the soil for 24 hrs and then drain
- ✓ The above process is repeated for 2 more times.
- ✓ The land is allowed to dry and ploughed. Allow the soil to undergo natural oxidation.
- ✓ Apply well decomposed FYM or composted pressmud @ 5 t / ha as a source of microbial inoculam.
- ✓ Take up the sowing / transplanting after 60 days of spent wash application.

Spent wash application on soil properties

- The pH, EC and organic carbon contents of the soils were increased with spent wash application.
- The available N, P, K Ca and Mg contents of the soils were significantly increased.
- The population dynamics of bacteria, actinomycetes, fungi are considerably increased.
- Spent wash application reduces the nematode population by 46-78%.

PAPER MILL

Paper industry represents an important segment and is playing important role in social, cultural and industrial growth of a country. Any paper mill effluent and its treatment plant produces solid wastes. During pulp and paper manufacturing 30-50% material as solid wastes are generated. The different types of solid wastes obtained are

1. Paper or cardboard sludge
2. Bagasse pith
3. Pressmud
4. Fly ash
5. Lime sludge

The paper mill sludge contains considerable amount of macro and micro nutrients and is one of the source of organic matter.

Composting – An eco-friendly alternative for solid waste disposal

Due to high content of organic matter in the solid wastes from paper mill, composting and land application are alternatives to disposal. Land application of solid wastes provide an effective and environmentally acceptable option of waste disposal which also recycles valuable nutrients into the soil plant system.

Composting is considered as one of the biotechnological approach of solid waste disposal known to man. The organic component of the solid waste during composting is biologically decomposed and stabilized and supply essential nutrients without adversely affecting the environment. Composting reduces the difficulties in handling due to stickiness, inactive pathogens and sterilizes the organic constituents thus producing uniform organic material suitable for soil application.

Bagasse pith composting

It contains high quality of cellulose and lignin. The activated sludge and effluent treatment plant sludge can be mixed with bagasse pith indifferent proportions in order to maintain the optimum C:N ratio and nutritional requirement of microorganism during

composting. In order to enhance the composting process, an external source of inocula viz., *Pleurotus sajorcaju* (250 kg), *Trichoderma viridie* (0.4%) can be added per 100 kg of substrate.

- Lime sludge, fly ash can be added to acid soil as ameliorative substances.

TEXTILE MILL

Textile mills are biggest consumers of water and consequently one of the largest industry causing water pollution. In textile mill the water consumption is about 2300-2720 m^3/day and the waste water let out is about 2050-2250 m^3/day. The entire quantity of waste water from industries are traditionally discharged into open land or into nearly water bodies.

Textile industry is broadly classified as cotton textile, woolen textile, synthetic textile. The effluents released from these factories contain considerably amount of organic and inorganic substances, starch, NaOH, detergents, dyes, gum dextrine, waste sulphides, sulphate soaps etc. Generally chemical measurements like pH (9.8-11.8), soluble solids (1300-2000 mg /lit), total solids (4900-7000 mg/lit), chromium (10-14 mg/lit) can be observed in effluents.

Some of the physico-chemical characteristics of textile mill effluent

pH	-	8.36
EC (dS m^{-1})	-	2.12
Ca (mg/lit)	-	48%
Mg (mg/lit)	-	4.64
Na (mg/lit)	-	32
K (mg/lit)	-	32
Chloride (mg/lit)	-	242
Sulphate (mg/lit)	-	78%
BOD (ppm)	-	1037
COD (ppm)	-	1940

The main heavy metal in textile mill effluent is chromium (5-20 mg/lit)

Recycling of textile mill effluents

Applications of effluent at different diluations (25%, 50% and 75%) with well water along with lime, FYM and their combinations can be used for irrigation of agricultural and horticultural crops.

Lecture 19: Polluted Soils- Mine spoils- Characteristics – Reclamation – Management

The soil pollution in the mine area is induced by mining wastes produced in each step of mining operations. There are various kinds of wastes such as solid and liquid wastes and air emission as dust. These wastes release metals into the environment. The Pb–Zn ores are almost produced from sedimentary rocks as well as volcanic, magnetic, or carbonate hosted deposits. Several elements such as Cd, Mn, Cu, Pb, and Zn were generated from Zn–Pb mining. The level of metal contamination could be different from site to site depending on the metal (metals) extracted from each site and may be linked to the climate and the geology of the site area.

Impact of metals in mining wastes on plants

The first direct effect of mining industry on the environment is the deforestation of the mining area. To establish a mine, the company needs to destroy trees and plants and clean the topsoil, which leads to a high destruction of landscapes, forests, and wildlife habitats at the site.The second effect is the high amount and various wastes generated from this kind of industry. The most important wastes are solid wastes, cyanide, acid mine drainage, and metals. These metals are non-biodegradable and can persist in the environment,which make of them a serious danger to plants since they are too sensitive to environmental changes, especially to metals contamination. Metals in polluted soils may induce changes in the physiological, biochemical, and metabolic activities of the plant which led to a negative effect on plant growth and inhibition of root growth. It was reported that metals can cause oxidative stress, by the production of reactive oxygen species (ROS). These produced ROS can lead to lipid peroxidation, biological macromolecule deterioration, membrane dismantling, ion leakage, and DNA-strand cleavage. Metals such as Ni, Cu, Zn Cd, Cr, and Pb may induce chlorosis, necrosis, and inhibition of root and shoot growth, disruption of photosynthesis, reduction of chlorophyll production, and disturbance of enzyme activities.

Impact of metals in mining wastes on Soil microorganisms

Soil is the natural habitat of different microorganisms such as bacteria and fungi that are playing an important role in enhancing the soil quality by their role in soil formation, organic matter production, and soil fertility. These microorganisms can be found in nature associated

with clay particles, organic matter, and rhizosphere of plants or also in small colonies in the pores among the particles. With these associations, microorganisms are highly sensitive to the ecosystem changes and can be affected by extreme pH, temperature, and chemical pollution. Besides the microbial biomass, the enzymes activities are also sensitive to metals and are significantly reduced when high levels of metals are present in soil. It was reported that metals induced a decrease in microbial biomass and lowered enzymes activities especially dehydrogenase, acid-phosphatase, and β-glucosidase activity. Several researches indicated that enzymes activities (especially dehydrogenase), microbial biomass, soil respiration, counts of N-fixing bacteria, were too sensitive to metals in polluted soils, which makes of them a great indicator of soil polluted by metals . Also, the metal pollution can cause molecular effects by inhibiting the DNA-mediated transformation in microorganisms.

Techniques of remediation of contaminated soils by metal mining wastes

In the recent decades, the high interest given to the environmental pollution and humanity safety has raised the challenges to remediate the contaminated soils and minimize their effects on the ecosystem. Generally, to achieve this goal and remediate the soils, two types of methods can be used: ex-situ and in situ methods, which have five objectives: isolation, immobilization, toxicity reduction, physical separation, and extraction of the toxicity. In these methods, we can find the engineering remediation (physicochemical methods) and the biological remediation (bioremediation). The use of any of these methods is based on several parameters such as the nature of the contaminants, the soil type and the characteristics of the contaminated site.

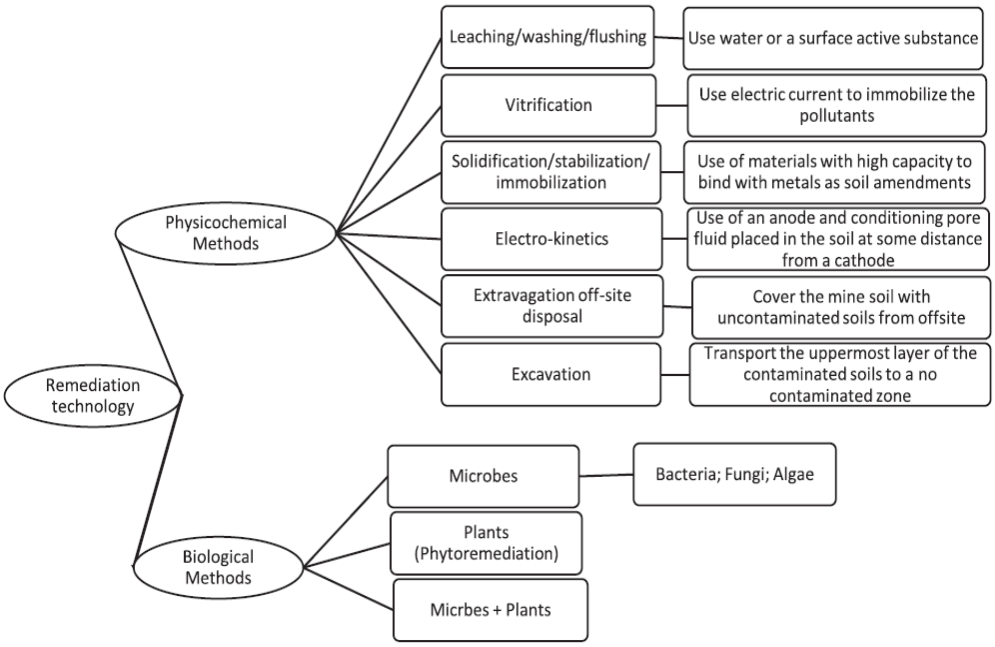


Fig. 1. Remediation technologies of metals in mine soils

Physicochemical methods

The chemical approach is based on extraction or immobilization of the pollutants from the mine soils using chemicals such as lime, phosphate, and Calcium carbonate ($CaCO_3$) While, the physical approach involves methods such as physical and mechanical techniques in order to reduce the soil toxicity by mixing contaminated and uncontaminated soils or transporting contaminated soils to hazardous waste landfills. The physicochemical approach includes several techniques such as:

• Leaching/washing/flushing soil: using water or a surface active substance

• Chemical reduction/oxidation,

• Vitrification: using electric current to immobilize the pollutants

• Solidification/stabilization/immobilization: the use of materials with high capacity to bind with metals as soil amendments.

• Electro-kinetics: based on the use DC-voltage to form the electric field gradient using an anode and a cathode on both sides of the electrolytic tank which contains the contaminated soil

• Extravagation off-site disposal: covering the mine soil with uncontaminated soils from off-site

• Excavation: transporting the uppermost layer of the contaminated soils to a no contaminated zone

These techniques are inadequate since they are too expensive in terms of time and money and they have limitations. After the use of these techniques, soil physical, chemical, and biological properties are affected and secondary pollution is produced, especially when they are used separately. Thus, gives the idea of using them in combination to reduce the cost and the contamination

Biological techniques

Due to several limitations of the physicochemical approaches, scientists in the end nineteenth century proposed the biological remediation as a new technology to clean-up the contaminated sites

Bioremediation

Bioremediation is the use of microorganisms (e.g., yeast, fungi, or bacteria) to perform the absorption, precipitation, oxidation, and reduction of the soil metals due to their capacity to degrade and detoxify some pollutants. These microorganisms must have certain characteristics such as enzymes to degrade the pollutants in question and a high resistance to relevant heavy metals present in the contaminated area reported that the bioremediation is subdivided into seven categories: biodegradation, bioimmobilization, biolixiviation, bioslurry, biorestauration, bioaugmentation, and biostimulation. The success of the pollutant degradation using these methods is linked to the good microbes in the right place and in the favorable environmental conditions

Techniques Description

Bioremediation: The use of microorganisms or microbial processes to degrade environmental contaminants

Phytoremediation: The use of specific plants (metal accumulator plants) to extract contaminants from the top soil layer

Bioventing: Method of treating contaminated soils by drawing oxygen through the soil to stimulate microbial activity.

Land forming: Solid-phase treatment system for contaminated soils: may be done in situ or ex situ.

Bio stimulation: Stimulation of indigenous microbial populations in soils or ground water by providing necessary nutrients

Phytoremediation

Phytoremediation is defined as the removal of pollutants (e.g., metals) from contaminated soil using green plants.The first mine site covered with plants in order to decontaminate the soil was in 1970s and the first plant species documented as metal accumulator plants were *Thlaspi caerulescens* and *Viola calaminaria*. Recently, several plants have been used as phyoremediator of metals in polluted sites such as *Polygonum hydropiper* and *Salix atrocinerea* for Zn, *Hirschfeldia incana* for Pb, and *Echinops spinosus* L for Zn. The planting of eucalyptus or pines was found to reduce the quantity of Cu and Cr in mine soils. Other plant species such as *Pteris vittata* L. and *Alternanthera bettzickiana* have judged as a good phytoremediotor plant for As, Cd, and Pb. The phytoremediation includes different categories such as phytoextraction, phytofiltration, phytostabilization, phytovolatization, and phytodegradation.

Technology	Technique description	Cleanup goal	Chemical treated
Phytoextraction	The ability of plants to extract and accumulate metals into their aboveground biomass	Remediation by removal of plants	Cd, Cr, Pb, Ni, Zn, radionuclides BTEX, pentachlorophenol, and short chained aliphatic compounds
Phytostabilization	The ability plants to decrease the mobility or/and bioavailability of a metal either to prevent its leaching to ground water or its entry into food chain	Remediation by Complexation of contaminant	Heavy metals in ponds, phenols and chlorinated solvents
Phytovolatalization	The conversion of metal into volatile form and its release into the atmosphere	Remediation by removal through plants	Chlorinated solvent, Hg, Se
Phytodegradation	Degradation of organic xenobiotics by plant enzymes within plant tissues	Remediation by destruction	Nitrobenzene, nitroethane, nitrotoluene, atrazine, chlorinated solvent, for example DDT, chloroform, etc.,

Fig. 2 Summary of phytoremediation techniques and phytoremediated chemicals

Lecture 20&21

Quality of Irrigation Water and Quality Criteria

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Quality irrigation water is an essential resource for better growth of plants and to get better yields. Poor quality water often results in water logging, salinity and sodicity problems in soils. The quantity and quality of water available in rivers, tanks and ground water is significantly influenced by rainfall. Generally rainwater is as pure as distilled water since it is practically devoid of any salts. Only in certain industrial areas the rainwaters during the first showers may contain some pollutants like NO_3, SO_2 etc., depending upon the nature of industries functioning in that area. In agriculture, the quality of irrigation water is the prime factor that decides the plant growth and soil fertility status than the quantity of water available for irrigation.

Irrespective of the source, whether it is surface water or ground water, it contains some amount of soluble salts. The continuous use of irrigation water of varying quality in terms of its higher salt content (saline), high electrical conductivity and Sodium Adsorption Ratio (Saline-Sodic) or high Residual Sodium Carbonate (Sodic water) affect the physical and chemical properties of soil over a long period of time. Ultimately these soils become unfit for any agricultural operation. Hence it is essential to assess the quality of irrigation water before using them. The use of various types of water for irrigation made it necessary to create a system of classification.

Criteria for evaluation of irrigation water

Water quality is determined according to the purpose for which it will be used. For irrigation water, the criteria used include salinity, sodicity and element toxicities. The other criteria involved in assessing water quality includes taste, colour, odour, turbidity, temperature, hardness, pH, BOD (Biological Oxygen Demand) or COD (Chemical Oxygen Demand), nutrient elements like N, P etc., and other pathogenic organisms.

Following are the characteristics on which irrigation water quality is judged.

1. Presence of total soluble salts in irrigation waters (EC).
2. Proportion of carbonates and bicarbonates in relation to calcium and magnesium in irrigation waters (RSC).
3. Proportion of sodium in relation to other cations in irrigation waters (SAR).
4. Puri Salt index
5. Soluble sodium percentage
6. Potential salinity
7. Permeability Index
8. Concentration of specific ions in irrigation waters like boron, chloride, magnesium, nitrate, lithium etc.

Several parameters are employed to judge the quality of irrigation waters. No parameter is complete in judging the quality of irrigation waters. Further the quality of water, should be judged by considering the crop and soil factors. Any judgement without considering the above factors may not be useful. Following are the parameters commonly adopted in judging the quality of irrigation waters.

1. Salinity hazard or total soluble salt concentration

The concentration of soluble salts in irrigation water can be classified in terms of electrical conductivity (EC) and expressed as dSm^{-1} at 25ºC. Based on electrical conductivity, the following classification of waters are made as given by USDA (United States Department of Agriculture)

EC at 25ºC ($dS\ m^{-1}$)	**Salinity Class**	**Remarks**
< 0.25	Low (C1)	Low salinity water - can be used for irrigation with most crops on most soils.
0.25 – 0.75	Medium (C2)	Medium salinity water - can be used if a moderate amount of leaching occurs. Moderate salt tolerant crops can be grown.

0.75 – 2.25	High (C3)	High salinity water - cannot be used on soils with restricted drainage. Special management for salinity control may be required. Good salt tolerant crops can be grown.
>2.25	Very High (C4)	Very high salinity water - unsuitable for irrigation under ordinary conditions. Soil must be permeable with good drainage. Salt tolerant crops can be grown. Irrigation water to be applied in excess.

Classes C1 and C2 are considered suitable (no problem) for irrigation purpose. C3 and C4 classes of water are not suitable for irrigation purpose (severe problem)

The effect of salt on crop growth is largely osmotic in nature. Higher proportion of soluble salts in soil solution increases its osmotic potential and thereby decreases water absorption, leading to physiological drought. If excessive quantities of soluble salts accumulate in root zone, the crop has difficulty in extracting water from the salty solution, thereby affecting the yields adversely. The osmotic pressure of one atmosphere is equal to 0.36 dSm^{-1} of Electrical Conductivity (EC) of irrigation water.

2. Total dissolved solids (TDS)

EC in dSm^{-1} X 640 = Total Soluble Salt Content (ppm)

Total dissolved solids (TDS) – ppm	**Salinity hazard and effects on management**
< 160	Very low hazard. No detrimental effects on plants and no soil build up expected
160 -480	Low hazard. Sensitive plants may show stress; moderate leaching prevents salt accumulation in soil
480 – 1280	Medium hazard. Salinity may adversely affect plants. Requires selection of salt tolerant plants, careful irrigation, good drainage and leaching
1280 – 1920	Medium-high hazard. Requires careful management to raise

	crops
>1920	High hazard. Generally unacceptable for irrigation, except for very salt tolerant plants when there is excellent drainage, frequent leaching and intensive management.

3. Sodicity hazard

High concentration of sodium is undesirable in water because sodium adsorbs onto the soil cation exchange sites, causing soil aggregates to break down (deflocculation) sealing the pores of the soil and making it impermeable to water flow. The tendency of sodium to increase in proportion on the cation exchange sites at the expense of other types of cations is estimated by the ratio called Sodium Adsorption Ratio (SAR).

$$SAR = \frac{Na^{+}}{\sqrt{(Ca^{2+} + Mg^{2+}) / 2}}$$

Sodium hazard	Sodicity class	SAR value	Remarks
Low	S_1	0 -10	Little or no hazard
Medium	S_2	10-18	Appreciable hazard, but can be used with proper management
High	S_3	18-26	Unsatisfactory for most of the crops
Very high	S_4	> 26	

The sodium hazard of irrigation water as expressed through SAR does not take into account of the anionic composition. **Adjusted SAR** is a value corrected to account for the removal of calcium and magnesium in their precipitation with bicarbonate and carbonate ions in the irrigation water added. It is used to predict sodicity hazard more correctly for those water

which contained appreciable amounts of HCO_3 but no RSC. Adjusted SAR should be evaluated for water which has EC higher than 1.5 and lesser than 3.0 dSm^{-1}.

$$\text{Adjusted SAR} = \frac{Na^+}{\sqrt{\frac{Ca^{2+} + Mg^{2+}}{2}}} [1+ (8.4-pHc)]$$

$$pHc = (pk_2' - pkc') + p[HCO_3^-]+p[Ca^{2+} + Mg^{2+}]$$

$(pk_2' - pkc')$ = Concentration of Ca^{2+}, Mg^{2+} and Na^+ ions in mel^{-1}

$p(HCO_3^-)$ = Concentration of carbonate and bicarbonate in mel^{-1}

$p(Ca^{2+} +Mg^{2+})$ = Concentration of Ca^{2+} and Mg^{2+} ions in mel^{-1}

4. Puri's salt Index (PSI)

This explains the relationship between sodium and calcium concentration in waters. It is used for predicting sodium hazard.

PSI = (Total Na^+ - 24.5) - (Total Ca^{2+} - Ca as $CaCO_3$) x 4.85

Where all the quantities are being expressed in ppm and all values of magnesium are reckoned as calcium.

If negative : Good quality water

Positive : Poor quality water and harmful for irrigation.

5. Alkalinity hazard

a. Residual Sodium Carbonate (RSC)

Bicarbonates occur in low salinity water and its concentration usually decreases with an increase in EC. The proportion of bicarbonate ions higher than calcium ions is considered to be undesirable, because after evaporation of irrigation water bicarbonate ions tend to precipitate calcium ions. Hence the effect of bicarbonate together with carbonate ions is to be evaluated through RSC. . The residual sodium carbonate (RSC) is used to evaluate the quality of irrigation water and is expressed in meL^{-1}.

$$RSC\ (meL^{-1}) = (CO_3^{2-}+ HCO_3^-) - (Ca^{2+} + Mg^{2+})$$

RSC value (meL^{-1})	Water quality
< 1.25	Water can be used safely
1.25-2.5	Water can be used with certain management
>2.5	Unsuitable for irrigation purposes

b. Residual Sodium Bicarbonate (RSBC)

The bicarbonate (HCO_3^-) anion is important in irrigation water. This brings about a change in the soluble sodium percentage (SSP) in the irrigation water and therefore an increase of the sodium hazard. Since carbonate ions donot occur very frequently in appreciable concentrations and because bicarbonate ions donot precipitate magnesium ions, Gupta suggested that alkalinity hazard should be determined through the index called Residual sodium Bicarbonate (RSBC).

RSBC = HCO_3^- - Ca^{2+}, all ions expressed as mel^{-1}

RSBC value (meL^{-1})	Water quality
< 3	Water can be used safely
3-9	Water can be used with certain management
>9	Unsuitable for irrigation purposes

Based on RSC / RSBC ratio there are six alkalinity classes proposed

1. Non alkaline water (-ve)
2. Normal water (0)
3. Low alkalinity water (2.5)
4. Medium alkalinity water (2.5 – 5.0)
5. High alkalinity water (5.0 – 10.0)
6. Very high alkalinity water (>10.0)

6. Soluble sodium percentage

This explains proportion of sodium in relation to total cations in water

Soluble sodium percentage (SSP) = $\frac{Na^{+} \times 100}{Ca^{2+} + Mg^{2+} + Na^{+}}$

where all the soluble cations are in meqL^{-1}.

Irrigation waters having SSP of 60 and above are harmful.

7. Potential salinity

This refers to judging quality of irrigation waters considering chloride and sulphate ion concentration in waters.

Potential salinity (PS) = ½ SO_4^{-} + Cl^{-}

If PS between 5 and 20 m.e L^{-1} – can be used only in sandy soils

If PS between 3 and 5 m.e L^{-1} – can be used in medium textured soils.

If PS between 1-3 m.e L^{-1} - can be used in fine textured soils.

8. Permeability Index

This refers to proportion of sodium and bicarbonate in relation to cations in water.

$$\text{Permeability Index (PI)} = \frac{Na + \sqrt{HCO_3}}{Ca^{+} + Mg + Na} \quad X\ 100$$

If Permeability Index is 60, it is unsuitable for irrigation.

9. Magnesium adsorption ratio

One of the important qualitative criteria in judging the irrigation water is its magnesium content in relation to total divalent cations, since magnesium adsorption by soils affect their physical properties. A harmful effect on soils appears when Ca: Mg ratio declines below 50.

$$\text{Mg adsorption ratio} = \frac{Mg^{2+}}{Ca^{2+} + Mg^{2+}}$$

Magnesium hazard on irrigation water is expected having Mg:Ca ratio more than 1.

10. Sodium ratio

$$\text{Sodium ratio} = \frac{Na^{+}}{Ca^{2+} + Mg^{2+}} \quad \text{(Concentration in meqL}^{-1}\text{)}$$

For good quality water, the ratio should not exceed one.

11. Sodium to calcium activity ratio (SCAR)

The application of SAR to the group of water which have EC greater than 5 dSm^{-1} and Mg / Ca ratio greater than 1 is obviously questionable. For the ground water having EC greater than 5 dSm^{-1} and dominance of magnesium over calcium, SAR is calculated as Na^{+} / Ca^{2+}

$$\sqrt{\quad}$$

The classification of SAR / SCAR was given by Gupta (1986) and he proposed following six classes of sodicity.

Class	Particulars	Ratio
1	Non sodic water	<5
2	Normal water	5-10
3	Low sodicity water	10-20
4	Medium sodicity water	20-30
5	High sodicity water	30-40
6	Very high sodicity water	>40

12. Specific ion toxicities

Besides the above parameters, concentration of specific ion if exceeds certain limits in water then the water may be unsuitable for irrigation. Excess of any ion (CO_3, HCO_3 Cl, SO_4, NO_3, B, Na, F) including heavy metals like nickel, lead, cadmium etc is injurious to plants. Boron is excessively available is some locations which makes the water quality poor.

a. Chloride

Since chloride has no effect on soil physical properties of a soil and is not adsorbed on the soil complex, it is not included in modern classification system. However it is used in regional water classification.

$$\text{Chloride concentration (m.e/L)} = \frac{Cl^- + NO_3^-}{CO_3^{2--} + HCO_3^- + SO_4^{2--} + Cl^- + NO_3^-}$$

Chloride concentration (m.e/L)	Water quality
<4	Excellent
4-7	Moderately Good
7-12	Slightly suitable
12-20	Not suitable for irrigation purposes
>20	Not suitable for irrigation purposes

b. Magnesium hazard

Magnesium hazard will be noticed if Mg: Ca ratio is more than 1.0

c. Sulphate

Eaton proposed three classes of sulphate

< 4 meq L^{-1} : Excellent water

4-12 meq L^{-1} : Good to injurious

> 12 meq L^{-1} : Injurious to unsatisfactory

Sulphate salts are less harmful compared to chlorides. This is because, when both the ions occur in this concentrations, only half of the sulphates get precipitated as $CaSO_4$ while the other half remains in soluble forms as Na-$MgSO_4$ in the soil.

d. Boron

Boron is essential for the normal growth of the plant, but the amount required is very small. The permissible limits of boron in irrigation water is presented below.

Boron class	Boron concentration (ppm)			Remarks
	Sensitive crops	Semi Tolerant crops	Tolerant crops	
Very low	<0.33	<0.67	<1.00	Can be used safely
Low	0.33-0.67	0.67-1.33	1.00-2.00	Can be used with management
Medium	0.67-1.00	1.33-2.00	2.00-3.00	Unsuitable for irrigation purpose
High	1.00-1.25	2.00-2.50	3.00-3.75	
Very high	>1.25	>2.50	>3.75	

e. Fluorine

Unlike chlorides, fluorides are only sparingly soluble and are present in most natural water in only small amounts. The concentration of fluoride ranges from traces to more than 10.0 ppm in ground water, although surface water donot exceed 0.3 ppm, unless they are polluted from other sources.

<1 ppm = Good

>1 ppm = Problematic

f. Nitrate concentration

Very frequently groundwater contains high amount of nitrate. When such water is used continuously for irrigation, physical properties of soil gets affected.

NO_3 ($meqL^{-1}$)

< 5 : No problem

5 – 30 : Intensity of problem is moderate

>30 : Intensity of problem is severe.

g. Lithium

Lithium is an important trace metal which may found in saline ground waters and irrigated soils. It has been found that small concentration of 0.5 to 1 ppm produce toxic effects on the growth of citrus crops. Saline soils of varying degrees found in India contain upto 2.5 ppm Li.

22. Factors affecting suitability of irrigation water and Management of poor quality water in agriculture

Factors affecting the suitability of irrigation water

The suitability of irrigation water is affected by the following factors

i) Chemical composition of the water
ii) Soil characteristics
iii) Plant characteristic and
iv) Climatic factors

The quality of water for irrigation is decided by the interaction of above four factors

Chemical composition of the water

Irrigated water containing appreciable quantities of HCO_3 tend to develop sodic conditions in soil, as carbonate salts of calcium and magnesium are of low solubility. The saline or highly saline water are largely composed of sodium and magnesium chlorides or sulphates. Chloride salts are highly soluble and salinise the soils in the proposition of their ion salt content, whereas sulphate salinise half of their concentration.

The high Mg content in irrigation water also affects soil properties. The Mg effect in irrigation water is calculated by the formula

$$\frac{Mg}{Ca + Mg} \times 100$$

A harmful effect on soil reportedly appears when the ratio of Ca + Mg exceeds 50. Pailwal et al. (1975) reportedly that the ratio of Mg: Ca in saline waters usually varies from 1-4. Pailwal and Gandhi (1976) suggested that adsorption of sodium in the soil slightly increased with the increase of salinity and more with SAR and decreased with the higher Ca: Mg ratio of the irrigation water. A lower Ca: Mg ratio reportedly enhanced soil sodicity and more so of the light textured soil.

Irrigation water with high SAR, increase the SAR, ESP and pH values of soil (Lal and Lal, 1988). This may be attributed to the increase in the proportion of sodium ion the soil solution with the application of water of higher SAR values. As a result of the rise in the SAR of the soil, the ESP increased resulting in an increase in pH and decreases in hydraulic conductivity.

Potassium usually occurs in very low concentration in irrigation waters and probably for this reason it has not been given any importance in deciding the quality of waters. But some saline ground water which contains appreciable amount of nitrate and potassium have been reported to cause beneficial fertilizing effects to plant growth and therefore, these ions must also be taken into consideration n while determining the quality of waters for irrigation.

Under identical condition of salt concentration and soil type, relative adsorption of cations would be Ca>Mg>k>Na.

Soil characteristics

Texture of the soils is one of the most important factors which affect the suitability of water for irrigation. The lighter the texture the greater the tolerance of saline water is the rule. Highly saline waters (EC 10-20 dSm^{-1}), high SAR (25-30) or high RSC (5-10 me l^{-1}) water have been successfully used on light textured soils in the arid and semi arid zones of India. In heavy textured soils, only waters with low salinity are usable (EC 3 dSm^{-1}) as even moderately saline waters (EC <3-5 dSm^{-1}) create severe problems. Saline ground waters containing salt to 8 gL^{-1} (EC 10 dSm^{-1}) could be successfully.

Presence of calcium carbonate in soil also plays a vital role in determining the suitability of saline water for irrigation. Gupta and Abichandani (1970) suggested that exchangeable sodium built up in the soil irrigated with saline water is reduced significantly after rainfall due to the presence of calcium carbonate in the soil. The rain water is highly effective in dissolving the calcium from the soil and thus replacing exhalable sodium from the exchange complex.

The nature of clay minerals also influence the quality of water used for irrigation. Montmorillonite soils allow use of irrigation water with adjusted SAR of 6 without any problem. Incase of illite, vermiculite and kaolinte mineral containing soils respective limits of adjusted SAR are 8 and 16 respectively.

Plant characteristics

Plant group response varied under the same quality of irrigation water on a similar type of soil. The suitability of water is based on

a) The tolerance of a specific crop or crops in relation to the cation concentration or specific ion concentration and
b) The specific variety of the particular crop

The tolerance should imply that a crop of its variety does not suffer significant reduction in the yield at a specific level of salinity, grown under saline water irrigation as compared to yields obtained with the best available water on the same location.

Climatic factors

Climatic factors play a major role in modifying the plant response to salinity. Plant tolerance to saline water irrigation is often greater in winter than in the summer season. Rainfall is the most significant factor responsible for the leaching of salts from the plant zone which accumulate under saline water irrigation. In fact the salts which accumulate due to saline water irrigation in the rabi season are leached down in the ensuing rainy season.

Effect of Poor Quality Waters on Soil and Crop Growth and Management

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Soil and water systems are inter related through a complex web of interactions and any change in one system will cause varying degree of changes in the other. Water quality refers to the characteristics of a water supply that influences its suitability for a specific use. Water quality can be evaluated by its aesthetic, physical, chemical and biological properties which gets influenced by means of agricultural, industrial and anthropogenic uses.

Aesthetic properties include colour, smell, odour and appearance. Physical properties that influence water quality include turbidity and clarity, dissolved solids, sediment load and suspended organic and inorganic materials. Important chemical properties in relation to water quality include pH, salinity, alkalinity, dissolved oxygen, dissolved load of chemical constituents especially nitrate, phosphorus, fluoride, arsenic, heavy metals, pesticides and other toxic compounds. Among biological properties, the presence and number of micro organisms, their biomass and the presence of pathogens, phytoplanktons and zooplanktons are used to assess the impact of water quality.

The original source of salts in irrigation water is the rock that forms a part of the earth's crust - it is constantly subject to weathering which releases salts to be carried away by water. When the soil becomes truly saline, the visible surface might be a white crust or dark, moist, oily looking patch. However, salt accumulation begins to affect crop yields long before visible signs of its presence appear.

Hazards associated with water quality

There are three principal problems that can arise from the quality of irrigation water delivered to the agricultural fields.

- **Salinity hazard**

This is directly related to the quantity of salts dissolved in the irrigation water which are left in the soil after the applied water is lost by evaporation from the soil or through transpiration by the plants. Unless the salts are leached from the root zone, they accumulate in quantities

which will partially or entirely prevent growth of most crops. When water with high salinity (>2.25 dSm^{-1}) is used continuously for irrigation, it causes **salinization of soil**. It results in **physiological drought** in plants, wherein though there is sufficient moisture in the field, plants are unable to absorb water and eventually wilt. Yield of sensitive crops are affected. Use of saline water results in **clogging of dripper systems** and causes **foliar injury** when used in sprinkler irrigation systems. Salts deposited on crops can cause leaf and fruit discoloration, reducing market value.

Most surface irrigation water, whose source is rivers, has a total salinity of less than about 0.5 to 0.6 dS/m. Groundwater in the semi-arid and arid regions has generally higher salinity ranging from one dS/m to more than 12 to 15 dS/m. Sea water is highly saline with an average total soluble salts content of about 35 g/l corresponding to an electrical conductivity of about 50 dS/m.

- **Sodicity (alkali) hazard**

Sodicity hazard develops when irrigation water contains relatively more sodium ions than divalent calcium and magnesium ions resulting in break down of the physical structure of the soil because of the deflocculation effect of sodium. When the soil dries, it becomes hard and compact leaving an impervious layer for water penetration. Fine textured soils, high in clay are most subject to this action. When the soil is high in calcium and magnesium, it may counteract the effect of sodium. Appropriate quantities of amendments, e.g. gypsum can be applied to combat the effect.

- **Toxicity hazard**
  - A high concentration of boron in the irrigation water can have a toxic effect on the growth of many plants. Sulfate salts affect sensitive crops by limiting the uptake of calcium and increasing the adsorption of sodium and potassium, resulting in a disturbance in the cationic balance within the plant.
  - The bicarbonate ions in soil solution harms the mineral nutrition of the plant through its effects on the uptake and metabolism of nutrients.
  - High concentrations of potassium may introduce a magnesium deficiency and iron chlorosis.

* Excess nitrogen in irrigation water may result in succulence and excess vegetative growth

Management practices for efficient use of high salinity water

- More frequent irrigation
- Selection of salt tolerant crops and varieties
- Use of extra water for leaching
- Conjunctive use of fresh and saline waters
- Cultural practices

Desalinization of water to remove soluble salts through membrane filters and reverse osmosis has often been referred to as a technical possibility but at the present stage of available technologies it is doubtful if this method can have any large-scale application in the utilization of saline water for irrigation of most agricultural crops, at least in the near future.

- **More frequent irrigation**

More frequent irrigations maintain higher soil water contents in the upper parts of the root zone while reducing the concentration of soluble salts. The sprinkler method of irrigation is generally more amenable for increased frequency of water applications.

- **Selection of salt tolerant crops and varieties**

Proper choice of crops can result in good returns even when using high salinity water. Similarly, selection and breeding of salt-resistant crop varieties offer tremendous possibilities of utilizing saline water resources for crop production. Induction of salt tolerance by soaking seeds for a certain period in salt solution is a method for obtaining increased yields in saline water irrigated soils, while there are suggestions that growing seeds obtained from parents that have been irrigated with saline water helps in obtaining higher crop yields.

- **Use of extra water for leaching based on leaching requirement**

To prevent excessive salt accumulation in the soil, it is necessary to remove salts periodically by application of water in excess of the consumptive use. This concept (Richards, 1954) is quantified in the term ‘leaching requirement’ often referred to by the abbreviation, LR. Leaching requirement (LR) is the fraction of total water applied that must drain below the root zone to restrict salinity to a specified level according to the level of tolerance of the crop.

$$LR = \frac{Ddw}{Diw},$$ where D is the depth of water,

and dw and iw refer respectively to the drainage and irrigation water. Assuming strict salt balance conditions in the soil-water system:

Diw x Ciw = Ddw x Cdw where C refers to the concentration of salts.

Therefore,

$$LR = \frac{Ciw}{Cdw} \text{ or } \frac{ECiw}{ECdw}$$

This would imply that the excess amount of irrigation water of a known EC that must be applied is determined by the maximum permissible EC of the drainage water specified for a particular crop. The values of EC_{dw} represent the maximum salinity tolerated by the species grown under particular conditions. For barley, where a value of ECdw = 8 dS/m can be tolerated, leaching requirement = ECiw/8. Thus for irrigation water with conductivities of 1, 2 and 4 dS/m respectively, the leaching requirement will be 12, 25 and 50 percent. In heavy textured soils and in soils having expanding type clay minerals applying 15 to 20 percent more water is often difficult during the crop season due to poor permeability and consequent aeration problems.

- **Conjunctive use of fresh and saline waters**

There are situations where good quality water is available for irrigation but not in adequate quantities to meet the evapotranspirational needs of crops. Under these conditions, mixing of high salinity water with good quality water to obtain irrigation water of medium salinity for use throughout the cropping season can be done. Alternatively, good quality water could be used for irrigation at the more critical stages of growth, e.g. germination, and the saline water at the stages where the crop has relatively more tolerance.

Cultural practices

Where foliar damage by salts in irrigation water is a concern, irrigation methods that do not wet plant leaves can be very beneficial. Furrow irrigation, low energy precision application (LEPA) irrigation, surface drip irrigation and subsurface drip irrigation (SDI) methods can be very effective in applying irrigation without leaf wetting. More advanced irrigation technologies

(such as LEPA or SDI) can offer greater achievable irrigation application efficiency and distribution uniformity.

- **Checking the salt index of fertilisers**

All fertilisers have a salt index which indicates what the fertiliser contributes to soil salinity. If the irrigation water or soils are saline, changing to fertilisers with similar nutrients but with a lower salt index may help. For example, potassium chloride has a salt index of 114 but potassium sulphate has a lower salt index of 46.

- **Critical crop growth stages**

Some crops such as barley, wheat and corn are known to be more sensitive to salinity during the early growth period than during germination and later growth periods. Sugar beet and safflower are relatively more sensitive during germination, while the tolerance of soybeans may increase or decrease during different growth periods depending on the variety.

- **Residue Management**

The common saying "Salt loves bare soils" refers to the fact that exposed soils have higher evaporation rates than those covered by residues. Residues left on the soil surface reduce evaporation. Thus, less salt accumulates and rainfall will be more effective for leaching.

- **Other techniques**

Others include application of organic manures, raising green manure crops and incorporation, optimum use of manures and fertilizers, providing better drainage facilities, adopting ridges and furrow and drip methods of irrigation, mulching with locally available plant materials

Sodicity (alkali) problem

The adverse effect of irrigation water quality on soil physical properties is associated with the accumulation of sodium ion on the soil exchange complex which imparts instability to the soil aggregates and whose disruption followed by dispersion of clay particles results in clogging of soil pores. Sodium adsorption ratio (SAR) (Richards, 1954) of the irrigation water is defined by the equation below,

$$SAR = \frac{Na^{+}}{\sqrt{\frac{(Ca + Mg)^{++}}{2}}}$$

For SAR values greater than 6 to 9, the irrigation water could be expected to cause permeability problems in soils which contain swelling type clay minerals.

Eaton (1950) suggested that Residual Sodium Carbonate (RSC) is a good index of the sodicity hazard of an irrigation water. The anions HCO_3^- and CO_3^- in the irrigation water tend to precipitate calcium and magnesium ions in the soil resulting in an increase in the proportion of the sodium ions. **Hard water makes the land soft and soft water makes the land hard**.

Magnesium hazard

High Mg / Ca ratio results in increased dispersion of clay particles and reduced hydraulic conductivity. Mg/ Ca ratio of more than 1 is not satisfactory.

Management practices for efficient use of water with sodicity hazard

- **Application of amendments**

Gypsum needed to decrease RSC by 1 mmol (+)/L works out to 850 kg per hectare metre of water. **Gypsum** can be either incorporated in the soil or lumps of gypsum can be suitably placed in the water channel **(Gypsum bed technology)** to dissolve gradually. **Sulphuric acid** has also been used to amend water quality and can be applied directly to the soil or in the irrigation water. Being corrosive, handling of sulphuric acid presents problems which must be overcome through proper application techniques.

Mixing with an alternate source of water

If an alternate source of irrigation water is available, mixing the two sources may be helpful in obtaining water which is acceptable for irrigation considering its sodicity hazard.

- **Irrigating more frequently**

Irrigating frequently with small quantities of water is an effective way to manage water with a sodicity hazard. Reduced permeability of the soils restricts water supply to the roots. Also applying large amounts at a time can result in surface stagnation which affects most crops adversely. Frequent irrigations could also reduce the precipitation of calcium by reaction with

bicarbonates in water by keeping the soils wet. Using sprinkler, the ability to supply controlled amounts of water at a time should be considered where everfeasible.

- **Growing crops with low water requirements**

Extra application of water with a sodicity hazard will further aggravate the sodicity problem. If feasible, growing crops and irrigating during periods of high evapotranspiration demands should be avoided.

- **Growing tolerant crops**

Growing crops tolerant of excess exchangeable sodium will help to obtain better returns

- **Organic matter application**

Heavy dressings of organic manures, regular incorporation of crop residues, application of organic materials as rice hulls, sawdust, sugar factory wastes, etc., have all been found useful in maintaining and improving soil physical properties and in counteracting the adverse effect of high levels of exchangeable sodium. Organic matter can contribute to a higher cation exchange capacity (CEC) and therefore lower the exchangeable sodium percentage and improve the water holding capacity of the soil.

Toxicity problems

Apart from the salinity or the sodicity hazard, the constituents of irrigation water viz., sodium, chloride and boron may cause toxicity problems when taken up by the plants in excess amounts. Fruit trees, vines and woody ornamentals are especially sensitive to sodium and chloride ions. Leaf scorch, and dead tissue along the outside edges of leaves are typical symptoms of sodium toxicity. Injury due to chloride toxicity, typically, starts at the extreme leaf tip of older leaves and progresses from the tip back as the severity increases.

A slight excess of boron in the irrigation water or in the soil solution can cause toxicity to a variety of crops. Boron is taken up by the crop and is accumulated. For example, as little as 0.6

mg elemental boron per litre in the irrigation water may produce toxicity symptoms in citrus leaves; 1 mg/l may reduce the yields of citrus and certain stone fruits and 4 mg/l is harmful to many crops.

Other constituents of some irrigation water, such as lithium, selenium, molybdenum, fluoride and chromium may have deleterious effects on plants or animals even at very low concentration; however their occurrence in irrigation water has only very occasionally been reported.

Management practices

- Field practices that can eliminate or reduce the hazard due to presence of toxic elements is irrigating the crops more frequently.
- Application of amendments, such as soluble calcium salts or sulphuric acid, can greatly reduce the toxicity hazard due to excess sodium.
- Blending of water supplies, planting less sensitive crops, improving drainage conditions through profile modification, use of fertilizers in optimum doses to obtain vigorously growing plants etc. are some of the other practices that will help overcome toxicity problems.
- When irrigation water has high nitrate concentration, nitrogen in irrigation water should be subtracted from the fertilizer schedule (N crediting). Crediting may be done for other nutrients also.
- Damage caused by high-chloride irrigation water can be minimized by planting a less sensitive crop; avoiding foliar contact by using furrow, flood, or drip irrigation; and rinsing the plants at the end of each irrigation event if a source of high-quality water is available.

Lime Deposition Potential

Lime deposition occurs when calcium or magnesium carbonates (lime) precipitate out of irrigation water, leaving white residues or deposits. Lime deposited in irrigation distribution systems causes plugging especially in micro irrigation systems. For fruit and vegetable crops, the presence of lime residue reduces marketability, because consumers often associate white residues with pesticide contamination. The presence of lime deposits in soil can reduce the solubility of some plant nutrients such as phosphorus, zinc, manganese, and iron.

Management

- Avoid contact of high lime water with leaves and fruits by irrigating under the crop canopy

- Applying overhead sprinkler irrigation when less evaporation takes place (at night or on cloudy days)
- Irrigating with more water per irrigation and decreasing frequency to reduce deposition events
- Water used for over-tree cooling (water applied as mist to cool fruit on hot days) usually must be treated with acid to remove lime and lower pH to 6.5.

Relative Tolerance Of Crops And Ornamentals To Boron

<u>Very Sensitive</u> (<0.5 mg/l)

Lemon	*Citrus limon*
Blackberry	*Rubus spp.*

<u>Sensitive</u> (0.5 - 0.75 mg/l)

Avocado	*Persea Americana*
Grapefruit	*Citrus X paradise*
Orange	*Citrus sinensis*
Apricot	*Prunus armeniaca*
Peach	*Prunus persica*
Cherry	*Prunus avium*
Plum	*Prunus domestica*
Persimmon	*Diospyros kaki*
Fig, kadota	*Ficus carica*
Grape	*Vitis vinifera*
Walnut	*Juglans regia*
Pecan	*Carya illinoiensis*
Cowpea	*Vigna unguiculata*
Onion	*Allium cepa*

<u>Sensitive</u> (0.75 - 1 mg/l)

Garlic	*Allium sativum*
Sweet potato	*Ipomoea batatas*
Wheat	*Triticum eastivum*
Barley	*Hordeum vulgare*
Sunflower	*Helianthus annuus*
Bean, mung	*Vigna radiate*
Sesame	*Sesamum indicum*
Lupine	*Lupinus hartwegii*

Strawberry	*Fragaria spp.*
Artichoke, Jerusalem	*Helianthus tuberosus*
Bean, kidney	*Phaseolus vulgaris*
Bean, lima	*Phaseolus lunatus*
Groundnut/Peanut	*Arachis hypogaea*
American elm	*Ulmus Americana*
Pear	*Pyrus communis*
Apple	*Malus sylvestris mill*

Moderately Sensitive (1-2 mg/l)

Pepper, red	*Capsicum annuum*
Pea	*Pisum saliva*
Carrot	*Daucus carota*
Radish	*Raphanus sativus*
Potato	*Solanum tuberosum*
Cucumber	*Cucumis sativus*

Moderately Tolerant (2-4 mg/l)

Lettuce	*Lactuca saliva*
Cabbage	*Brassica oleracea capitata*
Celery	*Apium graveolens*
Turnip	*Brassica rapa*
Bluegrass, Kentucky	*Poa pratensis*
Oats	*Avena sativa*
Maize	*Zea mays*
Artichoke	*Cynara scolymus*
Tobacco	*Nicotiana tabacum*
Mustard	*Brassica juncea*
Clover, sweet	*Melilotus indica*
Squash	*Cucurbita pepo*
Muskmelon	*Cucumis melo*
Ragged-robin rose	*Rosa spp.*
Olive	*Olea europaea*
Zinnia	*Zinia elegans Jacq.*
Pumpkin	*Cucurbita spp.*

Tolerant (4-6 mg/l)

Sorghum	*Sorghum bicolor*
Tomato	*Lycopersicon lycopersicum*
Alfalfa	*Medicago saliva*

Vetch, purple	*Vicia benghalensis*
Parsely	*Petroselinum crispum*
Beet, red	*Beta vulgaris*
Sugarbeet	*Beta vulgaris*
Athel	*Tamarix aphylla*
Palm	*Phoenix canariensis*
Date palm	*P. dactylifera*
Mangel	*Beta vulgaris*
Garden beet	*Beta vulgaris*
Gladiolus	*Gladiolus spp.*
Broadbean	*Vicia feba*

<u>Very Tolerant</u> (6-15 mg/l)

Cotton	*Gossypium hirsutum*
Asparagus	*Asparagus officinalis*

Lecture 23&24
Remote sensing and GIS in assessment, monitoring and management of wastelands and problem soils

Information on geographical location, aerial extent and spatial distribution of wastelands is essential for their effective management and sustainable development. Among the new technologies emerged for studying natural resources, remote and Geographical Information System (GIS) are effective technologies for detecting, assessing, mapping, and monitoring the wastelands. Space borne multispectral data, by virtue of providing synoptic views of fairly large areas at regular intervals, have been found to be very effective in providing the necessary information on salt-affected soils and waterlogged areas in a timely and cost-effective manner.

In the visual interpretation of satellite data, high soil moisture and surface waterlogged areas are identified as deep dark grey to light black in color described that areas with high water logging risk. To study the spatial dynamics of wastelands, and to evaluate the utility of high resolution satellite data for wasteland mapping, the IRS P6 satellite data(23.5x 23.5 m) and topographical maps are generally used (Scale = 1:50000). The dynamic nature of wasteland categories warrants the uses of multi-season satellite data for their accurate delineation. Hence, such images need to be geo-referenced in a common coordinate and projection system. The Planning Commission of India has recognized GIS as "an invaluable planning tool in land use and wasteland development for identifying treatment areas and models, making trade-off calculations in choosing from competing land uses, and carrying out simulations and impact assessments. The management of "wastelands" is a priority area in the context of national development. In 1991, the Ministry of Environment and Forest (MoEF) embarked upon an ambitious project to apply GIS technology for wasteland management. This was based on prior work carried in 1986 when the Department of Science (DoS) under the National Wasteland Identification Project developed detailed wasteland maps of 147 districts in the country with a 1:50,000 scale. In this project, a Task Force was identified to evolve a suitable wasteland classification system. The classification system evolved and approved by the Planning Commission comprises of the following categories (NWDB, 1987):

i. Gullied and/or ravenous land

ii. Upland with or without scrub
iii. Waterlogged and marshy land
iv. Land affected by salinity/alkalinity(coastal or inland)
v. Shifting cultivation area
vi. Sandy (desert or coastal)
vii. Mining/industrial wastelands
viii. Under-utilized/degraded notified forest land
ix. Degraded pasatures/grazing land
x. Degraded land under plantation crops
xi. Barren rocky/stony wastes/sheet-rock areas
xii. Steep-sloping areas, and
xiii. Snow covered and/or glacial areas

The application of remotely sensed data in mapping degraded lands space borne sensors started with the launch of the first Earth Resources Technology Satellite (ERTS-1 / Landsat-1). However, the satellites Landsat-TM, SPOT and Indian Remote Sensing (IRS) Satellites with better spatial and spectral resolution, enabled to map and monitor degraded lands more efficiently. GIS proved to be an effective tool in handling spatial data available at different scales, voluminous point data such as soil information, rainfall, temperature etc. and socioeconomic data and to perform integrated analysis of data on various resources of any region and to arrive at optimum solutions for various problems.

Remote sensing imagery is vital for the understanding of land cover change, and thus forms an essential element of any effort to track land degradation and desertification trends. Change detection is the process of identifying differences in the state of an object or phenomenon by observing it at different times. Change detection is an important process in monitoring and managing natural resources and urban development because it provides quantitative analysis of the spatial distribution of the population of interest. These studies may be helpful for the planners and stack holders in the process of overall sustainable development of the study area. For detection of temporal changes in the wastelands, two period data sets i.e., IRS, Linear Imaging Self-Scanning (LISS III) are used. The geodatabase generated using Arc GIS and on screen digitization techniques, shows the type, extent and spatial distribution of

different wasteland categories present in the area. The procedure involved in wasteland mapping using satellite data consists of input data, preparatory work and methodology. The methodology normally adopted for mapping at any scale consists of preparation of base map, on –line visual interpretation of satellite data, development of legend, ground truth collection, analysis of soil samples, classification of degradation classes and finalization of maps in light of field information and analytical data. Thus, geocoded FCC products on 1:50000 scale are used for visual interpretation. The image characteristics, such as colour, tone, texture, pattern, shape, size, location and association enable one to identify and delineate different types of wastelands. These delineations, however, are tentative and subject to confirmation in the field. Therefore, ground truth forms a vital input to mapping with remote sensing data. The key for interpretation is subject to changes depending upon the season, scale and resolution of the imagery. Certain categories of wastelands like salt-affected land, water-logged/ marshy land and sandy areas can be easily delineated by virtue of their spectral separability, pattern and location; whereas gullied or ravenous land, shifting cultivation etc. can be delineated with moderate success. However, undulating upland with or without scrub cannot be easily delineateddue to similar reflectance pattern with fallow land. The issue can be resolved to some extent using multidate images.

Using space-borne multi-spectral data, maps, showing extent, spatial distribution and magnitude of eroded lands, salt-affected soils, waterlogged areas, shifting cultivation, to name a few, at 1:250,000 and 1:50,000 scales have been generated. The saline soils appear in different shades of white tone with fine to coarse texture on the False Colour Composite (FCC) prints of the satellite data, owing to presence of the salts, and are recognizable under normal crop growth. For assessing these soils, the National Remote Sensing Agency has prepared maps on 1 : 250,000 scale using satellite data from Landsat Thematic Mapper (TM)/IRS sensors in association with other central and state government organizations. A digital atlas for India has been prepared. This information is being used for planning land reclamation and soil conservation programmes. Estimated area of wastelands in the country stands at 63.85 million hectares. Each maps shows village, forest compartment and micro watershed boundaries.

Image interpretation keys - Colour / Tone

White: Objects that reflect more light- Saline soil, sandy soil

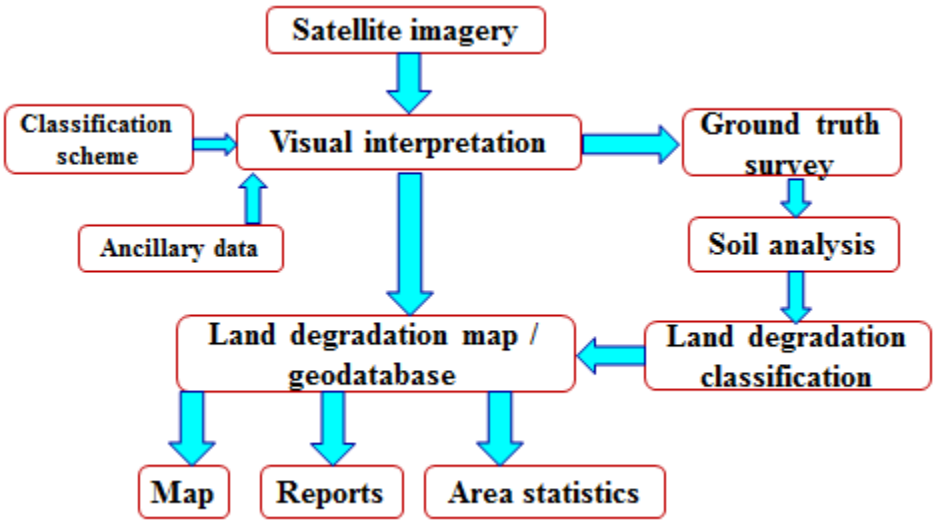
Black: Objects that reflects less light- Black soil, fire affected forest area

Green: Objects of red colour nature- Red soil

Red: Objects of green colour nature- Agricultural land, grasslands, forests

Blue: Objects of blue colour nature - Water bodies

Land degradation mapping using remote sensing and GIS approach



Land degradation assessment by remote sensing is carried out bases on the spectral reflectance of the surface soils. Degraded soils exhibits distinct spectral reflectance that are different from normal soils. This helps identification degraded soils easily from normal soil with remote sensing approaches. Eroded soils reflects more in the visible and NIR region. Saline soil exhibits bright white reflectance during summer but which is absent in monsoon season. Acid soil produces slight yellowish orange tone owing to higher iron content in soil samples. Surface ponding can be identified by the blue tone in imageries in different seasonal images continuously. Mining areas can be identified by the presence of blue tone in monsoon images (as similar to ponds) owing to stagnation of water during monsoon season. This can be confirmed by the landform and related information.

Sodic and calcareous soil do not produce distinct reflectance and hence it is difficult to delineate these degradation while visual interpretation. These soils are usually delineated with the help of previous soil maps and further ground truth verification. The factors such as

vegetation cover, management, tillage practices, etc. alters the reflectance value of the soil thus by hinders mapping degrade land precisely.

Procedures

Study area map and multi-seasonal satellite imageries of summer, monsoon and post monsoon season is essential requirement for assessing degraded lands. Degraded lands exhibits different colour/tone in satellite imageries and hence comparing multi-seasonal image is must. Multi seasonal data helps to identify the non- vegetated areas, agricultural areas and saline soils clearly as it compares the different seasonal images of the study area.

In addition to this, ancillary information such as topographical map, vegetation map, previous soil map, climatic map is also required to arrive concrete conclusion on degradation.

First prepare the study area vector map. Then multi seasonal images need to be visually interpreted on-screen by considering Survey of India toposheets, soil map, physiography map, land use / land cover map and waste land map available at specific scale (not more than 1:50,000 scale). The various degraded lands are delineated through this visual interpretation and preliminary map is generated suitably.

Field Service to be carried out towards confirmation the interpretation through ground verification. Soil samples are to be collected from the respective ground truth points information such as location, landform, current land use, cropping pattern,source of irrigation etc. is also observed. Physical degradation can be directly confirmed throw field observation

The collected soil samples were analysed in the laboratory with respected to chemical properties. Based on chemical characterization, chemical degradation is confirmed. Further the severity or degree of Degradation is able to assess from the laboratory analysis which is not possible with visual interpretation of satellite imageries alone.

The information gathered through ground truth survey and soil analysis is transferred to the initial interpretation map. From this transferred information, final degradation map is togenerated using suitable colour coding process at specific scales.

Mapping of sodicity, calcareousness, subsurface ponding and industrial effluent affected areas is not possible only with satellite data interpretation. It can be found out through more

ground truth data along with soil analysis. Assessment of land degradation in small patches is relatively difficult. But this can be overcome by using high resolution satellite data.

Geodatabase on land degradation is very useful for planning conservation and rehabilitation of natural resources in the context of food security and environmental safety.

In Remote Sensing studies, Normalized Differential Vegetation Index (NDVI) is widely used for assessing vegetation cover. Landsat TM/ETM imageries are useful for this analysis.NOAA AVHRR data is useful for study on dry land areas using NDVI. Analysis such as linear regression analysis, distributed lag models, multiple stepwise regression and dynamic factor analysis is useful in land degradation assessment Global Land Degradation Information System (GLADIS) provides information on land degradation with your spatial resolution of 8 km x 8 km

Lecture 25, 26 and 27- Multipurpose tree species for waste lands and Problem soils- Bioremediation through MPTs of Soil- MPTS- Nutrient Cycling under waste lands and problem Soils

Kinds of Problem Soils

Physical Problems	**Chemical Problems**
Fluffy Paddy Soil	Acidic Soils
Sandy Soil	Salt affected- Saline soil, Sodic soil
Sub soil Hardening	
Surface Crusting	
Water logged soil	
Peaty and Marshy soils	

Multipurpose trees (MPTs) are raised first for some periods then agricultural crops are raised in the inter space available. Extended growth of trees on alkali soils ameliorates by decreasing the pH and EC along with increasing organic matter and fertility. Trees such as *Eucalyptus tereticornis, Acacia nilotica, Albizia lebbeck, Terminalia arjuna, Prosopis juliflora* are selected on the basis of ability species tolerant to sodicity; however, agricultural crops are grown as intercrops on the basis of local need

Multipurpose tree species recommended for physical constraints

Tree species: *Eucalyptus robusta, Syzygium cumuni, Terminalia arjuna, Salix tetrasperma, Dalbergia latifolia, Eucalyptus camaldulensis, Eucalyptus grandis*

Grasses: *Brachiaria mutica, Dichanthium caricosum, Paspalum notatum, Brachiaria decumbens.*

Multipurpose tree species recommended for chemical constraints

Saline Soil

It is caused by the accumulation of soluble salts in the root zone. These excess salts reduce plant growth by altering water uptake and causing imbalance.

Mechanism for reclamation of saline soils

Halophytes are the native flora of saline soils. Few are suitable for reclamation. The basic principle of reclamation is the removal of excess salt to a desired level in root zone. Providing proper drainage, use of salt free irrigation water, acidic fertilizers, organic manures etc. are some of the mechanisms adopted. The process of salinization is accelerated by rapid evaporation from the surface. Leaching with water of good quality and adequate drainage of excess water from the soil is carried out. The selection of salt tolerant species is done which include suitable tree species such as *Prosopis juliflora, Tamarix articulate, Acacia nilotica* etc. Agricultural crops include barley, sugarbeet, cotton wheat, rice beans etc.

- The removal of excess salts to a desired level in root zone.
- Leaching and adequate drainage.
- Mulching to reduce salinity.
- Organic matter addition keeps the salts in dilute form and increases water holding capacity of soil.
- Green manuring, tree planting. Bioremediation through Agroforestry
- Promising woody species for saline soils are *Salvadora spp., Prosopis juliflora, Acacia nilotica, Parkinsonia aculeata, Butea monosperma, Terminalia arjuna, Salix spp., D. sissoo and Casurina equisetifolia.*
- Highly salt tolerant and high biomass producing grass species include *Aeluropus lagopoides, Sporobolus helvolus, Cynodon dactylon and Brachiaria ramosa.*

Sodic Soil

Alkali or Sodic soil is defined as a soil having a conductivity of the saturation extract less than 4dSm-1 and an exchangeable sodium percentage (ESP) greater than 15. The pH is usually between 8.5-10.0. Formerly these soils were called black alkali soils and the soil so formed is called solod, soloth or degraded alkali or sodic soil.

Areas: Haryana, Punjab, Uttar Pradesh, Bihar, Rajasthan & Madhya Pradesh.

Causes: Sodic soil may impact plant growth by – 1) Specific toxicity to sodium sensitive plants. 2) Nutrients deficiencies or imbalance 3) High pH.

Mechanism for reclamation of alkali soils

The reclamation practice include proper drainage of water to remove salts from the root zone use of salt free irrigation water, addition of organic matter, addition of molasses, alkali tolerant crops sush as paddy, cotton, mustard, wheat, tomato, onion etc. Green manuring of dhaincha has been found to be beneficial. Higher dose of N because of volatilization. Application of zinc in initial years of reclamation. Bioremediation through Agroforestry *Prosopis juliflora* and Karnal grass improves the soil condition to such an extent that after some time or years, less tolerant but more palatable fodder species such as Berseem (*Trifolium alexandricum*) senji (*Melilotus parviflora*) and shaftal (*Trifolium resupinatum*) can be grown under trees.

Saline and Sodic Soil

Soil that is high in both salt and sodium is considered to be saline - sodic soil. Due to high pH and the dominance of sodium (Na+) ions in alkali soil, part of the soil organic matter dissolves and comes into the soil solution. Extracts of such soils have a characteristic dark brown or black colour. The dissolved organic matter in the soil solution becomes deposited as a thin film on the soil surface. The prevalence of such black stains in such soils are called “Black alkali”.

Areas: Parts of Gujarat, Rajasthan, Punjab, Haryana, Maharashtra.

Mechanism for reclamation

Soils have the mixture of characteristics of both saline and alkali soils. Therefore soils showing high salinity should be reclaimed for both first for salinity and later for excessive exchangeable sodium. Growing of crops tolerant to high exchangeable sodium ensures reasonable returns during initial years of reclamation. Cropping practice including a green manure crop and/or legume is common. Low organic matter and high pH deters the biological activity and thus decreases the transformation of\and availability of nutrients, and causes significant volatilization of nitrogen from applied nitrogenous fertilizers. Tolerant crops such as

rice, sugarbeet and dhaincha are used. Trees species include *Prosopis juliflora*, *Acacia nilotica, Prosopis chinensis* etc.

Bioremediation through Agroforestry

Acacia auriculiformis, Azadirachta indica, Casurina equisetifoia, Dalbergia sissoo, Alianthus excelsa, Prosopis cineraria, Acacia tortilis and A. nilotica tree species are used for bioremediation of the saline and sodic soils.

Example of tree species as per tree classification (Tomar and Gupta, 1986) based on nature of tolerate/sensitive to different soils with varying pH

High tolerant (plant able to perform best in soil with pH>10.0)	**Tolerant (plant able to perform best in soil with pH 9.5-10.0)**	**Moderately tolerant (plant able to perform best in soil with pH 9.0-9.5)**	**Sensitive (plant able to perform best in soil with pH<9.0)**
Prosopis juliflora,	*Acacia catechu,*	*Acacia auriculiformis,*	*Bauhinia variegata,*
Acacia nilotica,	*Albizia lebbeck*	*Albizia procera,*	*Bombax ceiba,*
Tamarix articulata,	*Butea monosperma,*	*Azadirachta indica,*	*Cedrela toona,*
Casuarina equisetifolia	*Cordia rothii,*	*Bambusa* spp.*,*	*Ficus virens,*
	Eucalyptus tereticornis,	*Cassia siamea,*	*Hardwickia binata,*
	Pithecellobium dulce,	*Dalbergia sissoo,*	*Populus deltoids,*
	Pongamia pinnata,	*Grevillea robusta,*	*Tectona grandis,*

Tamarindus indica,	*Kigelia pinnata,*	*Thespesia populnea*
Terminalia arjuna	*Leucaena leucocephala,*	
	Melia azedarach,	
	Moringa oleifera,	
	Morus alba,	
	Sesbania sesban	

Example of tree species in different tree classification (Tomar and Gupta, 1986) on the basis of degree of tolerate to sensitive to soil with varying EC

Tolerant (EC 25-35 dS/m)	**Moderately tolerant (EC 15-25 dS/m)**	**Moderately sensitive (EC 10-15 dS/m)**	**Sensitive (EC 7-10 dS/m)**
Acacia farnesiana,	*Acacia pennatula,*	*Acacia auriculiformis,*	*Acacia deamii,*
Parkinsonia aculeata,	*Acacia tortilis,*	*Albizia caribaea,*	*Albizia guachapele,*
Pithecellobium dulce,	*Acacia nilotica,*	*Casuarina cunninghamiana,*	*Alelia herbertsmithite,*
Prosopis juliflora,	*Casuarina equisetifolia,*	*Eucalyptus tereticornis,*	*Caesalpinia eriostachya,*
Tamarix articulata,	*Casuarina obesa,*	*Leucaena shannoni,*	*Caesalpinia velutina,*
Tamarix troupii	*Casuarina glauca,*	*Pongamia pinnata,*	*Haematoxylon brasiletto,*
	Callistemon lanceolatus,	*Samanea saman,*	*Salix* spp,
	Eucalyptus camaldulensis,	*Terminalia arjuna*	*Syzygium cumini,*
	Leucaena leucocephala		*Syzygium fruticosum,*
			Tamarindus

Bioremediation through MPT's

Improvement of Soil physical properties

Litter added by tree species act as protective layer for maintaining soil physical properties such as soil moisture conservation, shielding against soil compaction and reducing soil losses from erosion and leaching.

Soil aggregates

Structural stability of the soils depends on the stability of soil aggregates and decline in soil aggregate stability leads to land degradation. Perennial plant species can improve soil aggregate stability because of addition of organic matter through litter and plant root turnover. The grass land system recorded the greater proportion of larger aggregate (> 5 mm) as compared to cultivable fields. Soils under MPT's increased 65% of (MWD) of aggregates and 21% increase in moisture as compared to other system.

Soil water holding capacity

The perennial tree species from natural forest and agro forestry system improves soil moisture content in each soil profile depth due to increasing organic matter and reduce erosion and evaporation losses

Eg: 74% of increase in soil moisture content under broad leaved tree species as compared to pine trees @ shilling.

Trees grown with agricultural crops in salt affected soils improve the physical properties of soil. For instance, tree minimizes the salt deposition in the upper layers of the soil, it prevents salt accumulation on the surface layer, it improves water permeability and it facilitate leaching of salts, it decreases the bicarbonate levels, it reduces soil pH and electrical conductivity (EC), it increases water holding capacity, as well as infiltration rate and hydraulic conductivity with soil fertility. Similarly, enhancing cation exchange capacity (CEC) reducing ESP and improvement

in desodification process all along the profile depth can be taken place through tree planning in salt affected soils

Improvement of Soil physical properties

Soil organic matter and soil organic carbon

38% increase in soil organic matter and soil organic carbon was recorded in poplar+ sugarcane crop as compared to poplar+ fodder wheat.

Soil Macro and Micro nutrients

The soil organic carbon, available phosphorus and available potash were increased. Among the tree species tested, *A. nilotica*, *Albizia procera*, *L. leucocephala*, *Azadirachta indica*, and Eucalyptus hybrid found most suitable tree species for rehabilitation of alkali soils.

Soil pH , EC and CEC

Tree species such as *T. articulata* followed by *P. juliflora* and *A. nilotica* reduced the soil pH and ESP. Organic carbon in the surface layer also increased under *T. articulata* by 0.23%

28&29. Interpretive grouping of soils- Land Capability classification

Land capability classification is the grouping of land units into defined classes based on their capability. Evaluation of land for land use planning is a consequent step following soil survey and mapping process. The capability of land units depends on their limitations and it is designed to emphasize on the hazards in different kinds of soils. It serves as a guide to assess the suitability of different land units for arable crops, grazing / forestry. This system however, maynot be suitable for intensively cultivated plain areas, such as Indo Gangetic plain (in India)

Depending on the nutritional and other properties soils behave differently. Some are suitable for cultivated crops, others for pastures and still others may be suitable only for forestry (or) recreation.

The grouping of soil units into capability classes and sub classes is done on the basis of their capability to produce crops and pasture biomass without adversely affecting the productivity over a long period of time. The criteria used for assessing a land unit are the physical land properties and the degree of limitation with which crop growth is inhibited. It is mainly based on

1. Inherent soil properties
2. External land features
3. Environmental factors that limit land use (climate vegetative)

The first two information can be derived from a standard soil survey report and the third information is collected from other agencies (or) sources.

The factors determining the capability of a soil

- Depth of soils stoniness, rockiness
- Texture and structure of soil
- Permeability (Movement of air and water through the soil)
- Relief (as expressed by slope)
- Extent of erosion
- Susceptibility to over flow, flooding and degree of wetness.
- Presence of toxic salts, alkali and other unfavourable chemical properties such as pH, gypsum salts, aluminium etc.
- Climatic variation (in respect of soil temper5ature and moisture regimes)

The capability classification does not suggest the most profitable use of land. Further, these grouping are subject to changes as new information about the soils and their response become available.

The capability classification consists of three categories namely

1. Capability classes
2. Capability sub-classes and
3. Capability units

Capability classes

Eight capability classes are recognized. The soils having greatest capability for response to management and least limitat6ions are grouped in class – I and those having least. Capability and greatest limitations are grouped in class-VIII. Each capability class is correlated with the cropping system as conditioned by LGP.

Land capability classifications (Eight classes)

Land suitable for cultivation				**Land not suitable for cultivation**			
Land capability class				**Land capability class**			
Class I Very good land (No limitation)	Class II Good land (Minor limitation)	Class III Moderately good (major limitation)	Class IV Fairly good (major limitation with occasional cultivation)	Suitable for pasture and grazing			Class VIII suitable for wild life and recreation
				Class V (No limi-tation)	Class VI (Minor limitation)	Class VII (Minor Limitation)	

Brief description / salient features of LCC

Name of the class	**Colour shown on land capability**	**Salient features**	**Crop suitability crops**	**Management practices**
Class – I	Green	Excellent cultivable deep, nearly level. Productive	Suitable for most crops like wheat,	Need no management

		land with no limitation (or very slight hazard)	barley, cotton, maize tomato and beat	practices for cultivation
Class – II	Yellow	Good cultivable land on almost level plain or on gentle slopes limitation of soil depth, salinity, texture, drainage / erosion that reduces the choice of plants	Suitable for wheat, barley, cotton Moderately suitable for maize, alfalfa, tomato Slightly suitable for beans	Cultivate with precaution need simple management practices.
Class III	Brown	Moderately good cultivable land on almost level plain or on moderate slope. Limitation :L Moderate erosion soil depth, soil salinity and soil texture. They may have vertic characteristics (or) drainage problem that reduces the choice of crops	Unsuitable for growing vegetable crops. Varying suitability for different crops	Cultivation with careful management practices need intensive care.
Class IV	Pink	Fairly good land on almost level plains or moderately steep slopes **Limitations** Strong / very strong soil salinity S3/S4 shallow soil depth, strong erosion, fine soil texture. Poor (or) excessive drainage.	Limited cultivation unsuitable for growing a variety of crops. Suitable for selected crops 8 for pasture. Not economical to cultivate	Need intensive soil conservation and management practices
Class V	Dark grey	**Limitations** Story, rocky nature (or) marshy-ness use of implements is difficult	Not suitable for arable farming suitable for grazing	
Class VI	Orange	Moderate limitation such as steep slope severe erosion, limiting soil depth, strongly gypsiferous, stony or sandy (sand – done	Non-arable land well suited for grazing (or) forestry	

		areas) eg. dense forest lands of the Himalays. Sandy (gypsiferous) and donal areas of Thar Desert in Rajasthan and SW Irag.		
Class VII	Red	Severe limitation such as steeply sloping land subjected to erosion (or) very shallow stony soils that have not	Fairly well suited for grazing or forestry but not cultivable	Need careful management for grazing and forestry
Class VIII	Purple	Severe limitation Very steeply sloping, highly eroded, rocky barren mountain landscape, hyper arid rocky undulating surface	Non-arable, suitable for wildlife (or) recreation. Extremely rough, rocky, arid, wet (or) extremely saline land	

Capability subclasses

The capability subclasses are based on kinds of dominant such as wetness or excess water (w), Climate (C), soil (S) and erosion (e). The subclasses are mapped by adding limitation symbols to the capability class number subscripts for example II e, IIIw etc. Therefore the subclasses indicate both the degree and kind of limitations. The capability subclasses provide information as to the kind of conservation problems or limitations involved. There are no sub-classes in capability class-I land, since there is no limitation in this class.

Capability units:

A capability unit includes soils which are sufficiently uniform in their characteristics, potential and limitations and require fairly uniform conservation treatment and management practices.

It the existing limitations can be permanently removed (or) reduced in extent by proper conservation measures (or) management practices, such as provision of irrigation, drainage, control of gullies and construction of flood – retarding irrigation structures, then the land capability class can be changed towards a better class. A further deterioration in the existing conditions will shift the capability to a worse class.

Land Suitability Classification for Field Crops, Horticultural Crops and forest Trees

Each plant species require definite soil and site conditions for its optimum growth. Although some plants may be found to grow under different soils and extreme agro-ecological conditions, all the plants cannot grow in the same soil and under the same environment. The

success or failure of a plant species to grow in a particular soil and site is determined largely by the climate, availability of water and plant nutrients. Water and nutrient availability are largely controlled by the soil physical, chemical and physico-chemical properties. Based on the soil properties and climatic data, the suitability of the land to support different crops can be worked out following the FAO guide lines (1976).

A. Suitability criteria

Most of the plant species need well drained, moderately fine to medium textured soils, free of salinity and having optimum physical environment. Soil resource maps help in predicting the behavior and suitability of soils for growing field crops and forest or plantation crops, once the suitability criteria is established. The suitability of a land to support a crop is classified into different categories consisting of Order, Class, Subclass and Unit.

Order

Land Suitability Orders denote the kind of suitability. There are two land suitability orders namely 'S' and 'N'. 'S' denotes that the land is suitable for crop cultivation and 'N' denotes unsuitable.

Class

The classes denote the degree of limitations. There are 3 classes under the order S (S_1, S_2 and S_3) and 2 classes under the order N (N_1 and N_2) reflecting the degree of suitability within the order.

S_1 - Highly suitable with no limitations (Green)

S_2 - Moderately suitable with moderate limitations (Yellow)

S_3 - Marginally suitable with severe limitations (Brown)

N_1 - Currently not suitable (Red)

N_2 - Permanently not suitable (Violet)

Sub class

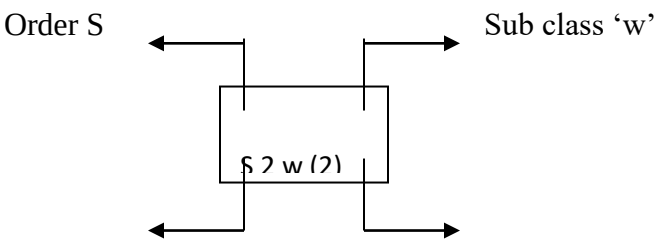
The sub-classes reflect the kind of limitations or the improvement measures required within a class. These are indicated by symbols using lower case letters following the arabic numerals used for classes. The following are the sub-classes:

1) Climate (c) (included different weather parameters)
2) Topography and landscape (t)
3) Wetness conditions of soil (w)
    - Drainage
    - Flooding
4) Physical conditions of soil (s)
    - Texture
    - Gravel / stoniness (Surface soil, subsurface soils)
    - Depth
    - Lime / Calcium carbonate
    - Gypsum
5) Soil fertility (f) (not easily correctable)
    - Organic matter
    - Cation exchange capacity
    - Base saturation
    - Nutrient availability
6) Salinity and alkalinity (n)
    - Salinity
    - Groundwater quality
    - Alkalinity / Sodicity

Unit

The land suitability unit is similar in management requirements. This suggests the relative importance of land improvement works. It is indicated by Arabic numerals enclosed in paranthesis, following the sub class symbol.

Example



(Suitable) (Wetness limitation)

Class 2 Unit 2

(Moderately suitable)

Capability index

Sys and Verhese (1975) proposed the following Capability Index (Ci) based on nine parameters for crop production in the arid and semi-arid regions.

$$Ci = ABCDEFGHI$$

Where

A - Soil texture

B - Calcium carbonate

C - Gypsum

D - Salinity

E - Sodium saturation

F - Drainage

G - Soil depth

H - Epipedon

I - Weathering stage / profile development

Evaluation based on degree of limitation

No limitation (0)	**:**	Characteristics are optimal for plant growth (Ci$\geq$ 80)
Slight limitation (1)	**:**	Characteristics are nearly optimal for the land utilization type and adversely affect productivity by not more than 20 per cent from the

		optimal yield (Ci : 60 to 80)
Moderate limitation (2)	:	Characteristics that have moderate influence on crop yield decline, but benefits can still be made and the land use remains economical (Ci : 45 to 60)
Severe limitation (3)	:	Characteristics that have such an influence on crop yield decline that use becomes uneconomical for the considered land utilization type (Ci : 30 o 45)
Very Severe limitation (4)	:	Characteristics that decreases the yield below profitable level and inhibits the use of soil for the considered land utilization (Ci : <30)

Soil unit evaluation

Phase 1. Collection of necessary characteristics or qualities which influence the productive capacity of considered land use type.

Phase 2. Determination of soil and climatic requirements of the land utilization types

Phase 3. Evaluation of the land utilization types

Each required parameter for a particular land utilization is compared with the actual character and the degree of limitation is tabled. The overall number and degree of limitation is determined and expressed in terms of orders and suborders.

Example : In the degree of limitation table, if the limitations are as under,

Situation 1: All the limitations are rated '0's and/or '1's : S_1

Situation 2: All are'1's one is '2' (correctable) : S_{1-2}

Situation 3: >four '1's and/or not more than three '2's (correctable) : S_2

Situation 4: > three '2's and/or one'3' (correctable) : S_3

Situation 5 : > one '3' correctable in future : N_1

Situation 6: one or > one '4' : N_2

The most limiting factor will be the sub class which suggests that improvement is required in that limiting factor

Steps for soil-unit evaluation

1. Prepare a table of site characteristics
2. Use the soil-site requirements of different crops/forest species and determine the suitability of the soil-site for that crop factor by factor
3. Prepare a table with the kind and degree of limitation ratings.
4. Determine the overall suitability
5. Prepare the suitability class for all the crops cultivated in the location and also for alternate crops suggested
6. From the overall suitability and from among the cultivated crops, select a crop which is having the highest suitability class or the lowest limitation as the best suited crop for the site. Also suggest the best suited alternate crops based on the suitability rating.
7. Compare the present and potential yields of the currently cultivated crops.
8. Compare the economics of the current crop and the prescribed alternate crop.

Lecture 30& 31

Problematic soils under different Agro ecosystems- coastal salinity, inland salinity, Marshy, swampy soils, red sand dunes (Theri soils) and Tsunami affected soils.

Salt-affected soil' refers to a soil having too high the amount of salt dissolved in the soil solution that consequently affects the growth and productivity of plants. Generally, if the electric conductivity (EC) of a soil solution extracted from a water-saturated soil is higher than 2 dS/m the soil is considered saline.The general characteristics of salt-affected soils are similar to non-problem soils but contain more easily soluble salts than normal. The result from an EC measurement will tell us whether the soil is a salt-affected soil. However, we can observe the condition of the area and the types of vegetation growing there. The area with salt-affected soils would show a thin film of white salt especially in the dry season. However, as the distribution of salt is not uniform, each area may possess different salinity levels. In the saltiest area the condition may not allow any crop to grow on or there may be some salt-tolerant plants only. For the areas with low salinity there may be some plants, but their growth is not so well. Therefore, for such a plot of land one may see many empty patches or some salt crusts appearing on the surface in some spots.

Occurrence of salt affected soils in india

Four major regions

- Semi arid indo-gangetic alluvial region(punjab, haryana, u.p, part of bihar)
- Arid tracts of rajasthan and gujarat
- Arid and semi arid of southern states
- The coastal alluvium

The salt-affected soils can be classified according to their origin into two types:

1. Coastal saline soils
2. Inland saline soils

Coastal saline soils.

They are mostly found along the coast in the areas that the tide can still reach or previously inundated by the sea that had caused salt accumulation in the soil.

In coastal areas, salinisation can be associated with the over exploitation of groundwater caused by the demands of growing urbanization, industry and agriculture. Over extraction of

groundwater can lower the normal water table and lead to the intrusion of marine water. Natural disasters in coastal areas, such as tsunamis, can cause severe salinisation problems. Originating from geological weathering, the salts are transported and accumulated in the profile as result of water movement. There are several mechanisms by which water trapped in sedimentary rocks can be altered into saline water. In many areas, saline and fresh subsurface waters exist in close proximity. When fresh groundwater is pumped from aquifers that are in hydraulicconnection with seawater, the change in gradients as a result of pumping may result in a flow of salt water from the sea towards the well.

The soils of coastal areas offer many unique problems like salinisation, water logging, clay pan formation and sea water inundation. The salt affected soils occur in the extensive areas all along the east coast from Chingleput, South Arcot, Tanjora, Ramnad to Tirunelveli. It has been estimated that about one lakh hectares in the coastal belt pose salinity problems. The low lying areas along the east coast of Thanjaur and Ramanathapuram district are saline or possible sodic. These localised areas are at or near sea level and may be influenced to some degree by sea water intrusion, but probably due to lack of internal drainage, salts have been accumulating over a long period of time from the irrigation water. About 8000 hectares adjoining kazhuvely swamp in South Arcot are affected by salinity. Similarly, the areas near Chindambaram in South Arcot district and Paramakudi in Ramanathapuram district are also affected.

Inland saline soils

The inland saline soils or salt-affected soils on the land. They are caused by the decomposition of sedimentary rocks interspersed with salt bearing rock, or from the salt brought up by groundwater that is brackish or salty, or caused by water that leached salt from other place and accumulated on the soil surface..

Soil salinity has caused heavy loss of national wealth in India. Out of 329 millionhectares of land in the country, about 175 million ha. (53 %) is suffering from degradation insome form or the other. There are 7.61 M ha of salt affected soils in India as per the Ministry ofAgriculture, GOI. The extent of this problem area as given by different sources varys from 8.56M ha to 10.9 M ha. Water logging affects another 8.52 M ha mainly in the irrigation commands,which includes some of the saline-alkali soils also. In Haryana, parts of Punjab, Rajasthan, UttarPradesh, Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Andhra Pradesh and Tamil

Nadu,substantial areas of good irrigated lands are affected by saline - alkali and water-loggingproblems

The problem of saline soils

Too much soluble salt in the soil can harm the plant growth because the plants can suffer the lack of water (dehydration) and from receiving excessive amounts of elements that are constituents of the salt that are accumulated in the soil, particularly sodium and chloride, making crop failure, crop yield reduction and low quality produces.

Utilization guidelines

The management of coastal saline soil may be carried out in two ways:

- Management to suit the existing natural conditions, such as preservation or rehabilitation of mangrove forest. For the area that has been transformed to be salt farm or shrimp farm, a preventive measure must be exercised so the salt water will not spread to other agricultural areas.
- Modification of natural conditions, such as building a dam to block the sea water so the area will become permanent farmland or raising the beds to grow crops and using water to wash the salt out and apply soil improving materials such as rice husk, compost and manure. If there is an acid soil layer beneath, the digging of thebed has to be very careful, not to dig the small canal down to the layer of acid soil.If the some part of the acid soil layer is dug up to fill the beds, the acidity of thesoil must be corrected with lime or marl.

(2) Management of inland saline soil can be done by using locally available technologies,e.g. washing the salt out of the soil with water.

Other methods include:

a. Addition of organic matter to the soil by plowing in the green manure and organic fertilizers
b. Addition of soil amendments such as rice husk to improve the structure of the soil to be more friable and allow more water to seep and leach the salts down to a deeper layer as well as to increase the soil fertility

c. Covering the soil with waste materials, such as rice straw, to preserve soil moisture all the time
d. Transplanting older rice seedlings than usual or planting a higher number of seedlings than normal
e. Select salt-tolerant crops such as acacias and eucalyptus. At the same time onemust be careful when performing some activities that may cause the spreading ofsalt to other vicinities such as large salt mining, deforestation or construction ofreservoirs in salt-accumulated areas.

Peaty and Marshy soils

These types of soils are found in Kerala, coastal track of Orissa, Sunderban area of W.B. When the vegetation growing in such wet places dies, it decomposes very slowly dues to excessive wetness of soils and after several hundreds of year a layer of partly decayed organic matter accumulates on the surface, giving rise to such peaty and marshy soils. These are black coloured, heavy and highly acidic soils. When properly drained and fertilized, these soils produce good crops of rice.

Peaty soils are found in the areas of heavy rainfall and high humidity, where there is a good growth of vegetation. Thus, large quantity of dead organic matter accumulates in these areas, and this gives a rich humus and organic content to the soil. Organic matter in these soils may go even up to 40-50 per cent. These soils are normally heavy and black in colour. At many places, they are alkaline also. Soils belonging to this group are found in Kottayam and Alappuzha districts of Kerala where it is called kari. Marshy soils with a high proportion of vegetable matter also occur in the coastal areas of Orissa and Tamil Nadu, Sunderbans of West Bengal, in Bihar and Almora district of Uttaranchal. They are the result of water-logging anaerobic conditions of the soils, and the presence of iron and varying amount of organic matter. The peaty soils are black, heavy and highly acidic. They are deficient in potash and phosphate. Most of the peaty soils are under water during the rainy season but as soon the rains cease, they are put under paddy cultivation.

Swampy soil

Swamp, wetland ecosystem characterized by mineral soils with poor drainage and by plant life dominated by trees. The latter characteristic distinguishes a swamp from a marsh, in

which plant life consists largely of grasses. Swamps are found throughout the world. They exist in areas with poor drainage and sufficient water supply to keep the ground waterlogged, and they have a high enough supply of minerals in the water to stimulate decay of organisms and prevent the accumulation of organic materials. They are often found in regions of low relief associated with rivers that supply the water.

A swamp is an area of land permanently saturated, or filled, with water. Many swamps are even covered by water. There are two main types of swamps: freshwater swamps and saltwater swamps.

Swamps are dominated by trees. They are often named for the type of trees that grow in them, such as cypress swamps or hardwood swamps. Freshwater swamps are commonly found inland, while saltwater swamps are usually found along coastal areas. Swamps are transition areas. They are neither totally land nor totally water.

Swamps exist in many kinds of climates and on every continent except Antarctica. They vary in size from isolated prairie potholes to huge coastal salt marshes. Some swamps are flooded woodlands. Some are former lakes or ponds overtaken by trees and shrubs.

1. **Freshwater Swamps**

Freshwater swamps form around lakes and streams. Rain and seasonal flooding cause water levels to fluctuate. In the wet soil, water-tolerant vegetation grows and helps maintain a moist, swampy condition.

2. **Saltwater Swamps**

Saltwater swamps form on tropical coastlines. Formation of these swamps begins with bare flats of mud and sand that are thinly covered by seawater during high tides. Plants that are able to tolerate tidal flooding, such as mangrove trees, begin to grow and soon form thickets of roots and branches. Mangrove trees often grow on tall, thin roots. The roots anchor sand and other sediments. The growth and decay of the roots increase the accumulation of soil.

Their soil

Red sandy dunal soils of Tamil Nadu is called Theri soils. Theri-soil occupy about 20,000 hectares in Tuticorin, Tirunelveli and Kanyakumari districts. Tuticorin district has the highest area 11,200 ha.Theri (mettu) lands are deep land zones. About 20,000 hectares of Theri-lands are left unused in the aforesaid three districts. These are considered to be unsuitable for continuous irrigation.

Theries are soils formed in aisomegathermic and ustic regimes from geogenic sand deposits under a climate with mean annual temperture of 28 to 32 C and 700 mm of annual rainfall. The soil colour is 2.5 YR with a variation from 5 YR to 10 R. The soils are extremely deep except due to lithological discontinuity. Texture ranges from single grain to sandy loam. Structure is fine weak sub angular blocky to very coarse strong angular blocky. Soils are moderately acidic with low organic carbon content. The soils are dominated by heavy minerals like limonite and zircon and a little rutile, monazite and garnet and quartz in light mineral fraction.

Tsunami affected soils

The coastal area of Tamil Nadu was stuck by tsunami in the morning of 26 December 2004. This caused huge loss of human and animal lives, damage to houses and agricultural lands. Based on the situation studied in the fields, various observations made at the sites and analysis of the soil and water samples collected from the affected areas, it was observed that soils of the area as well as water have turned saline due to inundation of sea water. The electrical conductivity of saturation extract of the surface soil varied from 3.1 to 62.0 dS m^{-1}, whereas that of surface crust showed very high values (110–117 dS m^{-1}), which indicated accumulation of salt on the surface. The chemical composition of soil saturation extract showed higher concentration of sodium, followed by calcium and magnesium as the major cations, whereas chlorides and sulphates were the major anions followed by bicarbonates, indicating that the soils turned saline, but not sodic. However, exchangeable sodium percentage (ESP) of soils varied from 10.2 to 27.4 for surface soil. These soils are low in available N and P (except few), and medium to high in available K which needs regular monitoring of fertility status and soil test-based fertilizer application.

Observations made at the field and analysis of collectedsoil samples after the tsunami showed that there is deposition of salts on the surface in all the areas where sea water intrusion

took place during the tsunami. However, in the coastal soils of TN, the tsunami caused three kinds of major damages:

Deposition of slushy greyish-brown clay material,

Deposition of sand and sea water intrusion,which receded leaving salts in the soil.

Deposition of slushy greyish-brown clay was exclusivelynoticed at Vallapalam and Vedharanayam areas in the southern parts of Nagapattinam district, where the thicknessof deposits varied from 5 to 30 cm under wet conditionand after drying it was a 2–5 cm thick layer of a slushybrown sea sediment (SD-1) over rice fields,with high salt concentration (17.6 dS m–1) compared with thesoil beneath. Cracks developed after drying,which may result in the formation of a physical barrierfor aeration and germination of seeds. Chemical compositionof these sea sediments showed that they were rich inorganic carbon (0.95%), consisting of higher amount ofclay resulting in increased water-holding capacity and CEC(26.78 cmol kg–1), with an ESP of 55.1. It was also observedthat this material is rich in N, P, K and CaCO3. The deposit can be mixed with soil by ploughing.It will improve soil texture, water-holding capacity andsoil fertility, especially in case of surface soil.The incorporation of deposited material will not act as aphysical barrier.

Considering the deposition of sand, the source of the sandy material is mainly coastal sand dunes/adjoining field which came with sea water and were deposited in the field. This kind of damage was noticed in the mid and northern villages of Nagappatinamdistrict and southern region of Cuddalore district. The thickness of the deposit variedfrom 2 to 5 cm, having salinity in the range of 3.1– 62.0 dS m^{-1}. This sand need not be scraped and removed, but should be ploughed back in the field itself. The accumulated salts will leach down during the rainy season due to the sandy nature of the soil.

Sea water had entered and receded leaving salt, this kind of damage was noticed in few areas of Nagappatinam and Cuddalore districts. The sea water entered into the fieldsdirectly as well as through backwater canals. The water that entered receded within a few hours (quickly) in some areas and five to seven days (slowly) in some other parts. Analysis showed that the soils have become saline, but not sodic.

Therefore reclamation of the soil is only possiblethrough leaching of salt by water. Thus, there is no need to have a blanket application of gypsum (the Government of TN had supplied

gypsum to farmers as relief measure for soil reclamation). However, its application should be location-specific, i.e. where pH is > 8.5, ESP > 15 and where the pH increases during leaching. It can be applied as a source of calcium and sulphur for oilseed crops like sesame and groundnut.In such cases, its dose should be 100 to 200 kg ha^{-1}

The soils are coarse-textured and mono-cropped with low CEC and organic carbon, farmers are advised to go in for green manuring, in addition to application of compost and FYM (10–12 t ha^{-1}), etc. These soils are low in available N and P (except few), and medium to high in available K, which needs regular monitoring offertility status and soil test-based fertilizer application.

In most of the areas, salinity of surface soil can be reducedthrough leaching with natural rainwater for at least one year and land can be initially used for cultivation of salt-tolerant rice varieties and other crops, viz. rice: CSR- 13, TRY 1, TRY(R) 2, Co 43, Co 47, ADT 43 and ADT 36; ragi: TRY 1; sunflower: CO4; groundnut: VRI 2; green gram: K 851; brinjal: PLR 1 and AU 1; clusterbean: PusaNavbahar; jack: Palur 1; pomegranate: Ganesh; sapota: PKM 1; cashew: VRI 2 and 3; amla: BSR 1, and tamarind: PKM 1.

Since in saline soils, young seedlings are sensitive tosalinity, only aged seedlings (one week more than normal age) may be planted with higher number (4–6) of seedlings per hill.

In case of fertilizer application, 25% extra N as basal dose should be added along with zinc sulphate@ 25–35 kg ha^{-1}. Phosphorus should be applied on thebasis of soil test or 15 kg ha^{-1} as maintainer dose. In case of delayed supply of canal water or rainwater, sowing of sprouted seeds should be resorted to. Major emphasis should be given to the management of the seafront of the land so as to protect the lands for agricultural production and inhabitation.

A natural shelter belt and eco-friendly approach can be made by the creation of a bio-wall growing mangrove trees, coconut (*Cocosnucifera*), casuarinas (*Casurinaequisetifolia*), pungam(*Pongamiapinnata*), punnai (*Calophylluminophyllum*), palmyrah (*Borassusflabelliformis*), poovarasu (*Thespesiapopulnea*), etc. along the coastal line. This will dissipate the energy of the sea waves and thus diminish their damaging force. It will act as sanctuary for fishes and other sea flora, which can be economically exploited. It will also prevent soil erosion and act as a sink for atmospheric carbon dioxide.

32. Agricultural Ecosystem Services – Soil Fertility improvement in Problem Soils

Ecosystem services are the many and varied benefits that humans freely gain from the natural environment and from properly-functioning ecosystems. Such ecosystems includes, agroecosystems, forest ecosystems, grassland ecosystems and aquatic ecosystems. These ecosystems functioning properly provides such things like agricultural produce, timber, and aquatic organisms such as fishes and crabs. Collectively, these benefits are becoming known as 'ecosystem services', and are often integral to the provisioning of clean drinking water, the decomposition of wastes, and the natural pollination of crops and other plants.

Supporting services

These include services such as nutrient recycling, primary production, soil formation, habitat provision and pollination. These services make it possible for the ecosystems to continue providing services such as food supply, flood regulation, and water purification.

Provisioning services

- food (including seafood and game), crops, wild foods, and spices
- raw materials (including lumber, skins, fuel wood, organic matter, fodder, and fertilizer)
- genetic resources (including crop improvement genes, and health care)
- water
- biogenic minerals
- medicinal resources (including pharmaceuticals, chemical models, and test and assay organisms)
- energy (hydropower, biomass fuels)
- ornamental resources (including fashion, handicraft, jewelry, pets, worship, decoration and souvenirs like furs, feathers, ivory, orchids, butterflies, aquarium fish, shells, etc.)

Regulating services

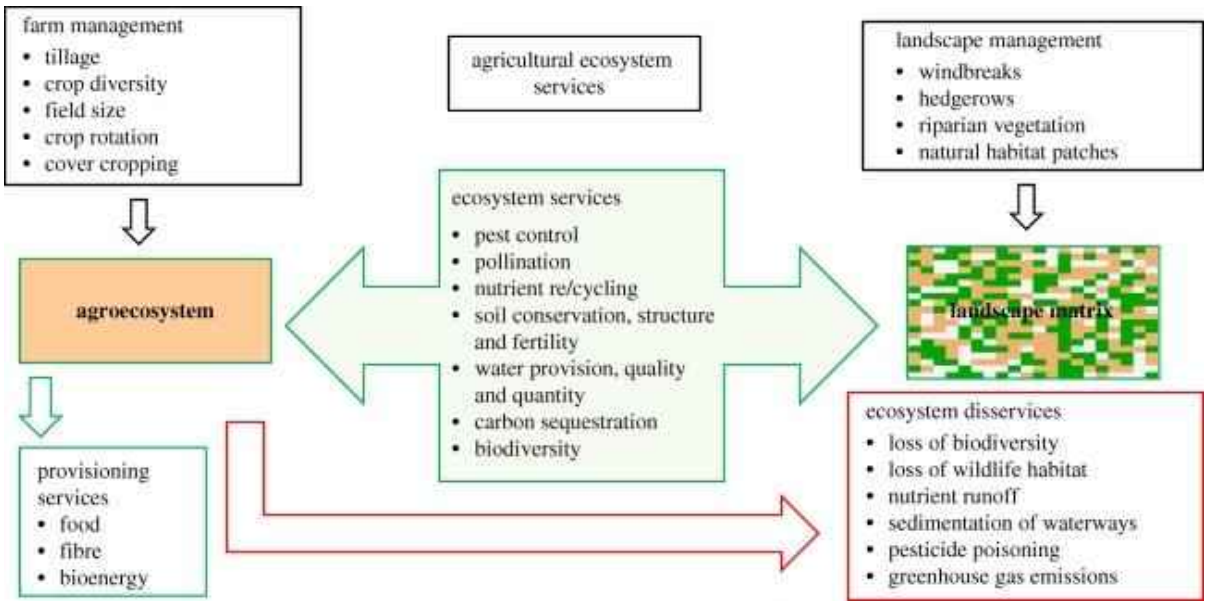
- Carbon sequestration and climate regulation
- Predation regulates prey populations
- Waste decomposition and detoxification
- Purification of water and air
- Pest and disease control

Cultural services

- Cultural (including use of nature as motif in books, film, painting, folklore, national symbols,architect, advertising, etc.)
- Spiritual and historical (including use of nature for religious or heritage value or natural)
- Recreational experiences (including ecotourism, outdoor sports, and recreation)
- Science and education (including use of natural systems for school excursions, and scientific discovery)
- Therapeutic (including Ecotherapy, social forestry and animal assisted therapy)

. There also is a fundamental *critique* of the concept of cultural ecosystem services that builds on three arguments.

1. Pivotal cultural values attaching to the natural/cultivated environment rely on an area's unique character that cannot be addressed by methods that use universal scientific parameters to determine ecological structures and functions.
2. If a natural/cultivated environment has symbolic meanings and cultural values the object of these values are not ecosystems but shaped phenomena like mountains, lakes, forests, and, mainly, symbolic landscapes.
3. Those cultural values do result not from properties produced by ecosystems but are the product of a specific way of seeing within the given cultural framework of symbolic experience.



Soil Fertility improvement by Agro ecosystem services

Soil structure and fertility:

Soil structure and fertility provide essential ecosystem services to agroecosystems. Well-aerated soils with abundant organic matter are fundamental to nutrient acquisition by crops, as well as water retention. Soil pore structure, soil aggregation and decomposition of organic matter are influenced by the activities of bacteria, fungi and macrofauna, such as earthworms, termites and other invertebrates. Micro-organisms mediate nutrient availability through decomposition of detritus and plant residues and through nitrogen fixation. Agricultural management practices that degrade soil structure and soil microbial communities include mechanical ploughing, disking, cultivating and harvesting, but management practices can also protect the soil and reduce erosion and runoff.

Conservation tillage and other soil conservation measures can maintain soil fertility by minimizing the loss of nutrients and keeping them available to crops. Cover crops facilitate on-farm retention of soil and nutrients between crop cycles, while hedgerows and riparian vegetation reduce erosion and runoff among fields. Incorporation of crop residues can maintain soil organic matter, which assists in water retention and nutrient provision to crops. Together these practices conserve a suite of ecosystem services to agriculture from the soil.

Nutrient cycling:

Nutrient cycling is the movement of nutrients through an ecosystem by biotic and abiotic processes.The ocean is a vast storage pool for these nutrients, such as carbon, nitrogen and phosphorus. The nutrients are absorbed by the basic organisms of the marine food web and are thus transferred from one organism to the other and from one ecosystem to the other. Nutrients are recycled through the life cycle of organisms as they die and decompose, releasing the nutrients into the neighbouring environment. "The service of nutrient cycling eventually impacts all other ecosystem services as all living things require a constant supply of nutrients to survive.

Primary production:

Primary production refers to the production of organic matter, i.e., chemically bound energy, through processes such as photosynthesis and chemosynthesis. The organic matter produced by primary producers forms the basis of all food webs. Further, it generates oxygen (O_2), a molecule necessary to sustain animals and humans.

Soil Fertility

Soil fertility is one of the most fundamental ES considering agricultural benefits, and has indirect value. Soil fertility is composed of many underlying ecosystem processes. Important indicators for soil fertility are SOM content and the ratio of carbon (C) (N). The natural supply of SOM predominantly results from the roots and stubble residues.

Temporary agricultural grasslands are known to increase soil fertility over time and result in aneconomical preceding crop effect (greater yield from successive crops). Even short-term (<3 years)grasslands in a crop rotation can provide benefits to soil quality .The empirical data from **legume–grass mixtures** optimized for carbon sequestration on fertileshow quantities of fixed carbon during the first year of 10.8 t C/ha during the mulch process, 3.6 t C/ha with cutting. After SOM formation, this corresponds to 2.4 t C/ha and 1 t C/ha,respectively. These results consider the roots at the time of harvesting, but only account for 30–50%of roots during vegetation, so the fixed carbon should be even higher. Standard values for soil of silage maize cropping estimate losses from 0.56 t C/ha to 1.04 t C/ha, compared with a supply from legume–grass mixture cropping between 0.6 t C/ha and 0.8 t C/ha. A veryhigh proportion of grasses in the mixture exhibits a high carbon capture rate, but results in a poortrade-off with the nitrogen

balance, whereas a mixture with a high proportion of legumes in the seed(80%) exhibits high carbon capture rates and fixes more nitrogen, up to 540 kg N/ha per year. N accumulation from the grass component of the mixture is also important for the soil fertilityincrease, and appears to be influenced more by the grass N demand than the legume N supply.The effects of perennial legume cultivation on the subsequent crops are apparent until the third andfourth year. Forage legumes have a high preceding crop effect due to the large amounts and residues left after harvesting.

The effects of cropping wildflower mixturesare favorable in two ways. First,the suspension of soil tillage for up to five years enables undisturbed development of soil biota. Second,the nutrient availability benefits from the increased supply of carbon after intensive perennial rootdevelopment and, in the case of legume-based mixtures, from the supply of nitrogen. However,there was no increase in SOM found after growing wildflower mixtures.

Soil Protection:

For this review of agricultural ES, soil protection was determined by the vulnerability of a soil tocompaction, contamination, and erosion. As the soil compaction sensitivity isdeterminedby the soil moisture content, conditions for rapid drying are beneficial. Erosion can primarily becontrolled by a protecting vegetation cover, which protects the soil from silting that disturbs infiltrationcapability and reduces the sediment transport capacity of the runoff. Further, the soil is stabilized atthe aggregate level by a high proportion of organic material in the soil particles, and at the field levelby intensive rooting.

Legume–grass mixtures provide increased resistance against soil erosion and soil compactionthrough a permanent ground cover, improved aggregate structure, high porosity and higher waterdemand from the culture. The intensive root development works against existing soil compaction; for example, the deep rooting alfalfa can break up compacted soil layers. The outreach of the roots and root mass of legume–grass mixtures increases with the proportion of legumes, while the overall root length increases with the proportion of grasses. The reduced need for fertilizers and pesticides also reduces the exposure to contaminants.

Lecture 33

Potential of agro forestry systems in management of problem soils

Agroforestry systems have the potential to make use of marginal and degraded lands through the soil improving effects of trees. Underlying all aspects of the role of agro forestry in maintenance of soil fertility is the fundamental proposition that trees improve soils. It would be useful to have guidelines on which properties of a tree or shrub species make it desirable for the point of view of soil fertility. This would help in identifying naturally occurring species and selecting trees for systems which have soil improvement as a specific objective.

Agroforestry is a sustainable land management system which increases the overall yield of the land, combines the production of crops (including tree crops) and forest plants and/or animals simultaneously or sequentially, on the same unit of land and applies management practices that are compatible with the cultural practices of the local population.

Nitrogen fixation and a high biomass production have been widely recognized as desirable. However, many properties are specific to particular objectives of systems in which the trees are used. Even species that are shunned for their competitive effects may have a role in certain designs. An example is the way in which Eucalyptus species with a high water uptake, which adversely affects yields in adjacent crops, have been employed to lower the water table and so reduce salinization

The agroforestry systems comprises Agrisilivicultural systems (improved fallow species in shifting agriculture (jhum), hedgerow intercropping (alley cropping), multispecies tree gardens, multipurpose trees/shrubs on farmlands, plantation and other crops, shade trees for commercial plantation crops, soil conservation hedges etc.), Agrisilvipastoral systems (tree-livestock-crop mix ground, home garden, woody hedgerows for browse, green manure, soil conservation etc., integrated production of crops, animals ,fuelwood, poles, etc.), Silvipostoral systems (multipurpose fodder trees on or around farmlands, living fences of fodder hedges and shrubs, trees and shrubs on pastures, integrated production of animals and wood products, etc.), aquaculture with trees, multipurpose tree lots, etc

How do we know that trees improve soils

1. The soil that develops under natural forest and woodland is fertile. It is well structured, has a good water-holding capacity and has a store of nutrients bound up in the organic matter. Farmers know they will get a good crop by planting on cleared natural forest.
2. The cycles of carbon and nutrients under natural forest ecosystems are relatively closed, with much recycling and low inputs and outputs.
3. The practice of shifting cultivation demonstrated the power of trees to restore fertility lost during cropping.
4. Experience of reclamation forestry has demonstrated the power of trees to build up fertility on degraded land.

The properties which are likely to make a woody perennial suitable for soil fertility maintenance or improvement are:

- A high rate of production of leafy biomass.
- A dense network of fine roots, with a capacity for abundant mycorrhizal association.
- The existence of deep roots.
- A high rate of nitrogen fixation.
- A high and balanced nutrient content in the foliage; litter of high quality (high in nitrogen, low in lignin and polyphenols).
- An appreciable nutrient content in the root system
- Either rapid litter decay, where nutrient release is desired, or a moderate rate of litter decay, where maintenance of a soil cover is required.
- Absence of toxic substances in the litter or root residues
- For soil reclamation, a capacity to grow on poor soils.
- Absence of severe competitive effects with crops, particularly for water.
- Low invasiveness.
- Productive functions, or service functions other than soil improvement.

Not all of these properties are compatible: for example, litter of high quality is not likely to have a moderate rate of decay. The last property, the existence of productive functions, is not directly concerned with soils but is of the highest importance if the tree is to be effective in fertility maintenance. A species needs to be acceptable and desirable in agroforestry systems from other points of view, especially production.

A tree might have all the desirable properties above, but, if it is not planted and cared for, it will not be effective in improving soil fertility. The capacity of trees to maintain or improve soils is shown by the high fertility status and closed nutrient cycling under natural forest, the restoration of fertility under forest fallow in shifting cultivation, and the experience of reclamation forestry and agroforestry.

Soil transects frequently show higher organic matter and better soil physical properties under trees. Some species, most notably *Faidherbiaalbida*, regularly give higher crop yields beneath the tree canopy. Trees improve soil fertility by processes which:

- Increase additions to the soil
- Reduce losses from the soil
- Improve soil physical, chemical and biological conditions

The most important sets of processes are those by which trees:

- Check runoff and soil erosion
- Maintain soil organic matter and physical properties
- Increase nutrient inputs, through nitrogen fixation and uptake from deep soil horizons
- Promote more closed nutrient cycling

Role of agroforestry in soil quality/heath

Compared to natural, a managed agricultural ecosystem has greater amounts of nutrient flowing in and out, less capacity for nutrient storage, and less nutrient recycling. The capacity of trees to maintain or improve soils is shown by the high fertility status and closed nutrient cycling under natural forest, the restoration of fertility under forest fallow in shifting cultivation, and the experience of reclamation forestry and agroforestry (Young, 2003). The processes by which trees maintain or improve soil fertility are given below:

1. Photosynthetic fixation of carbon and its transfer to the soil via litter and root decay,

2. Nitrogen fixation by all leguminous trees and in few non-leguminous species (e.g., Alder and Casuarinas),

3. Improved nutrient retrieval by tree roots, including through mycorrhiza and from lower horizon,

4. Providing favourable conditions for the input of nutrients from rainfall and dust

5. Control of erosion by combination of cover and barrier effect, especially the former,

6. Root uptake of nutrients that would otherwise have been lost by leaching,

7. Soils under trees have favourable structure and water holding capacity, through organic matter maintenance and root action,

8. Provision of a range of qualities of plant litter, woody, and herbaceous,

9. Growth promoting substances,

10. The potential through management of pruning and relative synchronizatoion of timing of release to nutrients from litter with demand for their uptake by crops, and

11. Effects of tree shading on microclimate

In agroforestry system nutrient addition takes place through leaf litter, pruning of woody compounds and atmospheric fixation. Some nutrients otherwise considered unavailable to crop because they are below the rooting zone of the annual crop, might be brought into the system from deeper layers in the soil with the the help of tree roots. Trees able to return nutrients through dead organic matter (leaf, branch, twig, fruits and flower) and thus helps in enrichment of top soil layer, available for the agriculture crops. Thus most important beneficial effect of the trees on the soil can include improvement of soil structure availability of nutrients. The objective, designing agroforestry system is to modify cycling in such a way as to make more efficient use of the nutrients whether these originate from natural removal process or from fertilizer. Trees in agroforestry system promote more closed nutrient cycling than pure agriculture systems.

Nutrient Recovery: Greater amount of nutrients that added to the soil, takes place through litterfall, trees translocate nutrients from deeper soil and deposit them on the soil surface via leaf sheading and other organic residues. The decomposition of organic matter residuces and

its mineralization results release of nutrients to the soil. However, the amount of nutrients released in agroforestry systems will be much smaller than tree monoculture plantations

Biological Nitrogen Fixing: Agroforestry trees, particularly leguminous trees, enrich soil through biological nitrogen fixation, addition of organic matter and recycling of nutrients. Some trees such as Leucaena species, Acacia species and Alnus species has been reported to fix as much as 400-500 kg, 270 kg and 100-300 kg nitrogen per hectare per year respectively. The fixed nitrogen may benefit symbiotically to the crops growing in its association and helps in soil fertility improvement.

The amount of nitrogen added from the legumes or pruning of trees species taken up by the first crop is reported quite low and large portion is left in the soil organic matter indicating a long term nitrogen benefit than immediate. Different tree components viz., leaf, twigs, fruit and wood have different decomposition rates which helps to distribute the release of nutrient over time. Some important nitrogen fixing plant species are given in Table 1. Biological nitrogen fixation takes place through symbioticand non-symbiotic means. Symbiotic fixation occurs through the association of plant roots with nitrogen-fixing microorganisms. Many legumes form an association with the bacteria Rhizobium while the symbionts of a few nonleguminous species belong to a genusofactinomycetes, Frankia. Non-symbiotic fixation is effected by free-living soil organisms, and can be a significant factor in natural ecosystems, which have relatively modest nitrogen requirements from outside systems (Nair, 1993)

Important N2 fixing plant species

Plant	**Botanical Name**	**Family**	**Nitrogen fixed (kg N/ha/yr)**
Black wattle	*Acacia mearnsii*	Mimosoideae	200
Beef wood, Saru	*Casurinaequisetifolia*	Casuarinaceae	60-110
Erythrina	*Erythrinapoeppigiana*	Pipil[onaceae	60
Apple ring, Areca	*Gliricidiasepium*	Fabaceae	13
Inga	*Inga jincicuil*	Mimosoideae	34-50
Subabul	*Leucaenaleucocephala*	Mimosoideae	100-500
Indian alder	*Alnusnepalensis*	Betulaceae	-
Horse bean	*Viciafaba*	Fabaceae	68-88

Nutrient Pumping: Tree root systems are involved in some favorable effects on soils such as carbon enrichment in soil through root turnover, the interception of leached nutrients, or the physical improvement of compact soil layers. Trees have deep and spreading roots and hence are capable of taking up nutrients and water from deeper soil layers usually where herbaceous crop roots cannot reach. This process of taking up nutrients from deeper soil profile and eventually depositing on the surface layers through litter-fall and other mechanisms is referred to as 'nutrient pumping' by trees. This process is mainly depends on characteristics of tree species and other soil, climatic and topographic factors.

Erosion Control: Loss of soil productivity and shortfall of food and cash crops are the immediate impacts of land degradation. India has 175 million ha degraded land. It is suffered from various problems of soil erosion and land degradation. These are major causes of land degradation and soil nutrient losses.

Amount of nutrient stored in above ground vegetation under different systems

Sl. No.	Species	DM (Ton/ha/yr)	Nutrient release (Kg/ha/yr)		
			N	P	K
1	*Delbergia sissoo*	4.2	67	3	16
2	*Eucalyptus globules*	8.5	58	5	48
3	*Eucalyptus hybrid*	4.6	22	9	7
4	*Populus deltoids*	4.5	25	16	26

DM= Dry matter

Soil:- Tree species selected for agroforestry taking into account soil type

Desert soil: *Prosopis cineraria, P. chilensis, Acacia tortilis, A. senegal, A. nilotica, Salvadoraspp*
Recent alluvium: *Acacia catechu, Dalbergiasissoo, Bombaxceibaetc*

Saline-alkali soils: *Prosopisspp, Acacia nilotica, Azadirachtaindica, Ailanthus spp, Eucalyptus spp, Tamarixspp,Pongamiapinnata*

Coastal and deltaic alluvium: *Casuarinaequisetifolia, Cocusnucifera, Areca catechu, Avicenniaspp*

Red soils: *Tectonagrandis, Madhucaindica, Mangiferaindica, Dalbergiasissoo, Acacia nilotica, Leucaenaleucocephala, Azadirachtaindica, Eucalyptus hybrid, Pterocarpusmarsupium, Adina cardifolia, Dendrocalamusstrictus*

Black cotton soils: *Acacia nilotica, A leucophloea, Tectonagrandis,Hardwickiabinnata, Adina cardifolia, Tamarandiusindica, Aeglemarmelos, Bauhinia spp, Dalbergialatifolia*

Laterite and lateric soils: *Tectonagrndis, Eucalyptus spp, Acacia auriculiformis, Azadirachtaindica, Tamarindusindica,Emblicaofficinalis*

Peaty and organic soil: *Syzygiumcuminii, Ficusglomerata, Bischofiajavanica, Lagerstromiaspeciosa, Glircidiasepium*

Hill soils: *Juglansregia, Alnusnitida, Toonaserrata, Cedrusdeodra, Quercusspp, Bombaxceiba etc.*

Lecture 34

Carbon sequestration and its role in problem soil management

Carbon sequestration

Carbon sequestration refers to the provision of long term storage of carbon in the terrestrial biosphere, underground/ oceans as long lived pools, so that the build-up of carbon dioxide (the principal greenhouse gas) concentration in the atmosphere will reduce

Carbon Sequestration

It is widely believed that reducing the atmospheric concentration of CO_2 could reduce future climate-related damage and would be more beneficial than costly. Options for achieving this goal include reducing future emissions as well as sequestering CO_2 that has already accumulated. Carbon sequestration involves "removing C from the atmosphere and depositing it in a reservoir" (UNFCCC 2006).

Carbon can be sequestered in two major ways: biological and geological. Geological sequestration, which has a fairly large technological potential, has not yet been demonstrated on a scale anticipated to mitigate CO_2 emissions. It is also more costly. Thus current discussions are focused on biological methods for sequestering C.

Atmospheric concentrations of carbon dioxide can be lowered either by reducing emissions or by taking carbon dioxide out of the atmosphere and storing in terrestrial, oceanic, or freshwater aquatic ecosystems. **A sink is defined as a process or an activity that removes greenhouse gas from the atmosphere**. The long-term conversion of grassland and forestland to cropland (and grazing lands) has resulted in historic losses of soil carbon worldwide but there is a major potential for increasing soil carbon through restoration of degraded soils and widespread adoption of soil conservation practices.

FAO is concerned with the **effect of agriculture on climate change**, the impact of climate change on agriculture and with the role that agriculture can play in mitigating climate change. Historically, land-use conversion and soil cultivation have been an important source of

greenhouse gases (GHGs) to the atmosphere. It is estimated that they are still **responsible for about one-third of GHG emissions**.

However, **improved agricultural practices** can help mitigate climate change by reducing emissions from agriculture and other sources and by storing carbon in plant biomass and soils. The work of FAO aims to **identify, develop and promote cultural practices that reduce agricultural emissions and sequester carbon** while helping to **improve the livelihoods of farmers**, especially in developing countries, through increased production and additional incomes from carbon credits under the mechanisms that have emerged since the Kyoto Protocol.

The **objective** is to **reverse land degradation** due to deforestation and inadequate land use/management in the tropics and sub-tropics through the promotion of improved land use systems and land management practices which provide win-win effects in terms of **economic gains** and **environmental benefits**, a greater agro-biodiversity, and improved conservation and environmental management and increased carbon sequestration.

The development of agriculture during the past centuries and particularly in last decades has entailed **depletion of substantive soil carbon stocks**. Agricultural soils are among the planet's largest reservoirs of carbon and hold potential for expanded carbon sequestration (CS), and thus provide a prospective way of mitigating the increasing atmospheric concentration of CO_2. It is **estimated that soils can sequester around 20 Pg C in 25 years, more than 10 % of the anthropogenic emissions**.

At the same time, this process provides other important benefits for soil, crop and environment quality, prevention of erosion and desertification and for the enhancement of bio-diversity. **Land degradation, does not only reduce crop yields but often reduces the carbon content of agro-ecosystems, and may reduce biodiversity**. It is therefore important to identify what important synergies can be found in the area of soil carbon sequestration between the three **UN conventions:** UNFCC, UNCCD and UNCBD.

Carbon sequestration activities have been supported through the CDM (Clean Development Mechanism) under the Kyoto protocol with a focus on afforestation and reforestation, seen as being the most effective and readily measurable means to sequester carbon

as biomass both above and below ground. In the post-Kyoto negotiations efforts are being made to give due attention to the huge carbon sequestration potentials in rangelands

Carbon stabilization: Prerequisite for Carbon(C) Sequestration

The stabilization of C is a prerequisite for C sequestration in soil for a long-term basis. The stabilization of C in soil can be categorized as: physical stability, chemical stability, biochemical stability and thermal stability. The physical stability of C mainly takes place inside the soil aggregates. The C stabilizes inside the aggregates mainly due to the physical protection that resists the microorganisms to cause decomposition. The C which is stabilized inside the aggregates is by nature and at various stages of decomposition products of plant origin. The macro aggregates are more potent for stabilization of C though this form of protection of C is transient in nature and largely affected by type of tillage operation. On the other hand, micro aggregate protected C are generally not affected by tillage operation and more of permanent in nature. The chemical stability of C is largely achieved by formation of clay-humus complex formation. In this respect, sesquioxides (oxide and hydroxides of Fe and Al) play a major role in the formation of stable clay-humus complex. The carbonaceous materials, i.e. polysaccharides, hemicellulose, uronides, levans, and numerous other polymers excreted as mucilaginous products by microorganisms are attached to clay surfaces by means of cation bridges, H bonding, van der waal's forces and anion adsorption mechanisms. The biochemical stability of C is directly related to its biochemical composition and ease of degradation by microorganisms. The lignin, polyphenol and quinone are inherently resistant to decomposition by microorganisms due to their complex aromatic ring structure. The thermal stability of soil C is related to temperature driven biochemical degradation and the rate of decomposition of soil organic carbon becomes double with each 10oC rise in soil temperature. Due to this reason, in tropical countries like India the cycling of C is very rapid and it takes enough time to increase in C content in soil even under continuous application of organic sources.

Soil management practices for Enhancing carbon sequestration

1. Conservation agriculture:

Conservation agriculture (CA) is an effective strategy of controlling erosion. Based on a long-term study in India, minimum tillage along with application ofcrop residue as mulch conserved rain water by 40% and 11% and soil by 69% and 28% over cultivated fallow

andconventional tillage, respectively, under maize cultivationat 4% slope. Similarly no-till with flat bed planting formaize showed the highest SOC status of top soil (0–15cm) Minimum tillage is generallyrecommended for soils in the Indian Himalayan regionbecause of reduced cultivation costs, better retention ofsoil water and physical protection of soil organic C (SOC)in aggregates. Carbon stock was found to be more under thenon-puddled/no-tilled treatments over conventional tillage in rice–wheat system. Leaving the crop residueson the field is another practice which could have animportant impact on the global C cycle.Retention of crop residues ofsorghum [*Sorghum bicolour*(L.) Moench.], and applicationof farmyard manure (FYM) equivalent to 25 kg N/haalong with 25 kg N/ha supplied through chemical fertilizersincreased and maintained the SOC stock in drylandVertisols of Solapur, Maharashtra.

2. Afforestation

Converting non-forest land to forest typically increases the chance of diversity of flora and fauna, except in situations where biologically diverse non-forest ecosystems are replaced by forests that consist of single or a few species (i.e. plantations of monocultures and especially exotic species). Where afforestation or reforestation is generally followed to restore degraded lands, it is also likely to have other environmental benefits, such as reducing erosion, controlling salinization, and protecting watersheds. In dry countries, expansion of forested areas can also be viewed as a desertification-reduction activity. Afforestation and reforestation potentially could achieve annual carbon sequestration rates in aboveground and below-ground biomass of 0.4–1.2 Mgc/ha/year in boreal regions, 1.5–4.5 Mgc/ha/year in temperate regions, and 4.0–8.0 Mgc/ha/ year in tropical regions. Forest ecosystems potentially stored more than 80% of all terrestrial aboveground C and more than 70% of all soil organic C.

Alternate land use

It has been observed that alternate land-use systems,viz. agro-forestry, agro-horticulture and agro-silvi culture,are more remunerative for SOC restoration than the solecropping system.Researchers reported that organic C content was about doublein agro-horticultural and agro-forestry systems comparedto the sole cropping scenario. Higher SOC and MBC were recorded in natural forest system, followedby cultivated and pasture ecosystem. Floor litter and humusC also varied significantly with land use.Total carbon (TC) content in floor litter ranged from 1.50to 5.20 Mg/ha. *Cardamom*-based agroforestry system hadsignificantly higher floor litter C concentration than otherstands.

3. Plantations and amendments

Reclamation of salt-affected and acid soils is an importantaspect of soil-quality enhancement and increasing Csequestration in above-ground biomass and in the SOCpool. *Prosopisjulifloral*is most adapted to alkaline soils and producedmost biomass hence contribute well to soil C sequestration.Among the other tree species, *Vachellianilotica,Casuarinaequisetifolia, Tamarixarticulata, Diplachnefusca*also caused a remarkable increase in SOC contents.Adaptable fruit trees for north-western regions in Indiainclude jamun (*Syzygiumcumini*), tamarind (*Tamarindusindica*), ber (*Ziziphusmauritiana*Lam.), and guava(*Psidiumguajava*L.). In north-central India, significantimprovements in SOC content of a sodic soil was observedthrough planting salt-tolerant tree species over aperiod of 8 years.

Application of manure and gypsum is also important inimproving soil structure and reclaiming sodic soils. Applications of farm byproductsand organic manures improved the quality ofsodicVertisols, enhanced SOC contents and yield of crop. Researchers observed the increase in SOC contentsignificantly owing to growing Karnal grass (*Diplachnefusca*) and of application of gypsum within three years.

Acid soils developed in high-rainfall area have verylow internal buffering capacity and low surface chargecharacteristics. These intrinsic soil attributes have a profoundimpact on agricultural productivity. If acid soils aremanaged properly, it could be productive and enrich thesoil with C. Scientists reported that growing ofGamba grass (*Andropogongayanus*Kunth) and stylograss (*Stylosanthesguianensis*) over a study period ofthree years showed a 6-fold increase in soil C at>30 cmdepth from the initial values, indicating a significant rate
of C sequestration.

Surface mining for coal drastically alters soil properties,destabilizes SOC and depletes SOC pools. Typical grasslandreclamation practices at Singer site included sowingrye grass (*Loliumperenne*L.), timothy (*Phleumpratense*L.), birds foot trefoil (*Lotus corniculatus*L.), orchard grass(*Dactylisglomerata*L.), blue grass (*Poapratensis*L.),alfalfa (*Medicago sativa* L.) and mammoth clover (*Trifoliumpratense*var. *perenne*L.) (Lorenz and Lal, 2007). After 45 years, the reclaimed soil under pasture at theSinger site stored the largest pool of C (79.9 Mg/ha), followedby Dyes Fork site under forest plantations (29.8Mg/ha) (Lorenz and Lal, 2007). Thus pasture may have ahigh potential for C sequestration in 0 - to 30- cm soildepth. Re-vegetation

with trees like *Acacia auriculiformis*and *Dalbergiasissoo*reported to had more C density inreclaimed coal mine soil in Jharkhand, India.

4. Use of biochar

In recent years, biochar gained lot of attention owing toits long-term C sequestration potential. Biochar is apyrolysed product of biomass which is produced in absence or presence of little amount of oxygen by thermaldecomposition. Agricultural grade biomass is producedbetween pyrolysis temperatures of 400 to 600° C. It is a matter of great concern that in north-west India,large amounts of rice straws are burnt annually, releasingCO_2 to the atmosphere, in addition to methane, nitrousoxide and air pollutants. In this scenario,biochar offers a significant, multidimensional opportunityto transform large-scale agricultural wastestreams of financial and environmental liability to valuableassets.

Maizebiochar showed the maximum potentiality to store morestable and least mineralizable C hence enhanced total soilC as high as 65% as compared to control (without biochar). The other kinds of biochar like pearl millet[*Pennisetumglaucum*(L.) R. Br. Emend. Stunz], rice andwheat biochars could enhance the total C in soil in thetune of 52%, 41% and 64% respectively. Biochar producedfrom peanut (*Arachishypogala*L.) shell could significantlycontribute to increase of SOC and MBC fractionsin degraded saline soils.

Biochar is highly stable against microbial decomposition and applying this to farmland has the potential to mitigate greenhouse gas emission. While corn stalk biochar significantly decreased CH_4 emissions from paddy soil, another recent report indicated that adding biochar to agricultural soil with mineral fertilizers can suppress N2O emissions without suppressing the activity of soil biota involved in N transformation processes such as mineralization or nitrification

Biological carbon sequestration

Biological C sequestration transfers C (from CO_2) in the atmosphere to biomass through photosynthesis and ultimately stores it in plants (foliage, wood, and roots) and soils. Biological sequestration encompasses various ways of using agricultural and forest land to enhance the natural storage of atmospheric CO_2. Examples include planting or preserving trees, altering crop production practices, planting vegetation in areas prone to soil erosion, and changing the way

grazing lands are managed. When forests, croplands, and grazing lands sequester C, they are referred to as C “sinks".

Forest and agricultural lands have received considerable attention as potential C sinks. In most cases, atmospheric CO_2 removal through C sequestration exceeds CO_2emissions from these land-use types, particularly from forests. According to the U.S. GHG Inventory 2006 (EPA 2008), land use, land-use change, and forestry activities comprise a net sequestration of 883.7 million metric tons (mmt) of CO_2-Eq. This represents an offset of approximately 14.8 percent of total U.S. CO_2emissions, or 12.5% of total greenhouse gas emissions. Over 84% of this net sink occurs on forest lands. Between 1990 and 2006, total land use, land-use change, and forestry net C flux resulted in a 20% increase in CO_2 sequestration, primarily due to an increase in the rate of net C accumulation in forest C stocks, particularly in aboveground and belowground tree biomass. In contrast, the agriculture sector is a net emitter of GHGs (EPA 2008). In 2006, CH_4 and N_2O were the primary greenhouse gases emitted by agricultural activities. Methane emissions from enteric fermentation and manure management represented about 23% and 7% of total CH_4 emissions from anthropogenic activities, respectively.

The use of biomass and biofuels to replace CO_2 emitting petroleum based fuels as energy sources has great potential for mitigating GHG emissions in the southeastern United States (EPA 2005). Over the coming decades, Florida forestry and agriculture could significantly offset and reduce the projected emission increases in the state. Florida’'s forest cover declined 36% between 1945 and 2002 (although it still covers 43% of the 34.3 million land acres); during the same period, crop and pasture lands increased by 22% (USDA/ERS, 2006). Improved management of forests and agricultural lands could provide an effective tool to help stabilize atmospheric GHGs.

Benefits of carbon sequestration in soils:

Soils are thelargest carbon reservoir of the terrestrial carbon cycle. It stores large amount of soil organic carbon (SOC), which is originated from plants and animal tissue that exist atdifferent stages of decomposition. According to Batjes and Sombroek (1997), soilscontained1550 Pg of organic C upto 1 m depth, 2500 Pg of organicC upto 2 m and 750 Pg of inorganic carbon at 1 m depth. This total soil C pool of 2300 Pg is three times the atmospheric pool of 770 Pg and 3.8

times the vegetation pool of 610 Pg. Carbon storage in soils is the balance between the input of dead plant material (leaf and root litter) and losses from decomposition and mineralization processes (heterotrophic respiration). Carbon is sequestered in soils in direct and indirect ways(Soil Science Society of America, 2001). Direct soil Csequestration occurs by inorganic chemical reactions CO_2 into soil inorganic C compounds such as calcium and magnesium carbonates. Indirect plant C sequestration occurs as plants photosynthesize atmospheric CO_2 into plantbiomass. Some of this plant biomass is indirectly sequestered as SOC during decomposition processes. Removing CO_2 from the atmosphere is only one significant benefit of enhanced carbon storage in soils. The benefits of atmospheric carbon sequestration into SOM are as follows:

Improved soil quality through enhanced fertility, soil structure and aggregate stability

- ✓ Improved soil quality through enhanced fertility, soil structure and aggregate stability
- ✓ Increased water holding capacity
- ✓ Decreased nutrient loss
- ✓ Reduced soil erosion
- ✓ Increased capacity to reduce the toxic elements from the
- ✓ Increased crop production